# OECD ECONOMIC SURVEYS

# GREECE

1997

**SPECIAL FEATURES**

• SOCIAL SECURITY:

PENSIONS AND HEALTH CARE

# OECD ECONOMIC SURVEYS

## 1996-1997

GREECE

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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# BASIC STATISTICS OF GREECE (1995)

## THE LAND

| | |
|---|---|
| Area (sq. km) | 131 957 |
| Cultivated area (sq. km) | 39 290 |

| | |
|---|---|
| Major urban areas, 1991 census (thousand inhabitants) | |
| Greater Athens (including Piraeus) | 3 073 |
| Greater Thessaloniki | 978 |

## THE PEOPLE

| | |
|---|---|
| Population (thousands) | 10 459 |
| Number of inhabitants per sq. km | 79.3 |
| Net natural increase (thousands), 1994 | 6.0 |

| | |
|---|---|
| Total employment (thousands), 1994 | 3 789.6 |
| by sector (per cent): | |
| Agriculture | 20.8 |
| Industry and construction | 23.6 |
| Other activities, mainly services | 55.5 |

## PRODUCTION

| | |
|---|---|
| Gross national product, GNP (billion Dr) | 27 084.4 |
| GNP per head (US$) | 11 181.3 |

| | |
|---|---|
| Gross fixed investment | |
| Per cent of GNP | 19.7 |
| Per head (US$) | 2 206.4 |

## THE GOVERNMENT

| | |
|---|---|
| General government (per cent of GNP) | |
| Current expenditures | 42.9 |
| Current revenue | 37.0 |

| | |
|---|---|
| Composition of Parliament (number of seats): | 300 |
| Panhellenic Socialist Movement (PASOK) | 169 |
| New Democraty (ND) | 109 |
| Political Spring (POLA) | 11 |
| Communist Party (KKE) | 9 |
| Independents | 2 |

Last general elections: October 1993

## THE CURRENCY

Monetary unit: Drachma

| | |
|---|---|
| Currency units per US$, average of daily figures: | |
| Year 1996 | 240.7 |
| May 1997 | 271.5 |

*Note:* An international comparison of certain basic statistics is given in an annex table.

*This Survey is based on the Secretariat's study prepared for the annual review of Greece by the Economic and Development Review Committee on 5 June 1997.*

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*After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 24th June 1997.*

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*The previous Survey of Greece was issued in July 1996.*

# Assessment and recommendations

***Accelerating the reform process is more pressing***

Since 1990, Greece has made continuing progress in correcting its long-standing macroeconomic imbalances. Inflation and interest rates have been reduced significantly, and fiscal consolidation has been sufficient to broadly stabilise the government debt at around 110 per cent of GDP. Nevertheless, the budget deficit as a percentage of GDP and the inflation rate remain 3 percentage points higher than the respective EU averages. With the pace of policy implementation having slowed in 1996, notably in the area of fiscal consolidation and structural reform, moving expeditiously ahead with reforms has become more pressing.

***Strong domestic demand drives output growth...***

Inflation has been brought down in the context of an acceleration of economic activity and this has provided a boost to consumer and business confidence. Output growth reached 2½ per cent in 1996, which was considerably above the EU average. Investment-led growth was driven by the expansion of EU-supported public sector investment, and residential construction recovered strongly following a five-year recession. Private consumption continued to gather strength, boosted by strong real wage growth and rising employment. Strong domestic demand was partly offset by weak exports. Despite brisk job creation, the unemployment rate rose to 10.4 per cent as the economic recovery drew more people into the labour force.

***... while cost-push pressures restrain disinflation gains and impair competitiveness***

Notwithstanding stronger activity and persistent cost-push pressures, the underlying inflation rate (excluding energy and fresh food products) has followed a gradual, yet steady, downward path, and declined by about 2 percentage points during the past twelve months to 6.6 per cent in May 1997. The headline Consumer Price Index (CPI) rate, which has been more volatile, fell further during the same period, to 5.4 per cent – the lowest rate since 1973. The stability of the exchange rate played a critical role in these developments as did the relatively small adjustments to administrative prices and excise taxes at the beginning of 1997. In the context of a broadly stable nominal exchange rate and unit labour costs outpacing those in major trading partners, exports and tourism suffered further losses of market share. Though the balance of payments under-records exports, especially to the Balkan area, they appear to be languishing in their traditional markets. With the difference in business cycle developments between Greece and abroad also playing a role, the current account deficit rose markedly, to 3.7 per cent of GDP in 1996. The enhanced credibility of the exchange rate policy combined with large interest rate differentials has prompted large private sector foreign currency borrowing and attracted foreign portfolio investment, so that the level of foreign reserves increased strongly over the same period.

***Prospects for 1997-98 remain positive...***

Output growth should continue to strengthen on the basis of strong domestic demand, especially for investment, and could be 3 per cent in both 1997 and 1998 according to OECD projections. The Greek authorities project even stronger output growth; 3½ per cent in 1997 and 3¾ per cent in 1998. The sources of growth will be as in 1996: lower interest rates, strong income gains, and augmented inflows of EU funds. Overall, strong domestic demand will be partly offset by the negative drag from net exports, as

export growth is projected to be hesitant compared to the expected recovery in world markets. Inflation should continue to decline gradually, as the stable exchange rate policy will continue to hold price increases down, but labour cost pressures will persist, though unit labour cost growth is expected to slow in nominal terms. The national collective agreement for 1997 provides for increases of 3 per cent above the government's inflation target, and private sector wages could be influenced by large increases in civil servants' pay. With firms aiming to contain labour costs by raising productivity, the unemployment rate should remain broadly stable at 10½ per cent. One important risk to the projections is stronger-than-envisaged wage push pressures during the recovery, with compressed business profits then dampening investment. Another risk is that, in the event of slippages in the implementation of the ambitious fiscal policy targets, confidence could be undermined. On the positive side, the recovery in tourism could be stronger than envisaged. Moreover, rapid implementation of structural reform, such as in the area of privatisation, and a positive conclusion of the social dialogue on employment policies would boost confidence.

***... but the policy mix needs rebalancing...***

The design of macroeconomic policies under the 1994-99 convergence programme consisted of a three pronged strategy. First, fiscal consolidation consistent with bringing the general government deficit down from 14 per cent in 1993 to below 3 per cent of GDP in 1998, which would reverse the adverse debt dynamics. Second, the use of the exchange rate as a nominal anchor to guide inflation expectations and pull down inflation. Third, civil servant wage norms to influence private sector wage negotiations and thus support the exchange rate policy. This strategy has undoubtedly proven successful in reducing inflation and interest rates while permitting an investment-led recovery. However, the

policy mix remains unbalanced. After several years of considerable real wage losses up to 1993, incomes policy – perhaps inevitably – has become more lax. In addition, following a string of years of budgetary consolidation, fiscal policy performance in 1996 was below expectations, with the primary balance remaining broadly unchanged. Without the support of incomes policies and fiscal retrenchment, a policy of keeping the nominal exchange rate broadly stable – though successful in containing inflation – resulted in a further appreciation of the real exchange rate, which adversely impacted on the current account. Specifically, relative unit labour costs have appreciated by 20 per cent during the past three years; they currently exceed the peak attained in 1990 by 3 per cent. This index, however, overstates the loss of overall competitiveness as it does not take into account the decline in financial and input costs. The Government's objectives for 1997 provide opportunities to redress the policy mix, with fiscal policy aiming at a significant primary adjustment.

***... and structural reforms need to be accelerated***

The Government has launched discussions with social partners on wide-ranging structural reforms with a view to improving the functioning of the labour market. Further reform is clearly needed, as the legislation passed in late 1996 does not address many of the labour market rigidities identified in *Implementing the OECD Jobs Strategy* that appeared in last year's *Survey*. The most important recommendations arising from that analysis were to improve wage flexibility, especially by lowering the minimum wage for the more vulnerable workers and simplifying the multi-tiered wage bargaining process, to reduce employment security provisions, and to focus active labour market policies on programmes for the long-term unemployed. Structural reform needs to be accelerated in several other critical areas so as to boost output, assist in reducing

inflation, and improve competitiveness. In particular, public sector management needs to be improved, especially as regards the operations of public enterprises and entities outside central government, and in state-owned banks (see below). The reform process would probably be greatly facilitated by an improvement in the quality and timely availability of national data.

***Capital inflows plagued monetary policy...***

Monetary conditions tightened considerably during 1996. Progress in reducing inflation was largely achieved by maintaining the exchange rate stable on average against the ECU. However, the Bank of Greece had to sterilise large capital inflows ensuing from this policy. The increase and broadening of reserve requirements for the second time in 12 months in mid-1996 was temporarily successful in mopping up the excess liquidity in the interbank market, but it had an adverse effect on banks' intermediation spreads. Moreover, with increased competition among banks and aggressive pricing of government paper, deposit rates, adjusted for inflation, have been pushed down to their lowest level since the liberalisation of interest rates. Lower rates may have been instrumental in propelling the expansion of the housing and stock markets, as well as consumption. In addition to the private sector sources of liquidity injection, the Bank of Greece had to absorb the liquidity created by the large scale conversion into drachma, concentrated at year-end, of the Government's accumulated foreign borrowing.

***... and resulted in policy adjustments***

Faced with these difficulties, the Bank of Greece's monetary strategy for 1997 contains several changes aimed at discouraging capital inflows, especially those of a short-term nature. Most importantly, exchange rate policy will be conducted more flexibly around an objective of broad stability, with the reference currency no longer the ECU but

an unspecified basket of European currencies. In this regard, monetary policy will be more responsive to progress made towards the government's target to reduce inflation below 3 per cent by the end of 1998. Finally, the Bank of Greece's new intervention policy attempts to discourage short-term capital inflows toward the interbank market. Henceforth, weekly repo auctions will replace the overnight facility as the main intervention instrument, and deposits at its overnight window in excess of a global banking system limit will be remunerated at a lower interest rate. While such innovations could provide short-run relief, the permanent solution nevertheless lies with progress in reducing the large interest rate differential, which partly reflects a public sector borrowing requirement which was of the order of 10 per cent of GDP in 1996. Meanwhile, a government policy of replacing external with domestic debt would work to partly offset the liquidity pressures arising from capital inflows. Increased credibility from granting independence to the Bank of Greece should also improve the effectiveness of monetary policy. The relevant legislation was submitted to Parliament in May 1997.

***Following a 1996 fiscal outturn below expectations...***

Rebalancing the policy mix will depend critically on the successful implementation of the 1997 fiscal policy. While it should be acknowledged that national elections and year-end strikes may have raised the deficit, the 1996 Budget clearly fell short of its objectives, and the envisaged primary surplus improvement of 1 per cent of GDP was missed by a considerable margin. The shortfall in tax and non-tax revenues from ambitious targets reflects the difficulty of tackling tax evasion, with the use of presumptive taxation and tax amnesties gradually providing diminishing yields. On the expenditure side, civil servant wages rose significantly after several years of losses in purchasing power up to 1993 and interest rates did not decline as

quickly as foreseen. While the general government deficit fell from 9.2 per cent of GDP in 1995 to 7.4 per cent of GDP in 1996, thereby attaining the convergence programme fiscal target, this result was achieved only after excluding from the Budget certain (previously included) payment items amounting to 1.3 per cent of GDP. These accounting changes comply with Eurostat recommendations for equal treatment of the EU member states. Despite a primary surplus of the order of 4 per cent of GDP, the debt to GDP ratio remained broadly unchanged as the Government continued to assume the debts of various loss-making public entities outside general government.

***... full implementation of the 1997 Budget is critical***

In 1997, fiscal policy aims at increasing the primary surplus by more than 2 per cent of GDP by broadening the revenue-enhancing strategy conceived over the past three years, and by adding critical new components which cut current primary expenditure. In addition, a more equitable tax system has been implemented, with a broader base for direct taxes and a reduced emphasis on indirect taxes. Thus, the 1997 Budget eliminated or reduced many tax expenditures, and introduced taxes on real estate and income from government securities. Regressive excise taxes, which have adverse affects on inflation expectations, were left broadly unchanged. A better management of public assets will be used to raise non-tax revenues. On the expenditure side, limits on hiring, increased use of means-tested welfare programmes, and the tightening of the budgets of public entities are expected to reverse the continued upward creep of current primary expenditures. In addition, lower interest rates and the use of zero-coupon bonds should provide interest savings exceeding 1 per cent of GDP. Acknowledging that the initial yield from the new measures is difficult to assess, but that the measures should gradually widen the tax base and reduce expenditure inefficiencies, the OECD

Secretariat projects that additional measures equivalent to 1 per cent of GDP will be required for the Government to attain this year's fiscal target of a general government deficit of 4.2 per cent of GDP. However, the Greek Government expects that the budget target will be met without additional measures, based on the fiscal performance over the first five months of the year, and the prospects for larger-than-envisaged savings on interest payments and unbudgeted non-tax revenues arising from the sale of government assets.

***Further fiscal adjustment needs to focus on expenditures...***

Following the measures introduced in the 1997 Budget, there appears to be limited scope to broaden the tax base expeditiously, as many avenues for tax avoidance have been closed, leaving the fight against tax evasion, where revenue gains are usually more gradual. The Government intends to place more emphasis on achieving a smaller but more efficient public sector. The recent initiatives to use current primary expenditures more effectively are appropriate steps in this direction and their rapid implementation appears critical. Nevertheless, larger funding by the Community Support Framework II requires maintaining the level of investment expenditures, and defence spending is unlikely to shrink. The choice is thus basically to limit the remaining bulk of expenditure, namely public workers' compensation, interest payments, and social security expenditure. Immediate results could be obtained through rationalisation, since personnel outlays account for one-third of current primary expenditure. The hiring norms announced in the 1997 Budget combined with the introduction of a new salary scale do not appear sufficient in this respect as they do not provide significant budgetary savings. With wage constraints having demonstrated that they provide only temporary relief, and the Government committed to maintaining the level of personnel in health, education, and

security, few options remain, apart from pursuing more fundamental public sector management reforms. These would probably lead to a reduction in the absolute number of personnel, and to performance-based pay, which could widen pay scales at both ends, accepting that to retain highly qualified staff requires commensurate compensation.

***... and debt reduction***

A rapid reduction in interest payments will require attacking the high level of general government debt which contributes to a large, albeit now declining, risk premium. A front-loading of fiscal adjustment, which has been successfully pursued by a number of smaller EU countries, would lead to a higher primary surplus, lower debt and interest payments. Other measures to lower debt directly also should be implemented, particularly where these would result in overall gains in economic efficiency. The Government so far has been rather hesitant to privatise, in part for employment considerations, and has preferred a strategy which aims to improve the efficiency of public enterprises and to reap the benefits in the form of non-tax receipts. However, EU-wide commitments to open the important utility and transportation sectors to competition will erode monopoly profits. Regarding loss-making public enterprises, most of which are involved in urban transport, little progress has been made in reducing losses and the level of subsidies. Finally, the privatisation of state-owned banks would improve their efficiency and thus reduce their wide intermediation spreads. Privatisation of public enterprises and banks would also eliminate the need for further large capital injections and debt assumptions, which have pushed up government debt significantly.

***Future problems for the budget will stem from pensions***

The pension system – described in detail in the special chapter of the *Survey* – is the largest item of the budget, accounting for 40 per cent of total current primary expenditure of the general government and its large and rapidly rising expenditure outlays crowd out other social expenditures. Total pension expenditures exceed contributions by some 5 per cent of GDP, with the gap covered by earmarked taxes and budgetary transfers. Moreover, demographic trends less favourable than in most other OECD member countries, combined with the pension system's continued maturation and inherent generosity will add enormously to prospective budgetary pressures. The OECD estimates that the pay-as-you-go system's unfunded liabilities are among the highest in the OECD. Early reform is indispensable, especially because changes to many parameters of the pension system can only be implemented gradually, often taking several decades. Delaying further reform much longer would necessitate far more abrupt and painful measures in the future than those that could be phased in now.

***The main problems of the system***

The current imbalances of the pension system stem from four long-standing principal characteristics, which apply in various degrees to the different institutions of a highly complex and segmented system. First, loose eligibility requirements have resulted in a large number of pensions. The number of primary pensions exceeds the population over 65 years old by 30 per cent. Second, pensions are generous in comparison to contributions, though most are not high in absolute terms, and there exist significant inequities because some schemes are more generous than others. Overall, the implicit rate of return on contributions is high, especially for workers with a short contribution history. Third, the system provides incentives for contribution evasion, most importantly through a combination of a

relatively high minimum pensions and low eligibility requirements. Fourth, segmentation in conjunction with poor administration permits abuse of the system and provide obstacles to the implementation of reforms.

***Some reforms have already been taken***

The 1990-92 reforms have provided temporary budgetary relief and under current policies, including the lagged effect of certain provisions of the reform legislation, should contain the system's gross financing needs to 6 to 7 per cent of GDP over the next ten years. The 1990-92 reforms provided for a substantial increase in contribution rates (and their introduction in the case of civil servants), revised the indexation system, tightened eligibility and raised retirement ages, especially for public sector employees, and introduced less generous and unified pension benefits for those commencing work after 1992. However, the resulting large increase in non-wage labour costs has been identified as an important disincentive to hiring. In response to a loss in pensions' purchasing power – amounting to over 20 per cent over the last six-year period – minimum pensions have been indexed to the CPI from 1996. Moreover, the Government introduced in 1996 a means-tested old-age minimum pension supplement for low income pensioners over 65 years of age based on uniform eligibility criteria. Finally, even the new regulations applying to individuals having started work after 1992 do not appear sufficient to return the system to viability. The authorities have acknowledged the need for reform to safeguard the system's continued ability to pay public pensions in the face of demographic and other challenges. To this end, a national debate has been launched to discuss reform measures.

***Options for further reform***

When investigating options for reform, the entitlements of the current generation of workers pose a major difficulty as they comprise approximately two-thirds of the pension system's unfunded liabilities. Most continental European countries, which face a similar yet less daunting task, have chosen to continue with the pay-as-you-go system and to implement gradual changes in its generosity so that the burden of adjustment is borne by both the current and future generations. Reforms to pay-as-you-go systems have concentrated on changes in eligibility requirements and generosity, as further increases in contribution rates are ruled out for employment considerations. Some countries have also opted for a partial funding of pensions. However, an overall systemic shift to a partially funded system is more readily implemented in an economic environment which can withstand the additional fiscal adjustment and the higher debt level resulting from the allocation of contributions to the funds, while current pensioners still receive their due benefits.

The most common reforms to a pay-as-you-go system involve a reduction of the gap between the value of individual pension payments and contributions by providing incentives to delay retirement. Such a delay is achieved by reducing the rate at which the pension rights are accumulated over a career. In Greece, the contribution period is short, averaging about 20 to 25 years for private and public sector employees as compared with up to 40 years elsewhere. Moreover, for private sector employees, the pension is based on earnings over the last five years of the career – the shortest pension base in the OECD – while for public servants, a pension is based on the last month's salary. The benefit accumulation rate, even after the full implementation of the 1990-92 reforms, will remain too high for the system to be sustainable. It will stand at 2.3 per cent per year of contributions, while it needs to be closer to two-

thirds that rate (and calculated on a more reasonable base) for pensions to come in line with contributions, according to OECD estimates.

In the case of Greece, eligibility can be tightened by eliminating some of the system's more blatant distortions. Perhaps most importantly, the eligibility requirement of 15 years for a pension in the private sector and the minimum pension need to be phased out and replaced by an extension of the recently instituted means-tested supplement that would protect low-income pensioners. The original rationale for a low eligibility requirement, which was based on the large number of non-qualifiers, especially due to urbanisation, is now less valid and the interaction with the high minimum pension is instrumental in providing an incentive for contribution evasion and early retirements. Though admittedly significant progress has been achieved in addressing abuses of the system (especially access to invalidity pensions), the 40 per cent of private sector employees classified as working under ''arduous and unhealthy conditions'' suggests continued abuse of this provision. In view of the introduction of new technologies in most professions, this provision should also be eliminated in all but truly exceptional cases. Finally, more generous eligibility conditions for married women, and especially those with under-age children (even after the full implementation of the 1990-92 reforms), conditions for receiving a survivor's pension, and highly subsidised farmers' pensions should all be means tested, which implies improving administrative capacity.

***The excess segmentation of the pension system should also be addressed***

A reform of the pension system will also need to address the issue of the unification of the large number of funds, as well as their benefits. Most arguments support consolidation of the number of funds as it enhances equity, facilitates workers' mobility, and lowers administrative costs signifi-

cantly. It would also reduce abuses of the system through improved monitoring, especially the receipt of multiple pensions. In view of the need to reduce the generosity of the system, the concept of a unified system across sexes, sectors and types of pensions could become more palatable. Unification would make the system more equitable, with the most generous pension benefits adjusting proportionally more.

***Dissatisfaction with the public health system***

Though the public health system does not face the same explosive situation as the pension system, outlays consume an increasingly larger share of government revenues, and are likely to rise further as the population ages. Despite the increased supply and quality of services and a health status of the population which does not compare unfavourably with the other OECD countries, the overall performance of the public health system, as expressed by public opinion surveys, is judged to be unsatisfactory. Problems are an insufficient amount of preventive care and a still unequal provision and coverage of services. Considerable imbalances have been created due to the lack of competition among suppliers of health services and the fixing of most prices for health care by government at unrealistically low levels. The current inefficiencies of the health system at a time when budget austerity is needed make reform of the health care system an immediate priority.

***... has induced new reforms, which aim to improve primary care...***

The implementation of the measures contained in the draft law presented to Parliament should address several shortcomings of the system. Most importantly, the introduction of a network of primary care centres should reduce the geographical inequities of the system and provide a missing gatekeeper to specialists and outpatient departments of public hospitals. However, the method of remuneration will be a critical component of the network's success, especially as

it requires attracting high quality doctors to rural and semi-urban areas. For this to occur, the career salary of primary care doctors should provide adequate incentives for them to practise outside the large urban centres. International experience suggests that a mixture of different forms of payment – capitation, salary, or fee-for-service – are best in providing incentives to perform high quality services.

***... and the management of hospitals***

The legislation will introduce professional management and global budgets for public hospitals. These innovations should improve management, which currently suffers *inter alia* from inflexible line-budgeting determined by public officials outside the hospital, soft-budget constraints, and poor procurement procedures. Nevertheless, it remains for future reform to address a main source of the inadequate service, which derives from the public hospital staff's job security – which is the same as for civil servants – and salaried compensation, which is unrelated to performance. More flexible salary and employment structures for medical employees should be considered. Acknowledging that public doctors bypass regulations that prohibit a supplementary practice in the private sector, permitting doctors to have a limited, and strictly enforced, private practice may provide the incentive for doctors to forfeit voluntarily their status similar to that of a civil servant. Moreover, it could compensate doctors for salaries below their private sector counterparts, and limit the recourse to unofficial payments by patients for obtaining access to hospital services. As with primary care, a combination of different methods of payments to replace a salaried remuneration should raise the quality of service.

An alternative to global budgeting would be to provide the budgetary resources currently targeted directly at hospitals to the social security funds to ''purchase'' services from the public hospitals, reflecting the system's original design.

Additional competition could be provided to this internal market, if prices were not fixed by the State. Though supplementary competition would be provided by the existence of more than one fund, this approach would require eliminating sectoral affiliation of the funds, a unification of minimum benefits provided, and the separation of health funds from their parent pension funds. In addition, measures to prevent funds from ''cream skimming'' for healthier (and thus less costly) members would be necessary. Such an alternative approach would require significant advances in administrative and regulatory capacity.

***High expenditure on pharmaceuticals has led to new measures***

The high level of expenditure on pharmaceuticals – the second highest in the OECD as a per cent of GDP – reflects a high prescription rate by doctors and high prices for drugs. The introduction of a unified positive list for all health funds, including daily treatment costs, is a first step in reducing consumption. Improved monitoring of individual doctors' prescriptions would also help in containing outlays on drugs. High drug prices could be reduced by the elimination of the large administratively-fixed profit margins for pharmacists and wholesalers as well as various earmarked taxes. Further reductions in prices could be attained through the promotion of over-the-counter and generic drugs which are not common in Greece. The existing mark-up pricing system whereby the price of domestically produced drugs is more stringently controlled has created a bias against domestic production and has promoted a surge in imports. The pricing system should be revised to eliminate price discrimination against domestic producers. Measures currently under consideration by the authorities move in the directions suggested above.

***Summing-up***

Having made commendable progress during the first half of the 1994-99 convergence programme, the economy is posi-

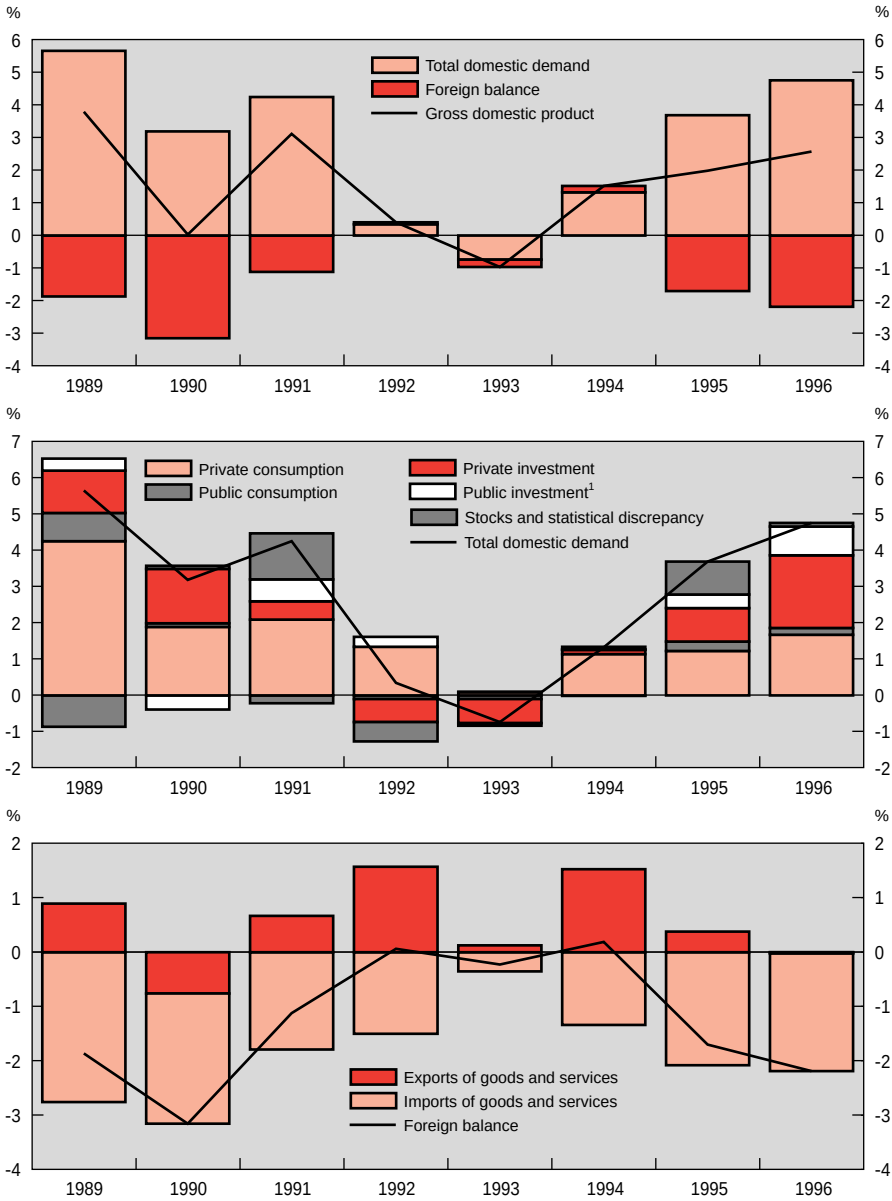
tioned to move towards the programme's final objective: sufficient convergence to gain entry into the European Monetary Union. With Greece still having a considerable distance to cover in this respect, the objectives of the second half of the convergence programme are necessarily more ambitious. Further progress in reducing large macroeconomic disequilibria will require removing the numerous long standing structural impediments in labour markets, financial markets, and public sector management that still remain. The new Government is addressing many of these issues but expeditious decisions and swift implementation will be needed for the adjustment strategy to remain viable.

# I. Recent trends and short-term prospects

The economic environment has improved during the past few years. Following a pronounced downturn in 1992-93, output has recovered, with growth performance gradually overtaking the EU average. Over the same period, disinflation gains have been substantial. Supported by the exchange rate policy, the headline inflation rate has been halved since 1993, and fell to single digits in 1995. It, nevertheless, remains significantly above the EU average. Reflecting both the relative price and demand factors, the current account has moved from near balance in 1993-94 into a large deficit position in 1995.

The Greek economy gathered strength in 1996, in contrast to most EU countries. The strengthening of both private consumption and public sector investment were the main driving forces, while an additional stimulus was provided by the revival in housing investment. As a result, total domestic demand grew by over 4 per cent in volume terms, the highest rate during the past six years.[1] However, the slowdown in export growth combined with the demand-induced surge in imports resulted in net exports acting as a large drag on GDP growth (Figure 1). The strengthening of the economy has been accompanied by solid employment growth, though the unemployment rate continued to rise as the labour force expanded during the recovery. Notwithstanding the growing slack in the labour market, wage growth in real terms exceeded productivity gains. The cost-push pressures from the labour market have been outweighed by subdued prices for imported inputs and declining financial costs, which permitted further progress in reducing inflation. The difference in business cycle positions between Greece and its main trading partners, combined with the losses in external competitiveness, resulted in a further widening of the current account deficit. This shortfall was, nevertheless, more than fully covered by capital inflows.

Figure 1. **CONTRIBUTIONS TO REAL GDP GROWTH**




1. General government and public enterprise investment.
*Source:* Ministry of National Economy.

## The strengthening of domestic demand was broadly based

Private consumption continued to recover strongly in 1996, prompted primarily by rapid real wage growth (a cumulated 7 per cent increase over the period 1994-96). Additional impetus was provided by job creation and declining interest rates on consumer loans. As a result, consumption of durable goods rose sharply, with purchases of new passenger cars (almost all of which are imported) increasing by around 10 per cent for the second consecutive year. Measures adopted in June 1996 succeeded in curbing the surge in consumer loans which had been supporting consumption.[2] Nevertheless, sales have remained strong. Indeed, registration of new cars was largely unaffected by the doubling of the registration tax on passenger vehicles. Strong household income gains also provided a boost to housing investment, which rose by 18 per cent in volume terms, following a five-year recession. This turnaround was supported by better mortgage credit conditions, which contributed to the 27 per cent expansion in housing credit. In addition, the sharp fall in yields on Treasury bills and time deposits in real terms during 1996 may also be increasing the attraction of housing investment.

In 1996, total business investment expanded at a two-digit rate in volume terms. Declining financial costs and low import prices permitted firms to maintain healthy profits despite the rise in unit labour costs exceeding inflation. Financial costs were eased by lower domestic interest rates, but perhaps more importantly by enhanced access to foreign borrowing. In addition, government financial subsidies to private business investment, though slightly lower than in 1995, still supported 4½ per cent of business investment in 1996. Nevertheless, public enterprise investment was the most dynamic component of business investment, with its share in total business investment jumping from 15.2 per cent in 1995 to 18.1 per cent in 1996, and accounting for one-third of business investment growth (Table 1).

Public demand continued to provide a critical stimulus to total demand. Public consumption grew for the second year in a row, after several years of contraction (from 1991 to 1994), reflecting a sharp rise in general government employment of the order of 2½ per cent in 1996. Continuing rapid growth of fixed investment in the general government sector was strongly supported by the Community Support Framework programme (CSF II) and large infrastructure building activity was in full swing. Efforts to facilitate the timely utilisation of

Table 1. **Public investment: key features**

Percentages

| | 1995 | 1996 | 1997 Budget |
|---|---|---|---|
| Public investment (as a share of total fixed investment)[1] | 23.5 | 24.2 | 25.5 |
| *Of which:* | | | |
| General government | 13.9 | 12.9 | 13.4 |
| Public enterprises | 9.5 | 11.3 | 12.1 |
| Share of public enterprises in business investment | 15.2 | 18.1 | 19.7 |
| Public investment as a share of GDP[1] | 4.8 | 5.2 | 5.8 |
| Public investment co-financed with the EU (as a share of total public investment) | 64.1 | 71.4 | 72.3 |
| EU funds directed at public investment (as a share of total public investment)[2] | 33.4 | 48.9 | 42.0 |
| *Memorandum items:* | | | |
| Subsidies to business sector investment (as a share of total business investment) | 6.0 | 4.5 | . . |
| Rate of absorption of EU structural funds | 71 | 75 | . . |

1. On the basis of national accounts data.
2. On the basis of Budget data. Includes fixed capital investment and capital transfers to public and private entities.

*Source:* Ministry of National Economy.

European Union (EU) structural funds assisted in raising their rate of absorption to an estimated 75 per cent of the annual target set out under the CSF II, somewhat higher than in 1995.[3] As a result, EU funds accounted for 49 per cent of total public investment in 1996, compared with 33 per cent in the previous year.

## Industrial production was weak

Despite the surge in domestic demand, industrial production stagnated in 1996 and even fell in the latter part of the year. The decline partly reflected the impact of road blockages by striking farmers in the wake of the 1997 Budget presentation. However, other factors, in particular the strong rise of relative unit labour costs in the context of the stable drachma policy, were playing an important role. Indeed, some labour-intensive industries, with heavy exposure to international competition, turned in a poor performance in 1996.[4] In contrast, construction activity was buoyant, reflecting the accelerated implementation of

public investment projects and a boom in residential construction. Overall, construction was the most buoyant sector of production, as manufacturing stagnated, agricultural production declined, and services remained weak.

Though total domestic demand grew by a firm 4.2 per cent in 1996, GDP growth was only 2.6 per cent due to the sizeable deterioration in net exports. The negative contribution to GDP growth of net exports amounted to 2.2 percentage points, partly due to the stronger business cycle position but also the real appreciation of the drachma which led to further losses in export market shares and largely explains the weak response of manufacturing output to rising domestic demand.

## Employment creation was strong

Net job creation accelerated in 1996 with total employment growing at 1.4 per cent compared with 0.9 per cent in 1995 (Table 2). Reflecting an

Table 2. **Labour market: recent developments**

Annual percentage changes[1]

| | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|
| Civilian labour force | 2.1 | 1.8 | 1.3 | 1.8 |
| Participation rate | 59.9 | 60.6 | 62.7 | 63.5[2] |
| Unemployment rate | 9.7 | 9.6 | 10.0 | 10.4 |
| Total employment | 1.0 | 1.9 | 0.9 | 1.4 |
| *Of which:* | | | | |
| Agriculture | –1.6 | –0.5 | –1.0 | –0.3 |
| Industry | –10.0 | –0.5 | –0.9 | . . |
| Services | 7.9 | 3.9 | 2.4 | . . |
| Dependent employment | 2.2 | 1.9 | 2.1 | 2.2[2] |
| Dependent private sector | 2.8 | 2.1 | 2.4 | 1.0 |
| Public sector[3] | 0.9 | 1.5 | 1.5 | 2.5 |
| *Of which:* General government[3] | 0.9 | 0.3 | 2.0 | 3.7 |
| *Of which:* Education[3] | 3.0 | 3.5 | 2.1 | 2.7 |
| Public enterprises and state owned banks[3] | 1.1 | 4.4 | 0.3 | –0.6 |

1. Except for the participation rate and the unemployment rate.
2. Estimates.
3. End of period.

*Source:* Data submitted by the National authorities and OECD estimates.

improved hiring climate in the private sector, manufacturing employment recorded a slight increase in the first half of 1996 following substantial labour shedding from 1989 to 1994 and a near stabilisation in employment in 1995. A large further impetus to job creation was hiring by general government. With total employment creation remaining brisk, productivity rose by only 1¼ per cent. Nevertheless, unemployment rose by almost one half of a percentage point in 1996 to 10.4 per cent.[5] Migration flows – partly related to social and economic developments in Albania and other parts of Eastern Europe – boosted the working age population while labour force participation increased in response to the recovery in output. The labour supply was possibly also influenced by the strengthening of active labour market policies, which provide training incentives. In the event, the participation rate rose from slightly below 60 per cent in 1993 to 63.5 per cent in 1996 – a level which remains low by international standards.

## Disinflation continues despite labour cost pressures

For the third year in a row unit labour costs increased in real terms, despite growing slack in the labour market, thus limiting progress in disinflation. In part, this development reflected the fact that civil servant wages accelerated and increased by a cumulative 8 per cent above inflation over the period 1994-96 (Table 3). For 1996, civil servants' incomes policy was initially designed to be fairly tight (an increase of 2.5 per cent in January and again in July, compared with a 5 per cent inflation target at the end of the year).[6] However, civil servants' wages rose far more steeply – by nearly 15 per cent – partly as a consequence of the granting of bonuses by certain ministries. During the same period, wage awards in public enterprises and state-owned banks followed those of civil servants' remuneration. In this environment and with healthy profits, enterprises in the private sector granted wage awards in 1996 that exceeded inflation by about 2 percentage points.

Largely reflecting labour market rigidities, the disinflation process has been slow but steady, with the underlying inflation rate (which excludes energy and fresh food products) falling from 9.4 per cent in 1995 to 8.0 per cent in 1996. The headline rate measured by the CPI was more volatile and did not decline until the last two months of 1996. As a result, headline inflation fell to 7.3 per cent during 1996 compared with 7.9 per cent during 1995, and overshot the

Table 3. **Prices and wages**

Percentage changes

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|
| **Prices** | | | | | | |
| Price deflators | | | | | | |
| GDP deflator | 19.8 | 14.6 | 14.1 | 10.9 | 9.3 | 8.8 |
| Private consumption deflator | 19.7 | 15.0 | 13.7 | 10.8 | 9.3 | 8.5 |
| Consumer prices [1] | | | | | | |
| Period average | 19.5 | 15.9 | 14.4 | 10.9 | 8.9 | 8.2 |
| End of period | 18.0 | 14.4 | 12.1 | 10.6 | 7.9 | 7.3 |
| Underlying inflation [2] | | | | | | |
| Period average | . . | 16.2 | 14.9 | 11.0 | 9.4 | 8.0 |
| End of period | . . | 15.2 | 12.6 | 10.5 | 8.6 | 7.4 |
| Wholesale prices, total | 16.7 | 11.3 | 11.9 | 8.7 | 7.8 | 6.1 |
| *Of which:* | | | | | | |
| Exports | 10.7 | 6.4 | 11.3 | 8.7 | 10.2 | 5.6 |
| Imports | 14.6 | 12.6 | 12.2 | 9.2 | 7.0 | 1.6 |
| Industry | 17.7 | 14.7 | 13.4 | 7.5 | 8.1 | 7.1 |
| **Wages** | | | | | | |
| Total wage bill (national accounts) [3] | 14.4 | 10.6 | 11.0 | 14.0 | 15.1 | 13.2 |
| Unit labour cost (national accounts) [3, 4] | 11.0 | 10.2 | 12.1 | 12.3 | 12.8 | 10.3 |
| Manufacturing, blue collar [5] | 16.7 | 13.7 | 10.5 | 13.2 | 13.3 | 10.0 [6] |
| Civil servants | 10.6 | 11.7 | 12.2 | 9.4 | 11.7 | 15.3 |
| Banks [7] | 16.5 | 1.4 | 19.1 | 17.5 | 13.1 | 13.0 [8] |
| Public enterprises | 14.5 | 7.5 | 9.7 | 18.8 | 10.9 | 14.1 |
| Minimum wage, blue collar | 14.4 | 11.3 | 12.0 | 12.6 | 9.4 | 7.8 |
| *Memorandum item:* | | | | | | |
| Effective exchange rate | –11.8 | –8.3 | –9.2 | –7.1 | –3.5 | –1.1 |
| Real effective exchange rate [9] | –6.1 | –3.3 | –5.2 | 3.9 | 7.4 | 6.9 |

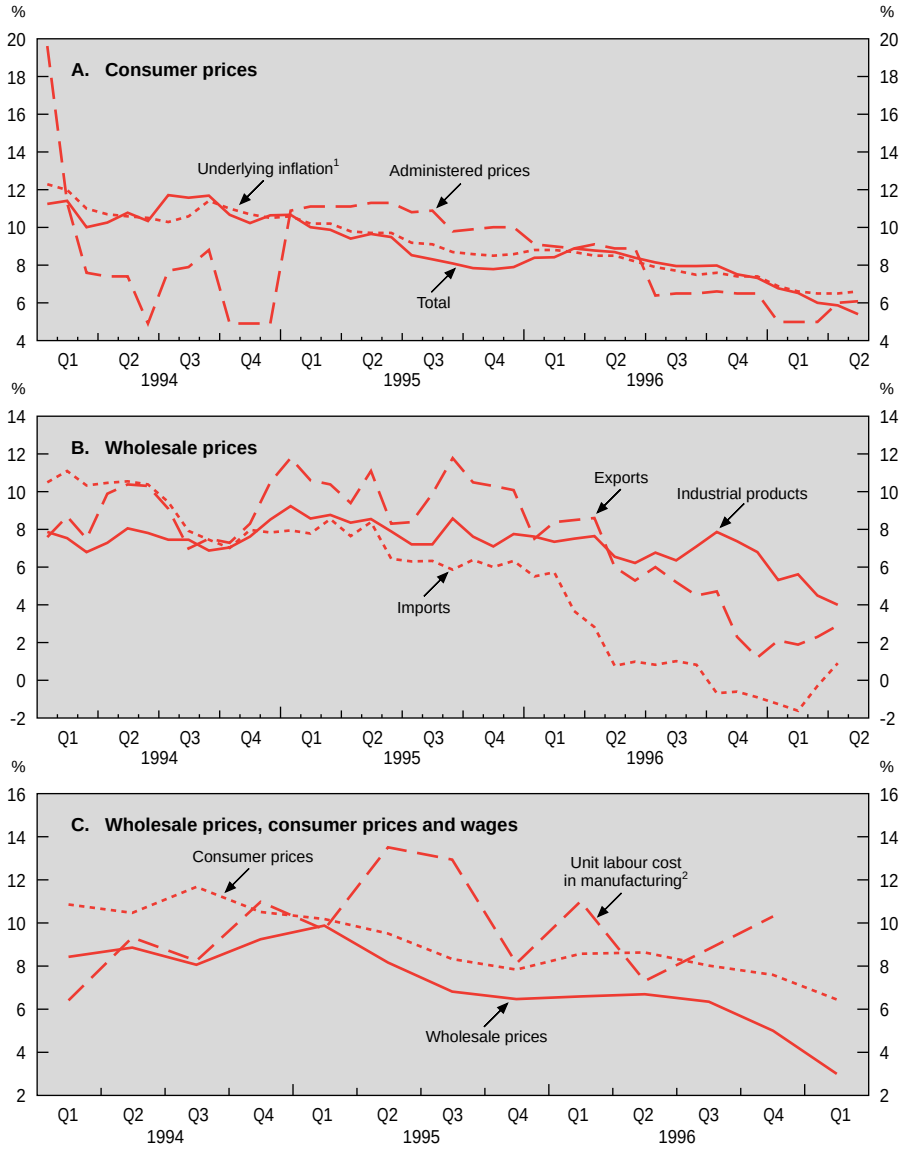
1. Based on 1988 weights prior to 1994 and 1994 weights thereafter.
2. Excluding energy and fresh food products. Based on 1988 weights prior to 1995 and 1994 weights thereafter.
3. Estimate for 1996.
4. Total compensation of salaried employees divided by real GDP.
5. Weekly earnings.
6. Annual rate for the period January to June 1996.
7. October on October.
8. Estimate based on bank's annual income statements compared with national statistical service data for previous years.
9. Unit labour cost weighted.

*Source:* Bank of Greece and Greek national statistical service.

government's target by 2.3 percentage points.[7] Apart from labour cost pressures, several transitory factors limited progress in disinflation. First, food and oil prices rose substantially in the course of 1996, thus pushing up the headline rate while the underlying inflation was receding. Second, rent increases continued to

Figure 2. **PRICES AND WAGES**

Change over corresponding period of previous year




1. Consumer price index excluding energy and fresh food products. Break in series in January 1995.
2. Bank of Greece estimates for the third and the fourth quarters of 1996.

*Source:* Bank of Greece and OECD Secretariat.

outpace inflation while the housing market recovered. To resist these forces, administrative price increases were restrained. Specifically, administrative prices were frozen during the second half of the year and rose by 6.5 per cent during 1996. In addition, a temporary reduction in fuel excise taxes was maintained throughout the last nine months of the year. Nevertheless, the fundamental factor pulling down inflation has been the ''hard drachma'' policy, which has helped to lower the cost of imported inputs. With low international prices, wholesale prices for imports rose by only 1.6 per cent in 1996, and ended the year with a negative 12-month inflation rate, (Figure 2). Reduced pressures on firms' margins also stemmed from the decline in domestic interest rates along with the easier access to cheaper foreign borrowing.

The disinflation process gathered momentum in early 1997, and in May the 12-month rise in the headline consumer price index dropped to a 25-year low of 5.4 per cent. This development largely reflected a downward correction of food prices from the peak in early 1996. Moreover, beginning-of-the-year excise tax and administrative price hikes were also lower than in 1996.[8] On the other hand, the underlying rate stood at 6.6 per cent in May 1997, one and a quarter percentage points above the headline rate, but almost 2 points lower than a year earlier.

## The current account deficit widened

The current account deficit rose significantly in 1996, climbing to 3.7 per cent of GDP from 2.5 per cent in 1995 and near balance in 1994. This deterioration reflected a widening of the trade deficit and a plummeting tourism balance. The current account deficit was far exceeded by private sector capital inflows, permitting a rapid build-up of foreign exchange reserves which covered almost eight and half months of imports at year-end 1996 (Table 4).

Exports of goods and services (national accounts basis) stagnated in volume terms in 1996, reflecting partly the economic slowdown in the main EU trading partners of Greece. However, Greece continued to lose large export market shares, both for goods and services, as a result of the loss of international competitiveness following the appreciation of the real effective exchange rate (deflated by relative unit labour costs) by a cumulated 20 per cent over the period 1993-96 (Figure 3).[9] Losses of export market shares for goods amounted to

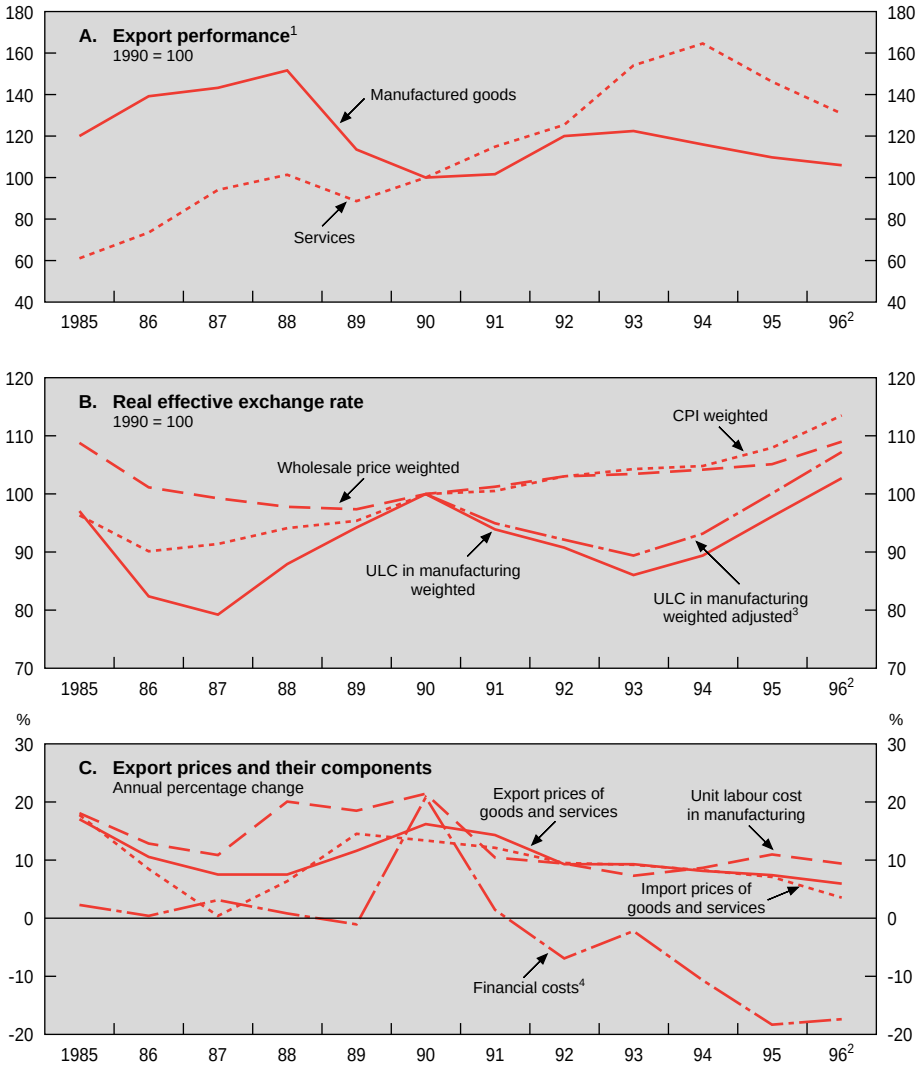
Table 4. **Balance of payments**

Million US dollars

| | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|
| **Trade balance** | –12 581 | –13 523 | –17 146 | –18 366 |
| (per cent of GDP) | (–13.7) | (–13.8) | (–15.0) | (–15.0) |
| Oil balance | –1 414 | –1 337 | –1 739 | –2 228 |
| Non-oil balance | –11 168 | –12 186 | –15 407 | –16 138 |
| Exports (f.o.b.) | 4 501 | 4 613 | 5 292 | 5 118 |
| Imports (c.i.f.) | 15 668 | 16 799 | 20 699 | 21 256 |
| **Invisible balance (net)** | 11 865 | 13 401 | 14 295 | 13 826 |
| Credits | 17 023 | 18 767 | 20 770 | 20 444 |
| *Of which:* | | | | |
| Tourism | 3 335 | 3 905 | 4 136 | 3 723 |
| Shipping | 1 920 | 1 957 | 2 190 | 2 263 |
| Migrant remittances | 2 431 | 2 657 | 3 071 | 2 996 |
| Net EU transfers | 4 085 | 4 307 | 4 968 | 5 057 |
| Debits | 5 158 | 5 366 | 6 475 | 6 618 |
| *Of which:* Interest payments, dividends and profits | 2 086 | 2 102 | 2 683 | 3 003 |
| **Current balance** | –716 | –122 | –2 850 | –4 539 |
| (per cent of GDP) | (–0.8) | (–0.1) | (–2.5) | (–3.7) |
| **Capital account (net)** | 4 400 | 6 904 | 3 162 | 8 657 |
| **Private sector** | 1 626 | 3 785 | 2 342 | 7 216 |
| Entrepreneurial | 1 981 | 2 757 | 5 396 | 10 177 |
| Real estate | 946 | 956 | 1 040 | 1 044 |
| Other long-term capital | –618 | 348 | –847 | 1 666 |
| Deposits in foreign currencies | 46 | 60 | –2 116 | –603 |
| Commercial banks and suppliers' credits | –351 | 311 | 555 | 285 |
| Amortisation | –379 | –647 | –1 687 | –5 353 |
| **Public sector** | 2 774 | 3 118 | 820 | 1 441 |
| Medium and long-term | | | | |
| Bank of Greece | 3 915 | 0 | 0 | 0 |
| Central government | 1 229 | 4 738 | 4 108 | 9 755 |
| Public enterprises and banks | 632 | 959 | 623 | 872 |
| Amortisation | –3 437 | –3 456 | –4 756 | –6 196 |
| Suppliers' credit | –11 | 3 | 0 | 0 |
| Short-term (net) | 446 | 873 | 845 | –2 990 |
| **Errors and omissions** | –663 | –415 | –342 | 77 |
| **Balance of payments** | 3 021 | 6 367 | –30 | 4 195 |
| Memorandum items: | | | | |
| Reserves (stock at year end) | 8 694 | 15 432 | 15 736 | 19 177 |
| Months of current year imports of goods and non factor services (including oil) | 5.0 | 8.4 | 7.1 | 8.3 |

*Source:* Bank of Greece.

Figure 3. **COMPETITIVENESS AND EXPORT PERFORMANCE**




1. Export performance is the ratio between export volumes and export markets on a customs basis.
2. Estimates.
3. Including social security contributions.
4. Changes in financial costs are proxied by changes in the 12 month Treasury-bill rate.

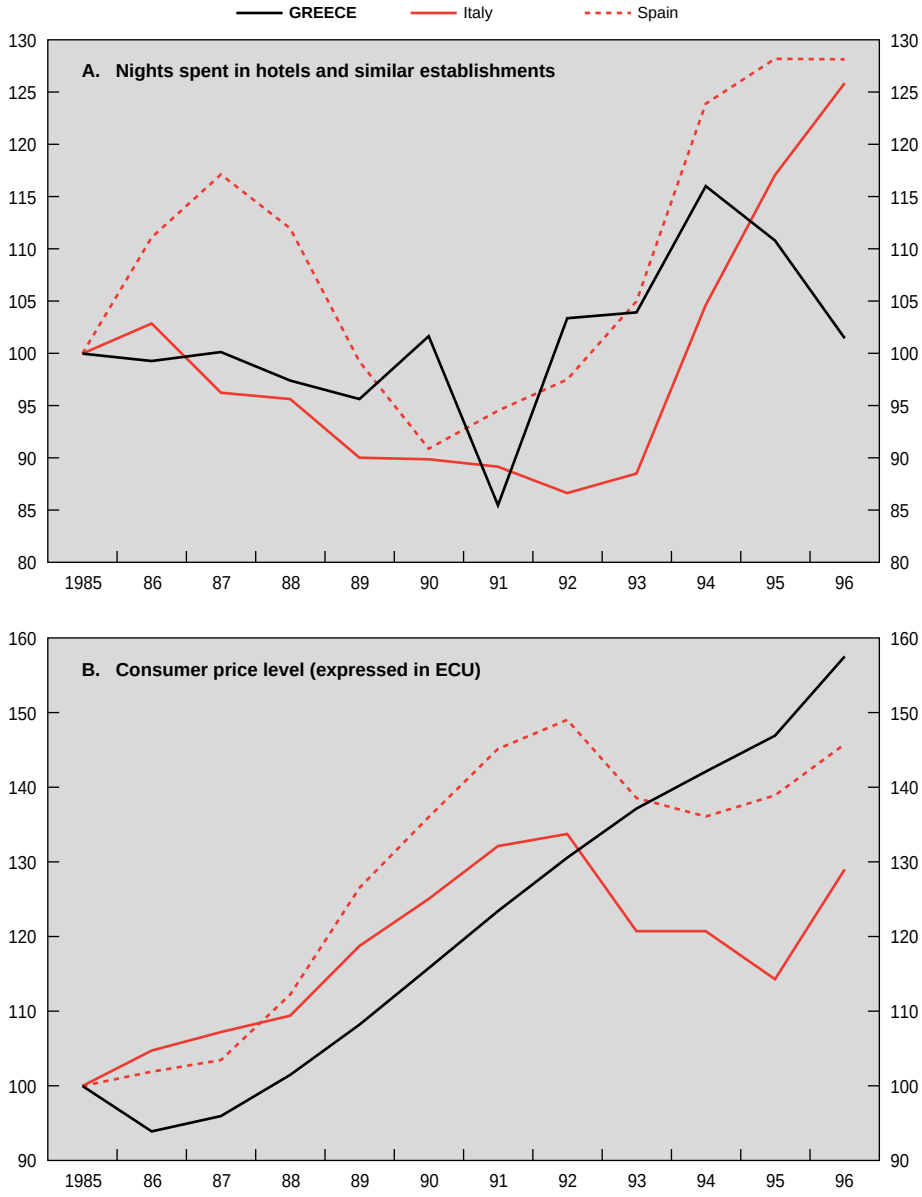
*Source:* Bank of Greece and OECD Secretariat.

around 10 per cent over the period 1993-96. When considered on a balance of payments basis, the export performance was worse. Merchandise export volumes declined by nearly 4 per cent in 1996, compared with a preliminary estimate of an increase of 2½ per cent on a national account basis. However, it is generally accepted that data on a settlement basis under-estimate exports especially as regards drachma-denominated trade with the Balkan countries and that this bias has grown steadily since the early 1990s.[10] Adding to the mixed view on export developments, the Greek authorities have indicated that fiscal sources for trade data point to substantially higher exports in 1996 than those included in the preliminary estimates of national accounts data. Though the overall level of exports may be in doubt, the composition appears to have shifted to less competitive markets outside the OECD.[11] In tourism, the drop in receipts accelerated in 1996 and amounted to about 14 per cent in real terms, almost twice the decline recorded in 1995. When the volume of tourism is measured by the number of nights spent in hotels and similar establishments, Greece recorded an estimated 8 per cent decline in 1996, compared with an 8 per cent increase in Italy and Spain considered as a group (Figure 4).[12] Tourist arrivals to Greece dropped faster, by 11.3 per cent over the seven months to July, compared with the same period in 1995.[13] Lower migrant remittances in absolute terms also contributed to the deterioration in the current account while net EU transfers remained broadly stable as a share of GDP.

On the payment side, the economic recovery and the shift in relative prices in favour of foreign goods provided a boost to goods imports.[14] The gradual acceleration of private consumption gave rise to large imports of cars and of other durable goods. And with investment in machinery and equipment rising for a second year, imports of capital goods also remained strong. Finally, the surge in international oil prices increased net oil imports by 0.3 per cent of GDP. Thus, imports of goods and services as a share of total domestic demand rose from 29.0 per cent in 1993 to 31.5 per cent in 1996 in constant price terms. Another significant contribution to the widening of the current account came from the large rise in payments for interest, dividends and profits, which increased by 16.8 per cent in drachma terms and reflected large foreign borrowing by the business sector. As a result, the ratio of interest payments to export receipts (excluding net EU transfers) rose to 13½ per cent in 1996 from 11½ per cent in 1995 (Table 5).

Figure 4. **TOURISM**

1985 = 100




*Source:* OECD (1997), Tourism Policy and International Tourism in OECD Member Countries 1995-1996, Paris.

Table 5. **External debt and debt service**

Million US dollars

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|
| | External debt | | | | | |
| Total | 29 401 | 29 745 | 32 269 | 39 538 | 41 973 | 42 474 |
| General government | 19 729 | 19 848 | 22 948 | 28 275 | 30 589 | 30 780 |
| *Of which:* | | | | | | |
| Long-term | 16 612 | 16 595 | 18 473 | 22 498 | 23 558 | 27 180 |
| Short-term | 671 | 454 | 1 446 | 2 272 | 3 150 | 160 |
| Military debt | 2 446 | 2 799 | 3 029 | 3 505 | 3 881 | 3 440 |
| Business sector [1] | 9 672 | 9 897 | 9 321 | 11 263 | 11 384 | 11 694 |
| *Of which:* | | | | | | |
| Banks and public enterprises | 7 016 | 6 803 | 6 645 | 8 321 | 7 915 | 7 966 |
| | External debt service | | | | | |
| General government [2] | 3 066 | 5 298 | 4 268 | 4 316 | 5 944 | 6 783 |
| Interest payments | 1 420 | 1 823 | 1 566 | 1 617 | 2 047 | 2 193 |
| Amortisation | 1 646 | 3 475 | 2 702 | 2 699 | 3 897 | 4 590 |
| Business sector | 1 454 | 1 492 | 1 532 | 1 773 | 2 989 | 7 584 |
| Interest payments | 516 | 439 | 418 | 369 | 443 | 625 |
| Amortisation | 938 | 1 053 | 1 114 | 1 404 | 2 546 | 6 959 |
| Total debt service | 4 520 | 6 790 | 5 800 | 6 089 | 8 933 | 14 367 |
| | Percentages | | | | | |
| *Memorandum items:* | | | | | | |
| Debt to GDP ratio | 32.9 | 30.3 | 35.0 | 40.3 | 36.7 | 34.6 |
| Interest service ratio [2, 3] | 10.7 | 11.8 | 11.0 | 10.1 | 11.5 | 13.3 |

1. External debt of banks and public enterprises and private non-guaranteed suppliers' credits, excludes a large share of private sector external borrowing.
2. Covers bonds and long-term loans.
3. Calculated as interest payments as a percentage of the sum of exports of goods and invisible receipts (excluding net EU transfers).

*Source:* Bank of Greece.

The deterioration in the current account was more than matched by private sector capital inflows while public sector net external borrowing remained low. The enhanced credibility of the stable exchange rate policy combined with large interest rate differentials contributed to attract large amounts of capital inflows, comprising mostly short-term borrowing by Greek firms. However, foreign portfolio investment in government securities and in the interbank market also surged. During 1995-96, net private sector capital inflows have come close to

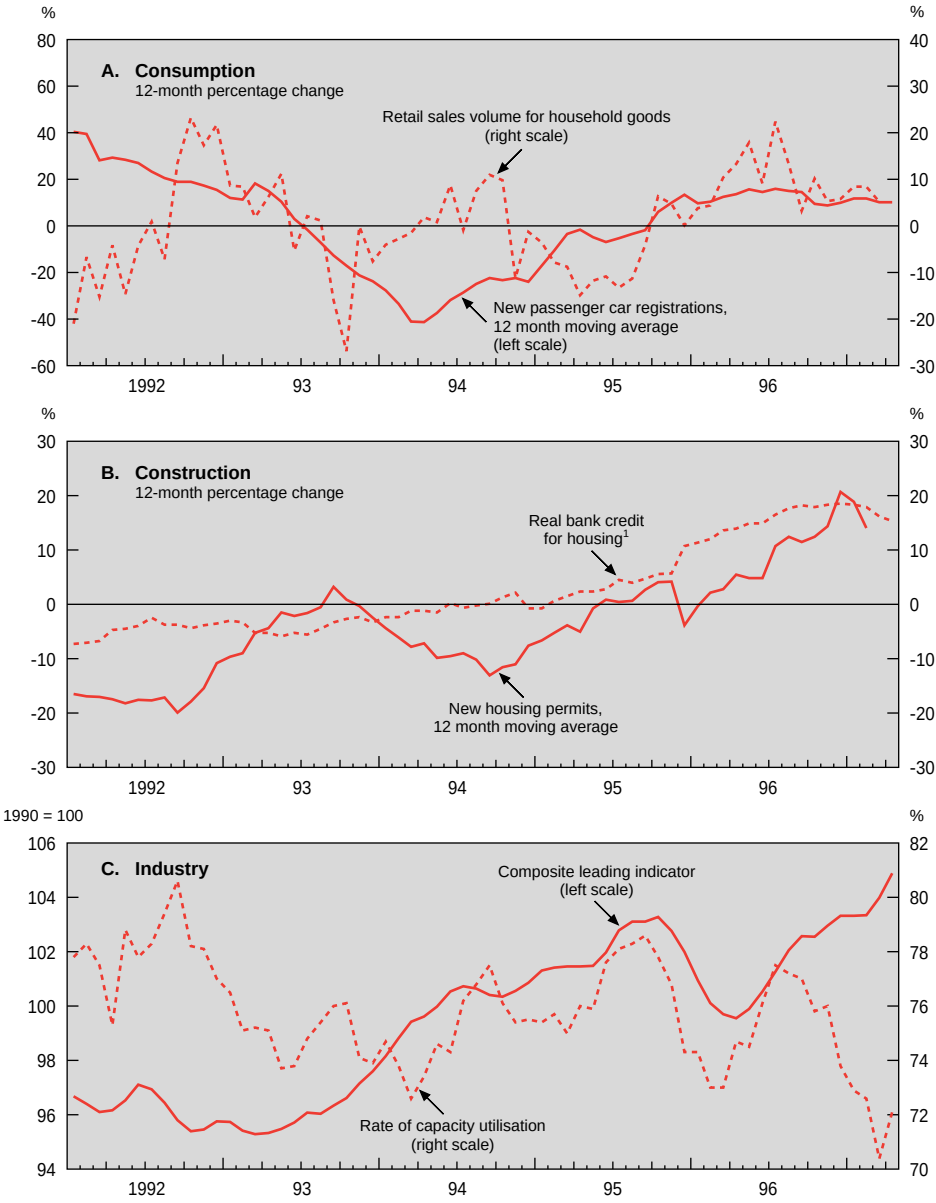
US$10 billion, equivalent to 7½ per cent of GDP. Public sector gross external borrowing rose modestly and reflected a lengthening of the term structure and, to a lesser extent, the increased resort to foreign borrowing by public enterprises. As a result of the capital inflows, foreign exchange reserves rose to US$19.2 billion at year-end 1996, from US$15.7 billion one year before, thus covering almost eight and half months of current year imports of goods and non-factor services. Nevertheless, the increased external borrowing has left the estimated total external debt equivalent to over one-third of GDP.[15]

## Prospects for 1997-98

Over the past three years, investment-led output growth has been boosted by lower financing costs and by an enhanced utilisation of EU structural funds. In turn, the improvement in the economic outlook has led to a revival in consumer and business confidence, and a booming stock market which has increased by about 70 per cent in the first five months of 1997. Most economic indicators point to a further gradual acceleration of output growth in 1997. Consumption continues to gather momentum as evidenced by the sharp rise in new car registrations and buoyant retail sale volumes for household goods. The sharp recovery in the residential housing market has not yet levelled off, with the growth rate for permits for new housing and bank credit, adjusted for inflation, remaining strong. Moreover, the rate of return on alternative (financial) assets for household savings continues to be low. The indicators for business investment are mixed, and suggest a further weakening in the tradeable sector where capacity utilisation has registered a sharp decline in the latter part of 1996 and early 1997. Nevertheless the composite leading indicators for business activity and the demand for imported intermediate goods remain buoyant (Figure 5). Additional support to output growth should be provided by an increasing disbursement rate for the public investment programme. The policy environment should be moderately restrictive in 1997 and 1998, with fiscal policy – based on the 1997 Budget and its extension into 1998 – remaining broadly neutral, and monetary policy based on the exchange rate policy being fairly restrictive.

Output growth should continue to strengthen on the basis of strong domestic demand, especially for investment, and should surpass 3 per cent in 1998 (Table 6). Projects financed by EU funds should provide a large boost to public

Figure 5. **LEADING INDICATORS**




1. Deflated by the consumer price index.
*Source:* Bank of Greece and OECD, Main Economic Indicators.

Table 6. **The short-term outlook**

| | 1996 | 1997 | 1998 |
|---|---|---|---|
| | Percentage changes | | |
| Private consumption | 2.2 | 2.4 | 2.7 |
| Government consumption | 1.4 | 1.0 | 0.5 |
| Gross fixed investment | 12.0 | 11.8 | 11.1 |
| Stockbuilding and statistical discrepancy[1] | 0.1 | 0.0 | 0.0 |
| **Total domestic demand** | 4.2 | 4.3 | 4.4 |
| Exports of goods and services | –0.1 | 4.0 | 4.3 |
| Imports of goods and services | 6.2 | 7.6 | 7.8 |
| **Foreign balance**[1] | –2.2 | –1.9 | –2.1 |
| **GDP at constant prices** | 2.6 | 3.0 | 3.1 |
| **GDP deflator** | 8.8 | 6.8 | 5.8 |
| Private consumption deflator | 8.5 | 6.0 | 5.4 |
| Unit labour costs, total economy | 10.4 | 7.9 | 6.4 |
| Total employment | 1.4 | 1.3 | 1.3 |
| **Unemployment rate (per cent)** | 10.4 | 10.4 | 10.5 |
| | Per cent of GDP | | |
| **Current account balance** | –3.7 | –3.5 | –4.0 |
| Excluding net official transfers | –7.8 | –8.0 | –8.6 |
| General government | | | |
| Net primary balance | 4.1 | 4.9 | 4.7 |
| **Net lending** | –7.4 | –5.2 | –4.0 |

1. Contribution to changes in real GDP, as a percentage of real GDP in the previous period.
*Source:* Ministry of National Economy and OECD projections.

investment, while lower interest rates should continue to spur housing investment. Lower financial costs and moderate prices for imported inputs should maintain healthy profits, and thus support business investment. Investment-led growth is likely to be complemented by the steady recovery in private consumption, which should be sustained by further disposable income gains. Regarding the external side, the international environment is set to improve, with export markets projected to grow by 7 per cent in 1997 and in 1998, and tourism is expected to recover from its 1996 nadir. Overall, strong domestic demand growth should be partly offset by the negative drag from the impact of the stable exchange rate policy on net exports. As a result, the current account deficit on a

payments basis should gradually widen to 4 per cent of GDP by 1998, and to approximately 8 per cent of GDP excluding net EU transfers.

Inflation should continue to decline at a moderate pace. Though the exchange rate policy will continue to pull down prices, cost-push pressures will be an opposing force. In the private sector, as set out in the collective agreement, minimum wages will rise by 3 per cent above the Government's 1997 inflation target, with the potential for further increases at the firm level. Moreover, generous public sector wage awards in 1996 and the transition to a new civil service pay scale in 1997 may influence private sector wage negotiations in 1998. The inflexibility of real wages in the face of large labour market slack should constrain employment creation as firms seek productivity gains, keeping the unemployment rate broadly stable at about 10½ per cent.

The projections are surrounded by several important risks. The main risk emanates from the labour market where the potential for larger than expected real wage awards, which could be in excess of productivity gains, could put further pressure on inflation. In combination with the exchange rate policy this would put pressure on firms' profit margins, especially for those in the tradeable sector, and dampen business investment and employment. Finally, slippage in fiscal consolidation could undermine consumer and business confidence, reverse the recent reduction in interest rate spreads, and brake the recovery in private consumption and housing investment. Over the medium-term, if these pressures accumulate, the current nexus of policies may not be sustainable. On the positive side, the recovery in tourism could be stronger than envisaged. Moreover, rapid implementation of structural reform, such as in the area of privatisation, and the positive conclusion of the social dialogue on employment policies would boost confidence.

# II. Macroeconomic and structural policies

The Government's economic policies are geared towards entering the European Economic and Monetary Union, while promoting economic development and social cohesion. The framework and orientation for policies are provided by the 1994-99 convergence programme. For 1998, it targets a general government deficit of below 3 per cent of GDP and inflation of 3½ per cent. With the first half of the convergence programme completed, the economic position has clearly improved. During the three-year period, the inflation rate has been halved while output growth reached its highest rate since 1991. Inflation reduction has been achieved by a primary balance adjustment at the general government level of over 5 per cent of GDP as well as the use of the exchange rate as a nominal anchor. These results should provide a springboard for achieving more difficult objectives in the second half of the convergence programme, when considerable further fiscal consolidation is planned in order to address a debt burden in excess of 110 per cent of GDP. The macroeconomic policy goals need to be supported by deeper structural reforms, which will reduce microeconomic rigidities and improve the performance of the economy. Without improved flexibility of labour markets, the inflation objectives will be much more difficult to attain and output growth could suffer. In addition, financial reform is necessary to reduce wide intermediation spreads which are constraining private investment, and improved public sector management and privatisation should assist fiscal consolidation, leaving more room for essential expenditures, and reduce the risk of crowding-out private expenditure.

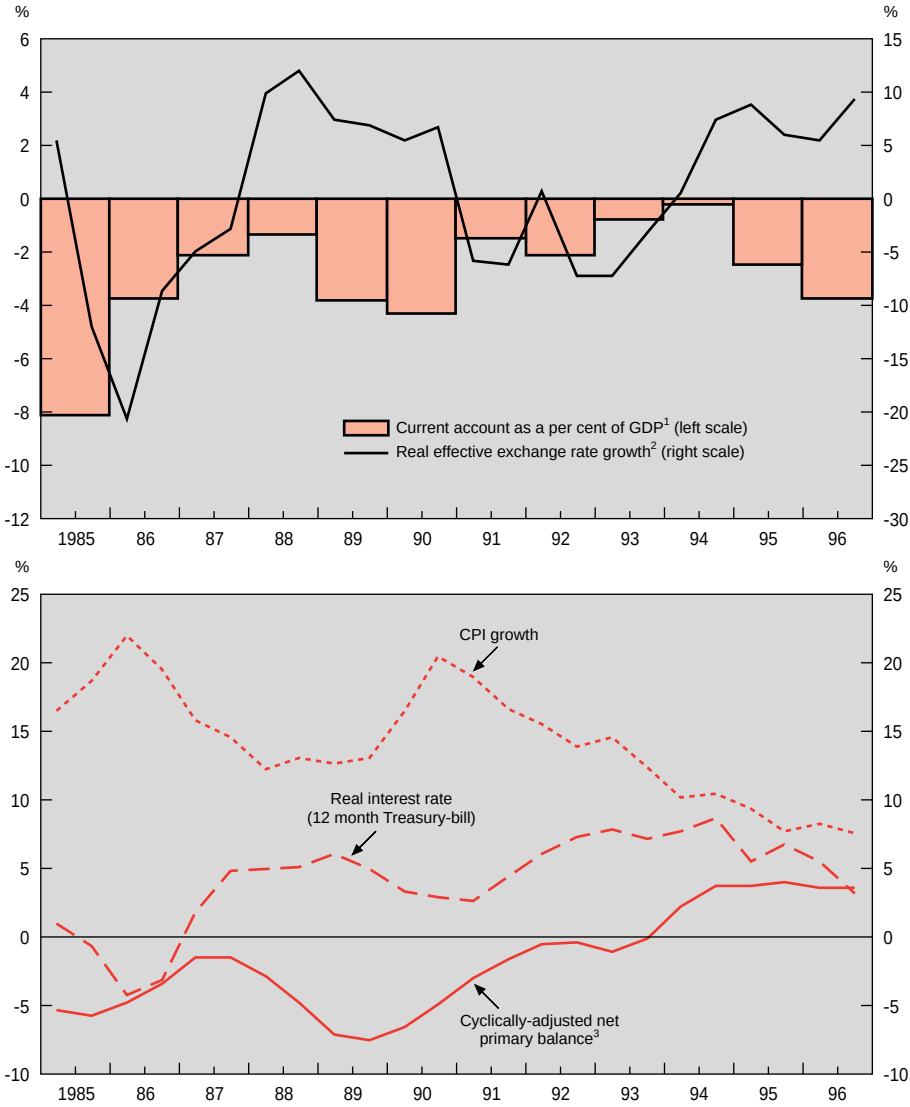
Maintaining momentum will depend critically on the successful implementation of fiscal policy during 1997. Since the introduction of the current adjustment strategy, inflation reduction has been the product of a substantial primary balance adjustment and a tightening of monetary conditions resulting from an appreciation of the real exchange rate and the maintenance of high real

interest rates (Figure 6). However in 1996, even allowing for the fact that national elections were held and that end-year strikes may have affected revenue collection, the performance of fiscal policy was below expectations, with the primary balance remaining broadly unchanged. The slippage in fiscal policy, combined with collective agreements which pushed minimum wages about 2½ percentage points above the inflation objective, are placing a heavy burden on monetary policy. The drachma ''hardened'' in 1996, and was broadly stable against the ECU. As a result, inflation reduction was largely due to a broadly stable exchange rate policy at a time of high wage pressures. This poor policy mix has had an adverse impact on the current account. In this situation, any evidence of deviations from the ambitious 1997 fiscal objective of a primary surplus improvement of 2½ per cent of GDP should trigger the timely implementation of additional measures in order to get the adjustment strategy back on track. The challenge for 1997 and the remaining two years of the convergence programme will be to re-balance the policy mix between monetary and fiscal policy and move ahead vigorously with the structural reforms necessary to gradually augment the economy's growth potential.

The correction of structural rigidities in the Greek economy should form a critical part of the adjustment strategy. A timely implementation of reforms would improve competitiveness and thus help the economy absorb part of the large appreciation of the exchange rate over the past three years. The past year has witnessed postponements of reform measures, notably those to improve public administration and of the privatisation programme. Nevertheless, progress was made in the implementation of projects financed by EU structural funds and in cleaning up the loan portfolios of state-owned banks. Looking forward, the presentation of a draft bill on health reform and the government-sponsored debate with social partners on reforms in the labour market and of the pension system open the door for essential reforms in three of Greece's most intractable structural problems. The problems facing the health and pension systems are reviewed in a special chapter; this chapter will cover other aspects of structural reform pursued since the last *Survey*, namely progress in the improvement of public administration, including public enterprises, and developments in labour market reform since the presentation of the *Jobs Study* follow-up in the previous *Survey*.

Figure 6. **MONETARY CONDITIONS**

Per cent change over the same period of previous year




1. On a payments basis.
2. Deflated by unit labour costs.
3. As a per cent of GDP, including payments on called government guarantees and capitalised interest.

*Source:* Bank of Greece and OECD Secretariat.

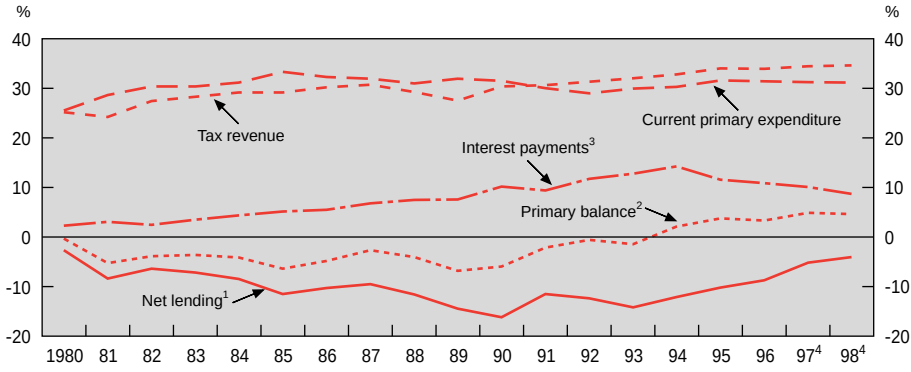
## Fiscal policy

### *Implementation of the 1996 Budget*

The 1996 Budget outturn fell short of its objectives. The Budget had envisaged a primary balance improvement of about 1 per cent of GDP, in order to boost the general government primary surplus to over 5 per cent of GDP. The adjustment effort was based on measures to enhance revenues, especially to broaden the tax base and to raise non-tax revenues. In the event, revenue measures yielded less than expected, and the primary surplus remained broadly unchanged (Figure 7). Despite slightly higher than expected nominal GDP growth, tax revenues fell short of budget projections, as ambitious improvements in tax efficiency did not materialise and tax elasticities remained broadly equal to one. In part, these developments reflect the declining yields from tax amnesties and the inability of the objective criteria to increase further the share of direct taxes paid by the self-employed. The shortfall in non-tax revenues, from very ambitious targets, arose from lower EU transfers for social programmes, lower

Figure 7. **MAIN COMPONENTS OF THE GENERAL GOVERNMENT BUDGET**

Per cent of GDP




1. Including capitalised interest and payments on called government guarantees.
2. Net lending excluding interest payments and including payments on called government guarantees.
3. Including capitalised interest.
4. OECD projections.

*Source:* Bank of Greece and OECD Secretariat.

dividends from public enterprises and lower than expected collections of unpaid fees and fines from amnesties.

Expenditure overruns were concentrated in interest payments on the government debt, the wage bill, and operating expenditures. Regarding the first of these, interest rates on government paper declined by one percentage point less than expected, though Treasury bill rates fell from 15.5 per cent in 1995 to 12.8 per cent in 1996.[16] Civil servant wages were higher than budgeted and increased by about 7 per cent in real terms. In part, these wage developments reflect the extended use of ''productivity'' bonuses.[17] The overrun in operating expenditures resulted partly from the un-budgeted cost of the early national election. In the past, Greek budget performance has followed a pronounced electoral cycle, with slippage occurring on both the revenue and expenditure side, and in this regard the 1996 Budget slippage was less acute than in past election years.

The Government has nevertheless met the 1996 convergence programme target for the general government deficit following two changes in accounting practices introduced in early 1997 which were in line with Eurostat recommendations. The deficit outcome of 7.4 per cent of GDP[18] excludes two previously included items, capitalised interest and payments to clear government guarantees that had been called, which amount to 0.6 per cent of GDP and 0.7 per cent of GDP, respectively (Table 7).[19] In addition, in line with European Commission directives, public enterprises' resources for investment partly took the form of equity participations rather than capital transfers as had been previously budgeted, which improved the primary surplus by 0.5 per cent of GDP. Without these three adjustments, the primary surplus would have been about 3 per cent of GDP relative to the 5 per cent envisaged in the 1996 Budget.

Reflecting the performance of general government, the public sector borrowing requirement (PSBR) as reported by the Bank of Greece on a cash basis increased to 11.1 per cent of GDP in 1996 from 9.4 per cent in 1995 (Table 8). With activity stronger, public enterprises in aggregate repeated their relatively good 1995 performance. Their operating accounts remained broadly balanced, despite tariff increases below the rate of inflation. However, similar to developments in general government, wages increased by over 5 per cent in real terms, pushing the wage bill to over one-third of operating expenses. Public enterprises' net borrowing requirement also remained broadly unchanged as investment increases of 25 per cent in nominal terms (bringing the total to 3.6 per cent of

Table 7. **General government accounts**[1]

Per cent of GDP

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 Estimates | 1997 Official projections |
|---|---|---|---|---|---|---|---|
| **Current revenue** | 32.9 | 33.7 | 34.4 | 35.9 | 37.8 | 37.3 | 39.2 |
| Direct taxes | 5.6 | 5.3 | 5.6 | 6.8 | 7.2 | 7.2 | 7.8 |
| Households | 3.8 | 3.7 | 3.8 | 4.7 | 5.0 | 5.0 | . . |
| Business | 1.7 | 1.6 | 1.7 | 2.1 | 2.2 | 2.3 | . . |
| Indirect taxes | 13.8 | 14.8 | 14.3 | 14.1 | 14.3 | 14.3 | 14.7 |
| Social security contributions | 11.2 | 11.2 | 12.1 | 12.0 | 12.4 | 12.4 | 12.7 |
| Other | 2.2 | 2.4 | 2.4 | 3.1 | 3.9 | 3.4 | 4.0 |
| **Current expenditure** | 39.4 | 40.7 | 42.7 | 44.5 | 43.9 | 42.8 | 41.4 |
| Consumption | 14.4 | 13.7 | 13.8 | 13.8 | 14.7 | 14.3 | 14.5 |
| Wages | 11.6 | 11.0 | 11.1 | 10.8 | 11.6 | 11.3 | 11.5 |
| Non-wages | 2.8 | 2.7 | 2.7 | 3.0 | 3.1 | 3.1 | 3.0 |
| Interest payments | 9.4 | 11.7 | 12.8 | 14.2 | 12.3 | 11.5 | 10.4 |
| Transfers and subsidies | 15.6 | 15.3 | 16.1 | 16.5 | 16.8 | 17.0 | 16.4 |
| Business | 0.6 | 0.5 | 0.6 | 0.5 | 0.5 | 0.4 | 0.3 |
| Households | 15.2 | 15.2 | 15.7 | 16.3 | 16.5 | 16.5 | 16.2 |
| Rest of the world | –0.3 | –0.5 | –0.2 | –0.3 | –0.1 | 0.1 | –0.1 |
| **Net saving** | –6.5 | –7.1 | –8.3 | –8.7 | –6.0 | –5.5 | –2.2 |
| Net capital transfers[2] | 1.9 | 2.1 | 3.0 | 0.7 | 0.3 | –0.8 | –1.0 |
| Fixed investment | 3.1 | 3.1 | 2.9 | 2.8 | 2.8 | 2.8 | 3.1 |
| **Net lending** | –11.5 | –12.3 | –14.2 | –12.1 | –9.2[3] | –7.4[3] | –4.2 |
| **Primary balance** | –2.1 | –0.6 | –1.4 | 2.1 | 3.1[4] | 4.1 | 6.2 |
| *Memorandum item:* | | | | | | | |
| Net lending | | | | | | | |
| Central government | –13.8 | –14.1 | –15.5 | –13.9 | –11.2 | –9.3 | –6.3 |
| Social security and hospitals | 2.3 | 1.9 | 1.5 | 1.9 | 2.0 | 1.9 | 2.0 |
| Regional government | 0.0 | –0.1 | –0.1 | –0.1 | 0.0 | 0.0 | 0.0 |
| Capitalised interest | 0.8 | 2.6 | 1.7 | 0.5 | 0.8 | 0.6 | 0.1 |
| Payments on called government guarantees | 1.5 | 2.2 | 3.0 | 0.7 | 0.2 | 0.7 | . . |
| Net lending excluding capitalised interest and payments on called government guarantees | –9.2 | –7.5 | –9.5 | –10.9 | –9.2 | –7.4 | . . |

1. On a national accounts basis.
2. Includes transfers from EU to support the investment budget.
3. Excluding capitalised interest and payments on called government guarantees.
4. The primary surplus was 4 per cent of GDP prior to revision of the accounts to include payments for judges' back-pay that will be paid during 1996-2000.

*Source:* Ministry of National Economy.

Table 8. **Budget plans and outcomes**

Billion drachmas

| | 1994 | 1995 | | 1996 | | 1997 |
|---|---|---|---|---|---|---|
| | Outcome | Budget | Outcome | Budget | Estimated outcome | Budget |
| | Administrative basis | | | | | |
| **CENTRAL GOVERNMENT** | | | | | | |
| **Current expenditure** | 8 251 | 8 846 | 8 880 | 9 572 | 9 751 | 10 225 |
| (Per cent of GDP) | 34.7 | 34.9 | 33.5 | 32.9 | 33.0 | 31.3 |
| Salaries and pensions | 2 269 | 2 537 | 2 564 | 2 843 | 2 984 | 3 294 |
| Goods and services | 460 | 508 | 497 | 593 | 709 | 576 |
| Grants and subsidies | 1 230 | 1 294 | 1 421 | 1 485 | 1 475 | 1 420 |
| Interest | 3 340 | 3 355 | 3 356 | 3 355 | 3 465 | 3 468 |
| Transfers to the EU | 309 | 352 | 312 | 385 | 348 | 397 |
| Restitution of revenues to third parties | 355 | 397 | 456 | 469 | 479 | 504 |
| **Current revenue** | 5 862 | 6 935 | 6 728 | 7 710 | 7 440 | 8 715 |
| (Per cent of GDP) | 24.7 | 27.4 | 25.4 | 26.5 | 25.2 | 26.7 |
| Direct taxes | 1 773 | 2 220 | 2 133 | 2 415 | 2 343 | 2 790 |
| Indirect taxes | 3 462 | 4 008 | 3 835 | 4 466 | 4 321 | 5 014 |
| Non-tax revenue | 627 | 707 | 760 | 829 | 776 | 911 |
| Transfers from the EU | 142 | 167 | 86 | 145 | 94 | 67 |
| **Investment budget balance** | –497 | –585 | –617 | –600 | –545 | –500 |
| Transfers from the EU | 288 | 451 | 322 | 575 | 548 | 700 |
| **Budget balance** | –2 887 | –2 496 | –2 769 | –2 462 | –2 856 | –2 010 |
| (Per cent of GDP) | –12.2 | –9.8 | –10.5 | –8.5 | –9.7 | –6.2 |
| **Primary deficit** | 453 | 859 | 587 | 894 | 609 | 1 458 |
| (Per cent of GDP) | 1.9 | 3.4 | 2.2 | 3.1 | 2.1 | 4.5 |
| **PUBLIC ENTITIES** [1] | 9 | –173 | –26 | –94 | –95 | –272 |
| (Per cent of GDP) | 0.0 | –0.7 | –0.1 | –0.3 | –0.3 | –0.8 |
| Excluding transfers and EU funds | –747 | –857 | –803 | –951 | –976 | –1 121 |
| (Per cent of GDP) | –3.1 | –3.4 | –3.0 | –3.2 | –3.3 | –3.4 |
| **PUBLIC ENTERPRISES** | –15 | –75 | –103 | –58 | –24 | –75 |
| (Per cent of GDP) | –0.1 | –0.3 | –0.2 | –0.2 | –0.1 | –0.2 |
| Excluding transfers and EU funds | –931 | –1 085 | –930 | –1 191 | –1 124 | –1 178 |
| (Per cent of GDP) | –3.9 | –4.3 | –3.6 | –4.1 | –3.8 | –3.6 |
| *Memorandum items:* | | | | | | |
| Debt take-overs and equity purchases | 231 [2] | | 163 | | 368 | 349 |
| Capitalised interest | 120 | | 211 | | 170 | 45 |
| Increase in Government reserves | 450 | | 463 | | 167 | . . |
| Total | 801 | | 837 | | 705 | 394 |
| (Per cent of GDP) | 3.4 | | 3.2 | | 2.4 | 1.2 |

Table 8. **Budget plans and outcomes** *(cont.)*

Billion drachmas

| | 1994 | 1995 | | 1996 | | 1997 |
|---|---|---|---|---|---|---|
| | Outcome | Budget | Outcome | Budget | Estimated outcome | Budget |
| | Cash basis | | | | | |
| **Central Government**[3] | –3 022 | | –3 047 | | –3 856 | . . |
| (Per cent of GDP) | –12.7 | | –11.5 | | –13.0 | . . |
| **Public entities** | 517 | | 546 | | 600 | . . |
| **Public enterprises** | –205 | | –1 | | –29 | . . |
| **Public sector deficit**[3] | –2 710 | | –2 502 | | –3 285 | . . |
| (Per cent of GDP) | –11.4 | | –9.4 | | –11.1 | . . |

1. Only the 7 most important entities.
2. Excludes the Government take-over of military debt amounting to Dr 924 billion.
3. Includes capitalised and accrued interest amounting to Dr 259 billion in 1994, Dr 84 billion in 1995, and Dr 179 billion in 1996.

*Source:* Ministry of Finance.

GDP) were mostly covered by equity participation and capital transfers from the budget. The national telephone company (OTE) whose tariffs were raised by 10 per cent at the beginning of 1996 was the most profitable among the public enterprises.

The financing of the 1996 PSBR favoured bonds over other instruments. As a result, approximately 60 per cent of the government's net financing needs were covered through floating rate bonds, especially those with a seven-year maturity. In addition, the government's choice of financing instruments was important in containing liquidity creation. With the rapid development of a secondary market, the Government was able to place most of its paper with the non-bank public and thus limit recourse to external debt financing. Nevertheless, the back-loaded financing needs of the budget, whereby about 30 per cent of the net financing is required in the last month of the year, produces large injections of liquidity into the financial system as the Government draws down the large stock of its reserves built up during the year. As a result, the Bank of Greece had to absorb significant amounts of liquidity, especially as the Government converted large amounts of its foreign currency denominated reserves to drachma in December 1996, as in late 1995.

The general government debt to GDP ratio stabilised in 1996 at 112 per cent. The overall public sector debt is 9 per cent of GDP higher at 121 per cent (Table 9).[20] As in past years, the debt burden was enlarged by several ''below the line'' items. In addition to the above mentioned capitalised interest, assumption of called government guarantees and equity participation, the debt was inflated by capitalisations for the Greek Bank for Industrial Development (ETBA) and the Bank of Crete (totalling 0.5 per cent of GDP).[21] Of larger concern, recent studies have revealed large amounts of inter-governmental arrears. The government has announced that arrears between entities of general government and public enter-

Table 9. **Public sector debt**

Billion drachmas

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|
| **Central government** | 13 473 | 16 969 | 24 078 | 27 910 | 31 781 | 35 795 |
| Domestic | 10 707 | 13 630 | 19 596 | 22 522 | 26 109 | 29 417 |
| *Of which:* | | | | | | |
| Treasury bills | 4 974 | 5 601 | 5 766 | 7 533 | 8 422 | 9 810 |
| Government bonds | 4 035 | 6 017 | 9 551 | 10 438 | 12 804 | 14 748 |
| *Of which:* | | | | | | |
| Rescheduled | 2 766 | 4 220 | 5 397 | 5 179 | 5 291 | 4 737 |
| Foreign currency denominated | 939 | 976 | 1 702 | 1 879 | 1 574 | 239 |
| Bank of Greece | 1 214 | 1 347 | 3 444 | 3 622 | 3 866 | 3 902 |
| Other | 484 | 665 | 835 | 929 | 1 017 | 957 |
| External | 2 766 | 3 339 | 4 482 | 5 388 | 5 672 | 6 378 |
| **Local authorities** | 63 | 71 | 80 | 87 | 93 | 121 |
| **Social insurance organisations** | 434 | 343 | 369 | 221 | 242 | 190 |
| Other | 112 | 118 | 96 | 96 | 93 | 92 |
| Inter-governmental debt | 555 | 794 | 1 031 | 2 091 | 2 608 | 3 123 |
| **General government debt** | 13 527 | 16 707 | 23 592 | 26 223 | 29 601 | 33 075 |
| Domestic | 10 751 | 13 358 | 19 100 | 20 825 | 23 919 | 26 687 |
| External | 2 776 | 3 349 | 4 492 | 5 398 | 5 682 | 6 388 |
| **Public enterprises**[1] | 1 621 | 1 790 | 2 021 | 2 491 | 2 435 | 2 691 |
| **Public sector debt**[2] | 15 148 | 18 497 | 25 613 | 28 714 | 32 011 | 35 766 |
| *Memorandum items:* | | | | | | |
| Debt as a per cent of GDP | | | | | | |
| Central government | 83.0 | 90.9 | 114.1 | 117.5 | 120.0 | 121.0 |
| General government | 83.3 | 89.4 | 111.8 | 110.4 | 111.8 | 111.8 |
| Public sector[2] | 93.3 | 99.0 | 121.4 | 120.9 | 121.0 | 120.9 |

1. Break in series in 1992.
2. Public enterprises inter-governmental debt has not been excluded from total debt.

*Source:* Ministry of Finance.

prises should be cleared by end-1997. Looking forward, since state-owned banks may be net creditors to loss-making public enterprises or entities, the Government may be forced to take over debt amounting to several percentage points of GDP.

### *1997 fiscal policy*

In 1997, fiscal policy is pursuing the revenue-enhancing strategy conceived over the past three years and adds a critical new component aimed at reducing current primary expenditure. The government's revenue objective aims at implementing a more equitable and efficient tax system, and at using public assets to raise non-tax revenues. Regarding expenditure, the Government is attempting to reduce the size of the public sector as well as promote the efficient use of resources so as to be able to reallocate the savings towards investment and the priority areas of education, social welfare and defence. The Budget was framed in a favourable macroeconomic environment, with output projected to accelerate to 3.3 per cent and inflation – as measured by the CPI – to decline to 4.5 per cent during the year. Interest rates were projected to fall by about 3 percentage points (a decline which exceeds that of inflation by one percentage point) and reach 9 per cent by end 1997. Overall, fiscal policy aims to achieve a primary balance improvement for general government of 2 per cent of GDP and to bring the deficit to 4.2 per cent of GDP in 1997, from 7.4 per cent in 1996.

The bulk of the improvement in the primary balance is expected to originate from an ambitious package of revenue measures. To this end, direct taxation was favoured over more regressive indirect taxation. Its full implementation would close important gaps in the taxation system. One of the most important innovations of the Budget was to eliminate or reduce many tax expenditures that created large inequities in the system of direct taxation, as they applied to upper-income classes, self-employed and special interest groups, but to maintain those with a social or developmental nature.[22] Other measures broadened the tax base to include two greatly under-taxed and thus traditional placements of wealth: a real property tax was introduced and a 7½ per cent withholding tax was placed on previously un-taxed government securities; the tax will apply to securities issued after 1 January 1997.[23] Nevertheless, a large share of the increase in personal income taxation should arise from bracket creep as nominal tax brackets were once more left unchanged at their 1992 level; during this five-year period the

price level will have increased cumulatively by about 60 per cent. Corporate taxes of the financial sector were increased from 35 per cent to 40 per cent and a 2 per cent tax on gross profits of mutual funds was introduced (to replace an ineffective withholding tax). The Budget continues to target tax evasion by raising presumptive taxation of the self-employed and imputed income from the ownership of luxury cars and boats. To dampen inflation expectations, excise taxes were left broadly unchanged in nominal terms (with small increases only in alcohol and tobacco excise taxes). Important increases in non-tax revenues are expected from larger dividend transfers from public enterprises and state-owned banks, as well as the improved management of leased publicly-owned buildings. Finally, tax amnesties for income and property taxes, as well as fines, are extended – though the continuing resort to amnesties is diminishing their returns.

The yield from several of the new measures to enhance direct tax collection is difficult to evaluate, especially the new real estate tax, the reduction of tax expenditures and the taxation of government securities. Regarding the first, a national land registry does not exist (which is partly to blame for past failures to implement a real estate tax). Meanwhile, the yield from the taxation of government securities may be fully offset by higher interest rates and will not apply to instruments issued prior to 1997. Nevertheless, the government projects the elasticity for direct taxes with respect to GDP resulting from the implementation of these measures to be two, which the OECD considers to be on the high side. For indirect taxes, the government's projection for the elasticity with respect to private consumption is about 1½, with reduced tax evasion, especially on VAT, compensating for the non-increase of excise taxes. Overall, the Government estimates current revenues to increase by nearly 2 per cent of GDP, with new measures accounting for about three-quarters of the improvement.

The expenditure strategy aims to improve spending efficiency in the three major expenditure components of the budget, namely personnel, interest payments, and expenditure of a multitude of public entities which receive transfers from the central government budget. In view of the large cost to the budget for wages and salaries, the budget strove to achieve expenditure savings by introducing a stricter hiring norm. Henceforth, only one in five retiring civil servants will be replaced, though the measure is diluted since the coefficient is lowered to 1:1 for employees in the education, health and security sectors, who account for about half of public employment. In order to eliminate inequities that have arisen

in the pay of civil servants, exacerbated during 1996 with the extended use of ‘‘productivity’’ bonuses, the 1997 Budget introduced a new, more transparent, wage structure which aims to rationalise the granting of all bonuses. The introduction of new pay scales could lead to upward pressure on wages, since reduction in nominal salaries is not used as an instrument for correcting the inequities. In the event, the wage bill is projected to increase by 11 per cent in 1997, approximately 5 percentage points above the government’s projection for the average increase in prices.

In a continuing effort to improve the means testing of welfare benefits, the 1997 Budget preserves the means-tested subsidy for low-income pensioners and introduces means testing for family benefits.[24] In addition, a new pension scheme will be introduced for farmers, which will gradually replace their low flat rate benefit (see Chapter III). In 1998, the new pension scheme will be in surplus, which will broadly offset an increase in the flat pension benefit for existing farm pensioners introduced in 1997.

Regarding the multitude of public entities, new legislation aims to rationalise their expenditure. As a result, transfers were frozen in nominal terms, except for those relating to personnel, pensions and health care. This caveat may limit this measure’s yield as it excludes most expenditure of these public entities. Nevertheless, any progress made by the newly formed committee to close such public entities and transfer their personnel to other parts of government will support the policy of containing public employment. Moreover, the new legislation encourages public entities to raise resources from the commercial use of their purportedly large real estate holdings. The yield from this measure will depend on how rapidly public entities can manage to implement this change, and raise non-tax revenue. There is in principle a strong incentive to do so because 25 per cent of earmarked taxes to such public entities will be henceforth kept by the State. As a result of these policies, the Government projects current primary expenditures to decrease by 0.3 per cent of GDP in 1997.

One-third of the deficit reduction will be due to lower interest payments, reflecting the gradual pass through of lower interest rate spreads after the 1994 currency turbulence. Moreover, the Government will push back interest payments through the introduction of new two-year zero coupon bond issues and the rescheduling of Dr 200 billion of payments (0.6 per cent of GDP). The development of the secondary market will be enhanced by the introduction of

several new instruments. In addition to the zero-coupon bond, the Ministry of Finance has introduced a three-year fixed rate bond and soon plans to issue five and seven-year fixed rate bonds and an inflation indexed bond. The new instruments should lengthen debt maturity, assist in gauging inflationary expectations, and help stimulate activity in the currently inactive corporate bond market. Regarding capital spending, a larger amount of investment is being pushed off-budget through the use of capital injections, especially to public enterprises. In this regard, the conversion of public enterprises during 1996 to *société anonyme* has made them eligible under European System of Accounts (ESA) accounting regulations for equity participations by the state in the place of capital transfers. During 1997, equity participations will reduce the deficit by 1 per cent of GDP.

Early indications on budgetary performance in 1997 indicate that ordinary budget revenues of the state budget are underperforming, increasing at a rate of the order of 12 per cent during the first five months of the year compared with budget projections of 17 per cent. The Government expects revenues to pick-up once the impact of the new measures begins to take hold later in the year. Moreover, expenditures are far below projections, though spending patterns are less predictable.

### *Medium-term fiscal adjustment*

Over the past years, much progress has been achieved in reducing the large fiscal imbalances and stabilising the debt burden. However, more remains to be done, and the Government has set ambitious targets for the future with the primary objective of joining monetary union. The fiscal strategy pursued by the authorities should gradually yield large dividends, especially as regards the widening of the tax base and the achievement of a more equitable tax system. Measures to reduce expenditure inefficiencies have only recently been enacted and their use on a broader front could produce substantial savings. Based on the policies announced in the 1997 Budget, the OECD Secretariat projects the primary surplus to exceed 4½ per cent in 1997 and the general government deficit to fall to 5.2 per cent compared with the government's projection of 4.2 per cent of GDP. The difference with the authorities' projection stems from a slightly less favourable macroeconomic environment and a more cautious estimate of the yield from the new budget measures. In any case, more fiscal adjustment will

clearly be required over the following years to attain the Maastricht criteria necessary for joining monetary union.

Following the measures introduced in the 1997 Budget, there appears to be limited scope to attain a further rapid broadening of the tax base, as many avenues for tax avoidance will have been closed, and in the fight against tax evasion gains are usually achieved only gradually. In this regard, the short-term benefits from the repeated use of tax amnesties for direct and indirect taxes may even be counterproductive insofar as they generate incentives for increased tax evasion. In addition, the continued resort to bracket creep probably increases the inequitable distribution of the tax burden between wage earners and the self-employed, while the low direct tax base precludes frequent recourse to restraining indirect taxes in order to dampen inflation expectations. The Government holds the view that more emphasis should be placed on producing a smaller but more productive state. The recent efforts to examine the efficient use of current primary expenditures through the improved targeting of entitlement programmes and by reducing waste at various levels of general government are appropriate steps in this direction. Nevertheless, accepting that the opportunity of the CSF II structural funds requires maintaining and possibly expanding, the level of investment expenditure, and that defence spending will continue to absorb a large share of expenditure (at least 4 per cent of GDP) these reform initiatives will need to be broadened. Future reforms will need to address the operation of the social security system which comprises about 60 per cent of current primary expenditures and – apart from contribution revenue – requires resources of the order of 5 per cent of GDP to cover its expenditures. In this regard, Chapter III of this *Survey* reviews options for reforming the health and pensions systems, which together account for over 90 per cent of social security expenditure.

Another third of current primary expenditures concerns outlays on personnel. Despite efforts to control this item better, its share of current primary expenditure has risen over the past two years, with general government employment increasing by 25 000 (5 per cent) – to account for one-fourth of dependent employment – and wages by nearly 10 per cent in real terms. While the bulk of the employment increase has been related to education and health services, hiring decisions have long-term effects on the budget and increased employment in certain areas should be more than compensated by cuts in other areas. In this

regard the hiring norms will not be sufficient. Additional measures will be required to reduce employment at a faster pace than that implied by normal retirement, especially as the 1990-92 pension reform will prolong civil servants' employment period. Moreover, staffing the enhanced regional governments has increased employment at the local level by 16 per cent over the past two years. Demands for staff at the local and regional level should be met by existing staff, which should also be part of the process of avoiding the duplication of services at the levels of central and local government. The intended consolidation of the number of municipalities – for which the relevant legislation is expected to be presented to Parliament during the summer of 1997 – should lead to efficiency gains at this level of government, including in the area of employment. Finally, the recent surge in civil servants' wages is unlikely to help dissuade the large demand for civil service jobs, while retaining highly qualified staff, especially at the higher echelons of the civil service, will require a further widening of pay differentials than that achieved by the new 1997 pay scale, and thus a reduction of real wages for some civil servants, unless employment reduction is achieved.

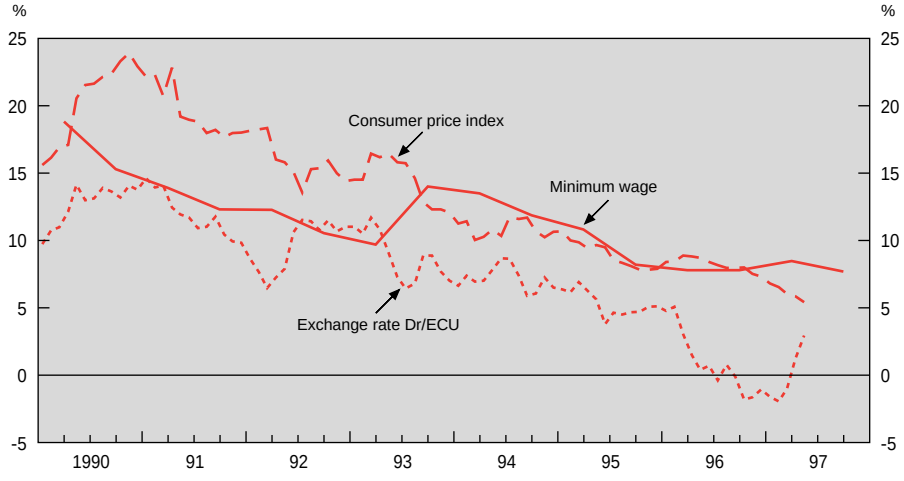
Finally, interest payments amount to over 10 per cent of GDP and consume over one quarter of current revenue. The scope for further declines in nominal interest rates depends in large part on further primary balance adjustment and thus a rapid reduction in the deficit may require attacking the high level of general government debt. However, the Government has so far been rather hesitant to relinquish control over large public enterprises and state-owned banks, in part due to employment considerations, and has preferred a strategy which aims to improve their efficiency and to reap the benefits in the form of non-tax receipts. However, as commitments to the EU will open the important utility and transportation sectors (electricity, telephone, and the airlines) to competition, these monopoly rents will eventually shrink as a source of revenue to the budget. Regarding loss-making public enterprises, most of which are involved in urban transport, improving the targeting of subsidies may reduce their losses and eliminate the rationale for keeping them as public monopolies. Finally, the privatisation of state-owned banks – in addition to the reduction in the public debt stock – should assist in the reduction of their wide intermediation spreads and help sustain private investment.

## Monetary policy

### *1996 outturn*

The primary objective of a reduction in inflation from 8 per cent during 1995 to the ambitious government target of 5 per cent during 1996 was missed by the sizeable margin of 2½ per cent despite a sharp tightening of the exchange-rate-based monetary policy. The effective exchange rate remained broadly stable in 1996 (Figure 8) in the face of about 6 per cent inflation differential *vis-à-vis* trading partners. The principal reason why monetary policy failed to attain the inflation objective was the inadequate support from other policies. Fiscal policy was significantly more expansionary than had been foreseen, and labour market rigidities resulted in important cost-push pressures. In addition, the effectiveness of monetary policy has been adversely affected by large capital inflows following the 1994 liberalisation of the capital account. These have been attracted by a substantial nominal interest rate differential in combination with the enhanced credibility of the exchange rate policy, especially after its crawl was basically eliminated in 1996. As a result, the Bank of Greece

Figure 8. **NOMINAL ANCHORS**
Monthly percentage changes




*Source:* Bank of Greece and OECD Secretariat.

has had to absorb large amounts of liquidity and introduce measures to discourage capital inflows.

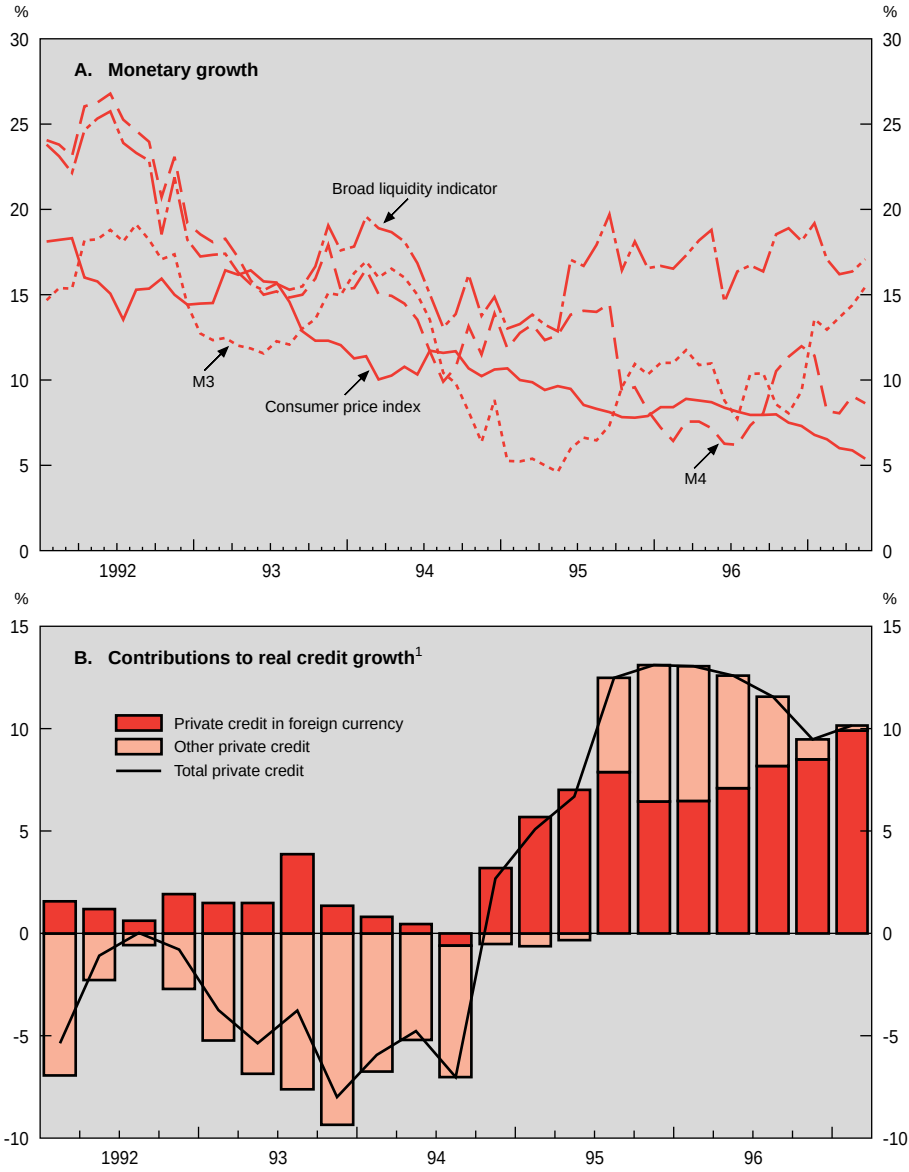
The principal intermediate target of monetary policy was a broadly stable average exchange rate for the drachma *vis-à-vis* the ECU during 1996. In the event, the exchange rate depreciated by 1 per cent *vis-à-vis* the ECU during the year and by 0.6 per cent on average compared with 1995. The monetary and credit targets were broadly observed, with the monetary targets at the upper limit of their respective ranges, and broadly in line with nominal GDP growth.[25] Nevertheless, money and credit aggregates were somewhat affected by portfolio shifts. For example, the rate of growth of M3 and M4 during 1996 were respectively decreased and increased by investors' heightened interest in Treasury bills prior to the introduction of a withholding tax on new issues in 1997. In addition, the deepening of the secondary market in government paper has increased holdings by the non-bank public (and thus reduced bank credit to government) and decreased the deposit base. The broader liquidity aggregate, comprising M3 plus non-bank holdings of government securities of all maturities, was more stable and rose by 18 per cent in 1996 reflecting the continued large financing needs of the budget (Figure 9, panel A).

The sources of the monetary expansion were both domestic and external. Regarding the former, credit in real terms to the private sector increased almost three times as fast as real GDP growth, with the bulk of the expansion arising – for the second year in a row – from foreign currency denominated credit.[26] Drachma-denominated credit exhibited a marked slowdown in the second half of the year, reflecting the measures introduced in June 1996 to limit consumer credit and restrict liquidity by raising and broadening reserve requirements.[27] However, the measures were less effective in stemming the expansion of foreign currency denominated credit (Figure 9, panel B). Overall, the most dynamic segments of the private sector credit market remained loans for housing purposes to consumers, and trade reflecting the economic recovery in these sectors, as well as improved credit terms especially in the case of housing. Bank credit to government remained subdued as its securities were directly or indirectly purchased by the non-bank public.

Attracted by large interest rate differentials and the enhanced credibility of the exchange rate regime, capital inflows provided a large impetus to liquidity during 1996 (Figure 10, panel A). For example, the interest rate differential

Figure 9. **MONEY AND CREDIT**

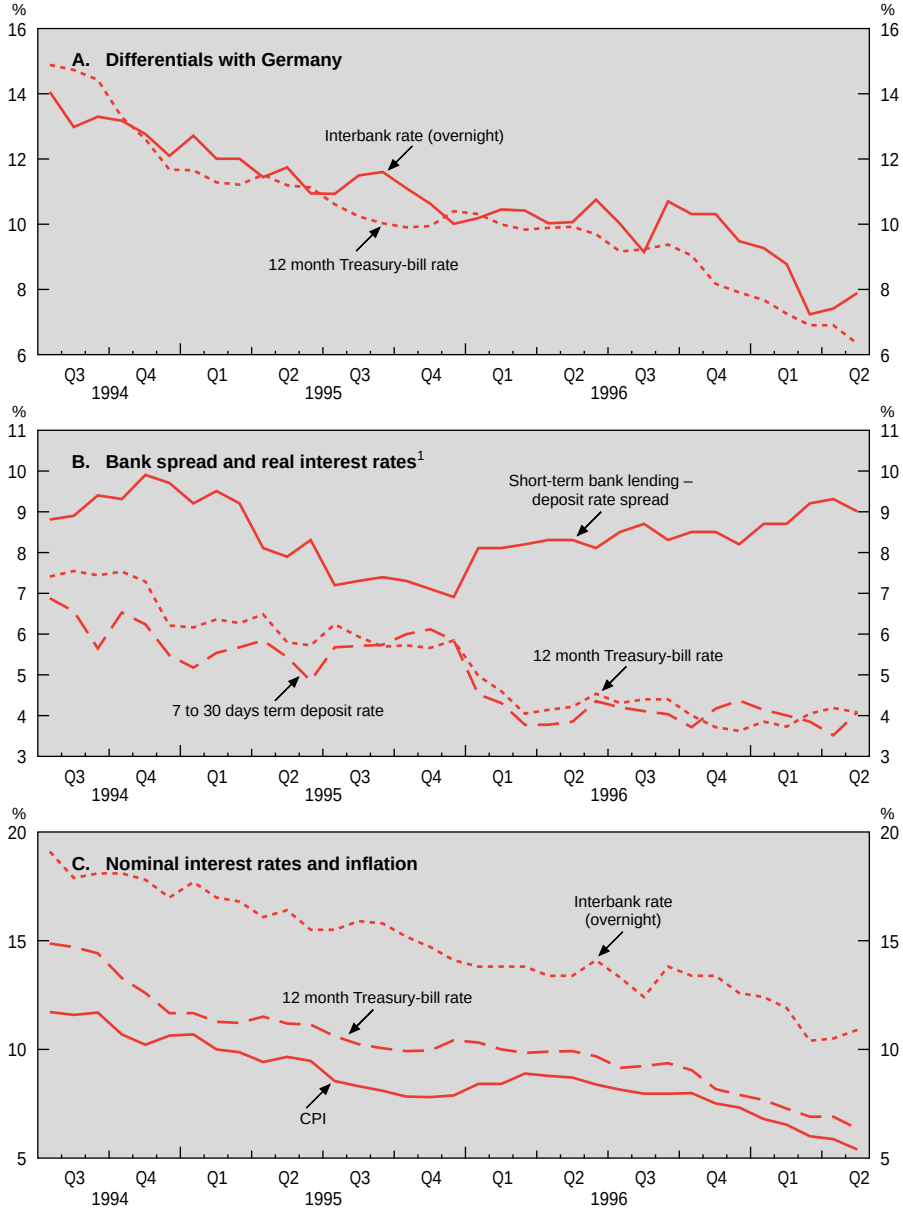
End of period, 12 month per cent change




1. Deflated by the consumer price index.

*Source:* Bank of Greece.

Figure 10. **INTEREST RATES**




1. Deflated by the CPI.

*Source:* Bank of Greece, Deutsche Bundesbank and OECD Secretariat.

relative to comparable German instruments was of the order of 10 per cent for most of 1996 while the exchange rate *vis-à-vis* the DM appreciated by 4 percentage points during 1996. Reflecting the attractiveness of foreign currency denominated borrowing, Greek entrepreneurs increasingly turned to foreign capital markets. In addition, foreign purchases of Greek fixed income instruments rose considerably, activity being concentrated in the interbank market. The upgrading by the rating agency Moody's of Greece's external long-term debt to Baa1 from Baa3 (the lowest investment grade) in February 1997 may provide an additional impetus to capital inflows. Overall, net foreign assets provided a much larger impetus to M3 growth during 1996 than in 1995.

The measures introduced in June 1996 provided some respite as they succeeded in absorbing the sizeable excess liquidity that the Bank of Greece had been sterilising in the interbank market. During the first half of 1996, the Bank of Greece had absorbed a daily average of Dr 450 billion (approximately one quarter of end-December 1995 required reserves), while subsequent to the introduction of these measures, contractionary interventions fell to a daily average of Dr 70 billion during the second half of the year. This reduction was partly facilitated by the budget's large recourse to non-bank financing during the second half of the year. Nevertheless, the increase in reserve requirements – on top of the increase of August 1995 – has not been without adverse consequences. These increases have probably been an important factor in reversing the rapid decline in the spread between banks' lending and deposit rates that had been observed in 1995 (Figure 10, panel B). Some of the factors leading to this large spread are, *inter alia*, the special bank tax (EFTE) of 4 per cent on a loan's interest cost, a 1.2 per cent fee on the value of a drachma-denominated loan used to subsidise certain favoured loans, the fact that only half of required reserves are remunerated at a fixed interest rate of 11 per cent, the large administrative costs and the large size of non-performing loans.[28]

The conduct of interest rate policy has been walking a tightrope between competing objectives. First, the maintenance of an anti-inflationary stance for domestic purposes. Second, in an environment of capital mobility and a basically fixed exchange rate, to discourage liquidity creating capital inflows attracted by the high interest rate differential. The strategy of the Bank of Greece during 1996 appears to have been to aim its interest rate policy at achieving the former objective, while sterilising or resorting to more drastic measures to absorb the

liquidity coming from the external accounts. Thus, the Bank of Greece's intervention rate reductions have followed the path of the underlying inflation rate during 1996. Nevertheless, the Government's more aggressive pricing of its securities – which dominate the financial market – appear to be placing downward pressure on deposit rates (Figure 10, panel C). The rapid decline in Treasury bill yields reduced their (tax adjusted) differential with deposit rates to just over 1 percentage point at end-1996 compared with 2.7 percentage points at end-1995. As a result, deposit rates adjusted for inflation fell in late 1996 to their lowest level since the early 1990s, soon after interest rates were liberalised in 1989. Subsequently, the reduction in deposit rates has lagged the sharp fall in inflation leading to a small recovery in deposit rates in real terms. Nevertheless, no discernible decline has been observed in the ratio of deposits to GDP.

### *Monetary policy and developments in 1997*

Further reduction in inflation remains the primary objective of monetary policy in 1997. Inflation is targeted to fall to 4.5 per cent by the end of 1997 and below 3 per cent by the end of 1998. The Bank of Greece's strategy for attaining the inflation objective will be similar to that of the past years and will be based on two intermediate targets; the stability of the exchange rate and the steering of the monetary aggregates within ranges consistent with the inflation target. However, several changes were announced in the operation of monetary policy which aim to make it more flexible and thus discourage capital inflows. First, the exchange rate objective has been linked to an (unspecified) basket of European currencies rather than the ECU and the stable exchange rate policy will be characterised by larger fluctuations around an objective of broad stability. Second, monetary policy will be more responsive to the progress made towards attaining the inflation target, as well as developments in capital and foreign exchange markets. Third, in gauging progress in disinflation the Bank of Greece will examine several indicators which affect core inflation and inflationary expectations. Fourth, monetary policy will respond more slowly than in the past to variations of monetary aggregates from their target ranges in view of their increased instability.[29]

The pressures on monetary policy recommenced during the first months of 1997. The primary source of liquidity was the government's drawing down reserves equivalent to Dr 1.3 trillion (4 per cent of GDP) during the final part

of 1996 of which about one-third were from foreign currency denominated sources. Private capital inflows remained relatively subdued. Reflecting the increased flexibility of exchange rate policy, the drachma depreciated by about 2 per cent against the ECU during the first five months of the year.

In March 1997, the Bank of Greece announced a new method for conducting interventions in the interbank market, which aims to discourage capital inflows into the interbank market. First, weekly auctions for 14-day repos and 14-day deposits (reverse repos) will replace the overnight facility as the main instrument for interbank interventions. The interest rate on these instruments will range between the rate on the Lombard facility and the overnight intervention rate. Second, the overnight intervention rate will be replaced by a two-tier system. Only deposits below a global banking system limit will receive the higher rate. The global limit – initially set at Dr 300 billion – has been distributed to individual banks according to their market size.[30] The interest rate on the lower tier deposits was reduced by two percentage points relative to those in the upper tier, which remained broadly at the level of the old overnight intervention rate. Other measures to discourage capital inflows were implemented in May 1997: the special bank tax (EFTE) was reduced by one percentage point to 3 per cent and the 1.2 per cent fee on the value of a loan was extended to foreign currency denominated loans.

## Structural policies

The government's priorities in pursuing structural reform remain unchanged: the improved functioning of product, labour and financial markets, the enhanced efficiency of public administration, and a reduced role for the State, except in sensitive industrial sectors (mainly military). During 1996 progress was quite slow, though the prospects for more substantial progress in implementing structural reform appear to have improved in 1997 (see Box 1). The new Government has commenced debate on several of the key structural problems facing Greece, namely the labour market and the pension system, and has presented to Parliament a wide-ranging reform of the health system.

The labour market legislation which was described in detail in the previous *Survey* passed into law in August 1996 and has only recently begun to be implemented. Its main objective was to create a more active labour market policy (Table 10). It introduced a multi-purpose voucher, which consolidates employ-

Box 1. **Main elements of the government's 1997 structural policies**

1. Privatise the following:
   - Public sector corporations
     National Telephone Organisation (OTE): sale of an additional 12 per cent
     National Petroleum Corporation (DEP): sale of a small percentage
     Airport duty-free shops
     Stock exchange operations: sale of 45 per cent
   - Industrial conglomerates under Industrial Restructuring Organisation (IRO) umbrella: (Softex, MEL, Eleusina Bauxite, Kerafina)
   - State-controlled banks
     Bank of Attica: completed in January 1997
     Bank of Crete
     Bank of Central Greece
2. Grant independence to the Bank of Greece.
3. Continue to restructure the portfolios of the large state-controlled banks.
4. Introduce legislation to modernise the capital and money markets and strengthen the Capital Market Supervisory Committee.
5. Implement fully recently adopted legislation to modernise the operations of the public sector, public enterprises, and the public health system.
6. Introduce new techniques to finance public works through increased participation of the private sector.
7. Carry forth social dialogues with social partners on reforms of the labour market and the pension system.

ment incentives and unemployment benefit programmes, expands a series of existing subsidy and training programmes, and reduces a component of non-wage labour costs in the case of youth hirings. Regarding product market competition, shop closing times were reduced by two hours on Saturdays and now end at 6 p.m. Finally, the Government is considering measures to improve competition in the dairy and fresh vegetable markets.

Though at a very early stage in the process toward labour reform, the government's dialogue with social partners includes some proposals presented by the follow-up on *Implementing the OECD Jobs Study* in the previous *Survey*. In a 19-point agenda for discussion among social partners presented on 14 April, the Government suggested *inter alia* the following axes for the creation of a labour market that is more in tune with the functioning of a modern economy:

- the replacement of daily and weekly working limits with annual ones with a view to reducing recourse to (more expensive) overtime work;
- an expansion of part-time work to the public sector;

Table 10. **Implementing the OECD Jobs Strategy – an overview**

| Proposal | Action | OECD assessment/recommendation |
|---|---|---|
| **I. Increase wage and labour cost flexibility** | | |
| *a)* Lower minimum wages for vulnerable workers (low-skilled, youth) | *a)* Small reduction in non-wage labour costs for youth hirings | Make-up the commensurate resources to the social security system |
| *b)* Simplify and decentralise the wage bargaining process | *b)* Social dialogue to examine the use of local collective agreements in high unemployment areas | Timely implementation. At a minimum, reduce the role of the middle tier (sectoral) of collective agreements |
| *c)* Eliminate the explicit inflation catch-up clause | | |
| *d)* Reduce the regressivity of the payroll tax system | | |
| **II. Reform employment security provisions** | | |
| *a)* Abolish administrative limits on dismissals | | |
| *b)* Reduce mandated severance payments for white collar workers | | |
| **III. Increase working time flexibility** | | |
| *a)* Permit limit to, and payment premiums, for overtime hours to be determined in individual collective agreement | Social dialogue to examine the use of annual working limits in lieu of daily or weekly limits | Timely implementation. |
| *b)* Promote the transferability of pensions | | Tighten the links between earning history and pension entitlements to strengthen work incentives |

Table 10. **Implementing the OECD Jobs Strategy – an overview** *(cont.)*

| Proposal | Action | OECD assessment/recommendation |
|---|---|---|
| **IV. Reform unemployment and related benefit systems** | | |
| *a)* Rationalise unemployment benefits | | Widen the use of other social security benefits as part of the reform of the pension system |
| **V. Expand and enhance active labour market policies** | | |
| *a)* Develop programmes for the long-term unemployed in lieu of employment subsidy | Consolidation of the system of employment incentives and unemployment benefits through the introduction of multi-use vouchers (for training, wage subsidy or unemployment benefits) | Accelerate the implementation of the multi-use vouchers |
| *b)* Allow for the creation of for-profit job placement agencies | | |
| **VI. Improve labour force skills and competencies** | | |
| *a)* Expand and consolidate public vocational training programmes | | |
| *b)* Gradually tighten time limits on students earning a university degree | | |
| *c)* Improve evaluation at pre-university levels | | |
| **VII. Technology and innovations** | | |
| *a)* Strengthen the links between universities and enterprises | *a)* Increase in the participation of enterprises in the vocational training system | |
| **VIII. Enhance product market competition** | | |
| *a)* Foster liberalisation in sectors dominated by the public sector | *a)* Progress in clearing non-performing loan portfolio in major public banks | Accelerate the process of deregulation and privatisation |
| *b)* Continue the privatisation process | *b)* Commenced introduction of management contracts for public enterprises | |
| | *c)* Partial privatisation of the national telephone company in April 1996 | |

*Source:* OECD.

– the use of special collective wage agreements in high unemployment areas;
– the introduction of special working conditions for a limited-duration in order to obtain experience and then facilitate the re-entry into the workforce of groups susceptible to long-term unemployment;
– a review of tax and pension regulations pertaining to pensioners' employment earnings; and
– granting work permits to illegal immigrant workers so as to place them on an equal footing with other members of the work force.

The agenda also supports the improvement of training.

Financial market reform during 1996 entailed further progress in restructuring the portfolios of state-owned banks and the introduction of various measures to improve the institutional framework at the Athens Stock exchange. The National Bank of Greece – which is by far the largest bank in Greece and controls about 30 per cent of the deposit base – made large strides in clearing its non-performing loan portfolio and debt equity participations with loss making enterprises. It wrote off Dr 117 billion by using provisions and not paying a dividend in 1997 and plans to clear the remaining non-performing loans amounting to approximately Dr 200 billion (0.7 per cent of GDP) before the end of 1997. Moreover, it continued to make progress in selling non-financial interests. The Greek Bank for Industrial Development (ETBA), under its restructuring and recapitalisation programme, is in the process of settling non-performing loans amounting to Dr 328 billion (1.1 per cent of GDP). However, little progress was made in selling the small state-owned banks, though the Bank of Central Greece should be sold during 1997 as the sale process has already begun. An auction for the restructured Bank of Crete did not succeed but a second attempt is underway, and the Bank of Attica was sold to a social security fund in early 1997 (though management will be maintained by the state-owned Commercial Bank for five years). Regarding the stock market, several regulatory measures have been introduced after a share-rigging scandal disrupted trading in November 1996. First, brokerage houses will only be able to trade up to the limit of their equity capital. Second, a guarantee fund to cover settlement risk will be created from transaction-based contributions. Third, control will be enhanced through internal auditing of brokers overseen by the Capital Markets Committee and traded firms will be required to issue timely audited half-yearly statements. Finally, lags in clear-

ing transactions – which is one of the root causes of the lack of transparency – will be greatly reduced following the full dematerialisation of shares by end-1997. In addition, derivative trading is envisaged to commence on the stock exchange during 1997.

Headway towards improving the effectiveness of public administration was uneven in 1996. The new regulation introduced in 1996 to control government guarantees appears to have slowed the issuance of new guarantees. It is too early to gauge the success of the measure, which requires public enterprises and public entities receiving transfers from the state budget to submit and justify budget proposals, as these will be received for the first time at the end of the first quarter of 1997. Moreover, it has proved difficult to find professional managers as envisaged under the 1996 legislation introducing management contracts for public enterprises. Managers have been placed in only few of the smaller public enterprises. Looking forward, the 1997 Budget introduces a new organisation which will assist public entities in rationalising their expenditures (System of Complete Accounting Surveillance). Regarding the Government's long standing fight against tax evasion, the plan to introduce computer verification of tax returns has been delayed further to 1998, though the tax police began operations in April 1997. Finally, the measures to improve the utilisation of CSF II funds and to facilitate foreign investment (the one-stop shop) have been put into place.

Privatisation has been disappointing since the last *Survey*, with little progress following the initial public offering of an 8 per cent share of the national telephone company (OTE) in April 1996. However, a second offering of OTE shares – of about 10 per cent has started and should be completed by September 1997 and the sale of Eleusis shipyards is nearing completion in June 1997. The Government is also committed to a small share offering of the national petroleum company (DEP) in the second half of 1997 and to close the remaining companies under the Industrial Restructuring Organisation (IRO) umbrella by end-1998. It has recently announced its intention to sell the airport duty-free shops, several units of the National Tourist Organisation, and operation of the Athens Stock Exchange. Regarding the restructuring of ailing public entities, the EU continues to hold up the release of the second and third tranche of state aid to Olympic Airways until its restructuring programme is back on track. However, the European Commission has approved the sale of Scaramanga shipyards to its workers, and thus cleared the way for the government's assumption of the company's debts.

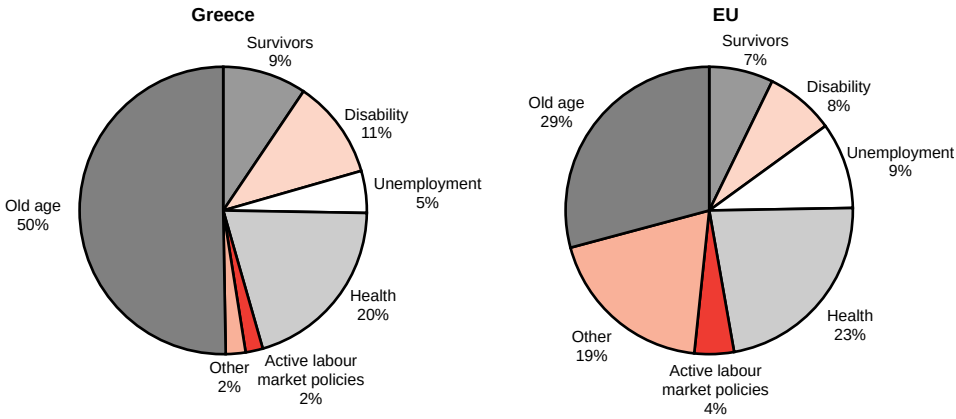
# III. Social security: pensions and health care

The substantial fiscal consolidation required in the period ahead necessitates a review of all social security entitlement programmes in order to assure an adequate but more efficient provision of services. Social security expenditure amounts to about 23 per cent of GDP, which is among the lowest levels for OECD countries, but nonetheless accounts for over 60 per cent of current primary spending of general government.[31] Moreover, expenditure pressures will mount considerably as the population ages and per capita income in Greece tends towards the levels of other OECD members. Social security reform implies essentially better control of pension and health care outlays, which together account for over 90 per cent of total social security expenditure, with most of the remainder directed towards labour market policies and a small part towards family benefits.

Though social security outlays have increased less rapidly than in other OECD countries, the composition of social security expenditure differs significantly, not only in the low emphasis on social programmes other than health care and pensions, but also in the concentration of expenditure on pensions (Figure 11). This is the only category of social expenditure for which Greece exceeds (as a per cent of GDP) the OECD average, despite a significantly lower per capita income. As for public expenditure on health care, Greek spending is not much different from that elsewhere in the OECD, after taking into account Greece's lower income level. As the Greek authorities are in the process of implementing a health reform and recognising the need for radical pension reform, this chapter investigates the situation in both sectors and discusses options for reform.

Figure 11. **THE DISTRIBUTION OF SOCIAL EXPENDITURE IN 1993**[1]

As a percentage of total expenditure




1. Greek data excludes certain categories of social expenditures included in the other EU countries. The Greek authorities are in the process of updating their data.

*Source:* OECD, Social expenditure statistics of OECD Member countries.

## The pension system

Critical reforms to the pension system were undertaken in 1990-92 to stem its rapidly deteriorating financial position. It was recognised at the time that these reforms would provide only some temporary breathing room until more fundamental changes to the pay-as-you-go system could be implemented. Indeed, unfavourable demographics – worse than in most industrial countries – combined with a still generous, but inequitable system, will soon expose the unsustainability of the situation again. The main shortcomings of the public pension system arise from loose eligibility conditions, introduced mostly during the period 1978-85, combined with individual pension benefits which in general far exceed the corresponding contributions and foster early retirement and contribution evasion. Furthermore, the complexity and excessive segmentation of the system, combined with poor administration, permit abuses of the system. As an illustration of the problem facing the Greek pension system, the OECD's long-

term projections presented in this chapter, which attempt to incorporate – to the extent possible – the full effects of the 1990-92 reforms, point to a present value of future pension liabilities, net only of employer and employee contributions, of the order of 200 per cent of GDP. These far exceed similar calculations for higher income OECD countries. It should be acknowledged that the complexity and data problems of the Greek pension system introduce a greater than usual degree of uncertainty around the central projections. Nevertheless, the order of magnitude of the imbalance indicates clearly the need for an overhaul of the pension system so as to better align individual pension benefits with contributions and at the same time provide adequate income security in retirement, especially for the elderly poor.

The Government has decided to delay the implementation of any radical reforms to the pension system in view of the priority placed on labour market reform and the perceived breathing room provided by the 1990-92 reforms. Nevertheless, the Government recognises that changes can not be postponed for long, because many of them could take several decades to be fully effective and if delayed they will be more onerous to implement. In this regard, it has commenced a social dialogue on the viability of the pension system. The discussion will focus in 1997 on improving the performance of the system as it currently stands, including the simplification of pension legislation, efforts to reduce contribution evasion, the enhancement of administrative capacity and perhaps a reduction in the number of funds, while deeper reforms will be considered at a second stage. With policy debate soon to come to a critical juncture, this chapter aims to provide an analysis of the institutional aspects which characterise the system and use the long-term simulations to assess options for reform. The first section describes the system, the second section analyses the microeconomic incentives imbedded in the system, while the third section explains the factors driving the unsustainability of the system and the effect of the 1990-92 reforms. The final section presents the projections and reviews reform options.

## *The structure of the pension system*

Pensions are provided almost exclusively by the public sector, and there is very limited recourse to private pensions.[32] Public pensions are provided through a plethora of funds, which have the status of public entities and form part of the general government sector (Box 2). The funds offer defined-benefits based on

Box 2. **The main pension funds**

The public pension system is highly segmented and complex, containing over 300 funds, with many different regulations for pension rights. The 28 primary pension funds and the system of public pensions can be classified by employment category into five groups whose funds have broadly similar regulations concerning pension rights, though the differences between the five groups are substantial. These groups cover respectively: *i)* private sector wage earners (principally IKA), of which seamen (NAT) are an important subset; *ii)* farmers (OGA); *iii)* non-agricultural self-employed (principally TEBE), of which another important subset covers professionals, such as doctors, lawyers, and engineers; *iv)* civil servants; and *v)* employees in public enterprises and state-owned banks. The large number of supplementary funds (over 200) were mostly created after the second world war to supplement the low level of primary pensions. Individually, these funds cover much smaller groups of workers than primary funds and are often based on enterprise agreements. Two large supplementary funds were created in the 1980s: one to cover approximately 50 per cent of private sector employees who did not have supplementary coverage, and a second for agricultural workers, who account for about 50 per cent of the self-employed. Consequently, only self-employed small entrepreneurs and traders do not have supplementary coverage. Separation pensions cover a smaller group of workers comprising mostly public employees, lawyers, doctors and since quite recently, a few groups of private sector wage earners. Table 11 presents the major characteristics of the primary, supplementary and separation funds that correspond to each of the five categories of workers.

employer and employee contributions. Though the funds are permitted to retain any existing cash surpluses – reflecting the system's original conception as a funded system – in total, these are not very significant and they generally operate as pay-as-you-go schemes.[33] The large number of funds arise from the historical development of the pension system, whereby funds were created by individual sectors. Partly as a result the benefit schemes offered by the funds, or even within the funds, differ. Workers are insured by at least one primary fund but usually have supplementary coverage (which may be provided by a separate fund). In addition, some workers contribute towards a lump-sum separation payment at the time of retirement. Thus segmentation of the pension system is both horizontal across economic sectors and vertical between three levels of pension payments: primary, supplementary and separation payments.

Table 11. **The structure of pension funds**[1]

| | Contributors (per cent of total) | Primary pensions (per cent of total) | Dependency ratio[2] | Main primary funds | Additional coverage by: | |
|---|---|---|---|---|---|---|
| | | | | | Supplementary fund | Separation fund |
| 1. Private sector employees | 39.5 | 37.6 | 1: 1.9 | IKA | Yes | Some |
| *Of which:* seamen | 0.8 | 2.7 | 1: 0.5 | NAT | Yes | Yes |
| 2. Public sector employees | 13.8 | 14.9 | 1: 1.6 | | | |
| General government[3] | 9.0 | 11.7 | 1: 1.4 | – | Yes[4] | Yes |
| Public enterprises and state-owned banks | 4.8 | 3.2 | 1: 2.6 | Eight funds | Yes | Yes |
| 3. Self-employed | | | | | | |
| Agriculture | 20.1 | 36.8 | 1: 1.0 | OGA | No[5] | No |
| Non-agriculture | 26.7 | 10.7 | 1: 4.4 | | | |
| Small entrepreneurs and traders | 22.0 | 9.0 | 1: 4.3 | TEBE | No | Some |
| Professionals | 4.7 | 1.7 | 1: 5.0 | Lawyers, doctors, engineers funds | Yes | Yes |
| Total | 100.0 | 100.0 | 1: 1.8 | | | |

1. Funds under Ministry of Welfare and Social Security and other ministries.
2. Primary pensions per contributor.
3. Excluding war pensions and pensions for participants in the national resistance.
4. The typical civil servant has two supplementary pensions.
5. The farmer's supplementary fund was turned into a primary fund in 1997.

*Source:* Ministry of Welfare and Social Security.

### *Problems of accountability and inequity arise from excessive segmentation*

The segmentation and complexity of the public pension system, in conjunction with poor administration, has led to a lack of transparency and inadequate monitoring.[34] As a result, pension fraud is difficult to detect and contribution evasion is sizeable. The lack of transparency is highlighted by the fact that most funds compile their accounts with delays amounting to several years, while accounting quality is poor. Few funds other than the large fund for private sector employees (IKA) collect anything more than the most basic data on their pensioners or contributors. This precludes a full compilation of pension system data, and raises the potential for abuses, including contribution evasion.[35] It is characteristic of the poor quality of the data that the total number of contributors across primary funds exceeds total employment – as measured by the labour force survey – by about 20 per cent. Moreover, for the largest fund (IKA), declared earnings for contributions fall far short of the respective total earnings data from the national accounts. As funds monitor pensions rather than pensioners, the total number of pensioners is not known, though the number of primary pensioners exceeds the population aged over 65 years by a wide margin. Without proper accounting and administration (*e.g.* a unique social security number), pensioners may receive more than one primary pension and exceed with impunity the existing overall cap on total pension receipts per individual, or continue to receive a pension while working and earning in excess of statutory limits.[36] In addition, little collaboration exists with the income tax authorities to verify if contributions are consistent with declared income.[37] Frequent recourse to contribution amnesties – four since 1985 – have further weakened incentives to contribute. In an effort to improve monitoring, the 1990-92 reforms set up a national registry. However, its completion has been delayed until well after the year 2000.

The segmentation generates additional problems. First, it is inevitable that any system where funds are based on economic sectors will eventually include funds covering sectors with a declining number of contributors especially when the system as a whole faces deteriorating demographics.[38] Even if the overall system is sustainable this will lead to financial difficulties unless there exists access to a common pool of resources which is funded by the surpluses of the more dynamic sectors.[39] Though there is no overall effect on the general govern-

ment deficit as the consolidated system balance remains unchanged, the prospect that liabilities will be absorbed by budget transfers could lead to a moral hazard problem with, for instance, too generous pension benefits being granted to the current generation of pensioners.[40] In Greece, bankrupt funds (and their liabilities) have been absorbed by the largest social security fund (IKA), which has necessitated budgetary transfers. Other sizeable concessions, which certain sectors have been able to extract, are earmarked taxes to support (or in certain cases practically replace) contributions.[41] Finally, the existence of many funds leads to high expenditure on personnel. Pension funds employ some 35 000 individuals (approximately 1 per cent of total employment) at a cost of Dr 70 billion (0.3 per cent of GDP). Personnel outlays relative to overall social security spending are twice the OECD average.

### *Microeconomic incentives: high pensions relative to contributions*

The complexity and segmentation of the pension system obscures the factors which are straining its finances. Not surprisingly the common theme to all groups of funds is the generosity of individual pension payments relative to contributions. Nevertheless, pensioners (especially from the private sector) complain about low primary pensions, which consist of slightly less than one-half the average wage. This paradox is resolved by the fact that contributions have been quite low in the past and often for short periods, while pensions start at an early age – for employees, usually before the age of 55 in the public sector and near the age of 60 in the private sector (Table 12). Moreover, the existence of supplementary and multiple pensions, as well as dependency allowances, often raise a pensioner's income substantially.

The existence of a relatively high minimum pension in conjunction with a low eligibility requirement appears to be playing a critical role in providing private sector salaried employees (mainly insured by IKA) with an incentive to evade contributions. During the period 1978-90, the minimum pension increased substantially, peaking at 46 per cent of the average monthly blue collar wage in manufacturing in the late 1980s (Figure 12, panel A). Raising further the adequacy of a minimum pension as retirement income, pensioners usually receive a supplementary pension (raising the total pension received by about 15 to 20 per cent) and, if eligible, dependency allowances. Over the same period, the number of old-age pensions which are based on or are slightly longer than the minimum

Table 12. **Main characteristics of different categories of pension funds (1994)**

| | Typical[1] retirement age | Typical[1] contribution years | Contribution base[2] | Average pension[3] | Minimum pension[4] |
|---|---|---|---|---|---|
| Private sector wage earners | 60 | 20 | 108 | 62 | 84 |
| *Of which:* NAT | . . | 13.5 | 108 | 62 | . . |
| Non-agricultural self-employed | | | | | |
| Small entrepreneurs and traders | 62 | 35 | 94 | 48 | 73 |
| Professionals | 65 | 40 | 147 | 124 | 64 |
| Agriculture | 65 | n.a. | 72 | 14 | . . |
| Public servants | | | | | |
| General government | 55 | 25 | 98 | 93 | 42 |
| Public enterprise and banks | 55 | 25 | 136 | 134 | 37 |

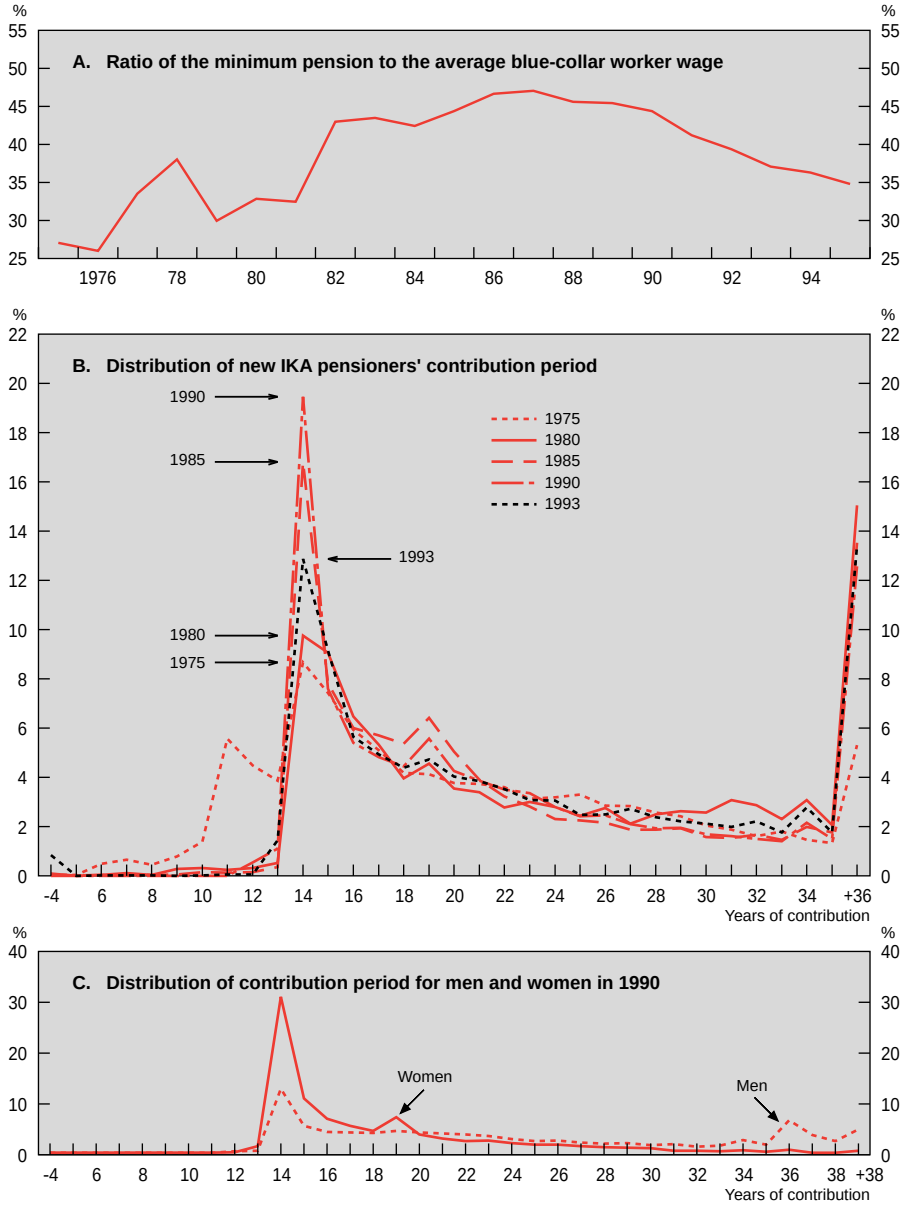
1. The typical characteristics refer to the situation of the most common pensioner as suggested by the data, which is acknowledged to be of a poor quality in certain cases. For public employees especially, the retirement age varies according to their date of permanent employment.
2. Earnings per capita for group of individuals belonging to the selected fund (OECD estimates derived from national accounts data) as a percentage of the national average earnings per capita. For the self-employed, the actual contribution base is far below earnings.
3. Old age pension per capita, including supplementary, for the specific fund as a percentage of the national average earnings per capita.
4. Ratio of minimum primary old-age pension to average old-age primary pension.

*Source:* Ministry of Welfare and Social Security and OECD estimates.

eligibility period (of 13.5 years until it was recently raised to 15 years) has also increased dramatically: the share of pensions based on the minimum eligibility period for IKA increased steadily to approach 20 per cent in 1990 (Figure 12, panel B).

It appears that once the qualifying period has been reached, the incentive to continue contributing has been greatly reduced. The minimum pension – despite its restrained rise since 1990 – continues to provide a high replacement rate (about 45 per cent), exceeding the one derived from the pension formula for over 25 years of contributions for the median income level (Figure 13).[42] This relatively high replacement rate has led to the startling number of pensioners that receive the minimum pension. For new pensioners, the number of retirees receiving the minimum pension rose steadily from some 14 per cent of total retirees in 1975 to near 60 per cent from 1985 onwards, so that 70 per cent of the total stock of pensions is currently at the minimum pension.

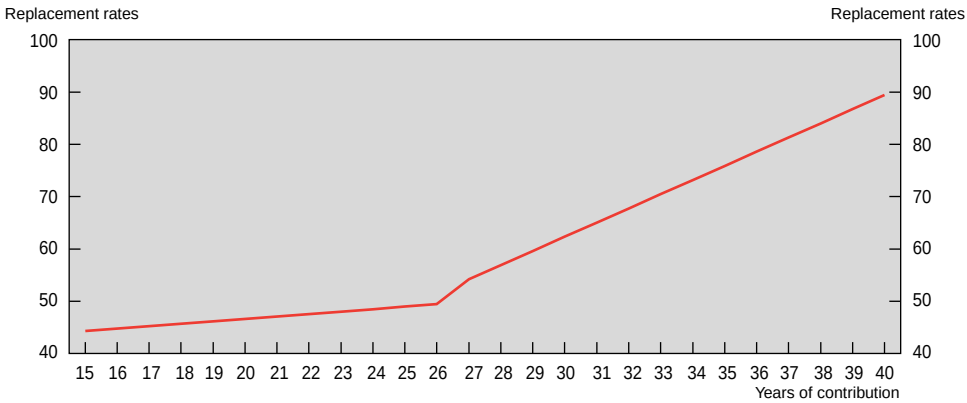
Figure 12. **MINIMUM PRIMARY PENSION FOR IKA AND LENGTH OF CONTRIBUTION PERIOD**




*Source:* National Statistical Service of Greece and Social Security budget.

Figure 13. **INTERACTION OF REPLACEMENT RATES AND THE MINIMUM PRIMARY PENSION FOR IKA IN 1995**[1]

13th bracket, per cent of pension base



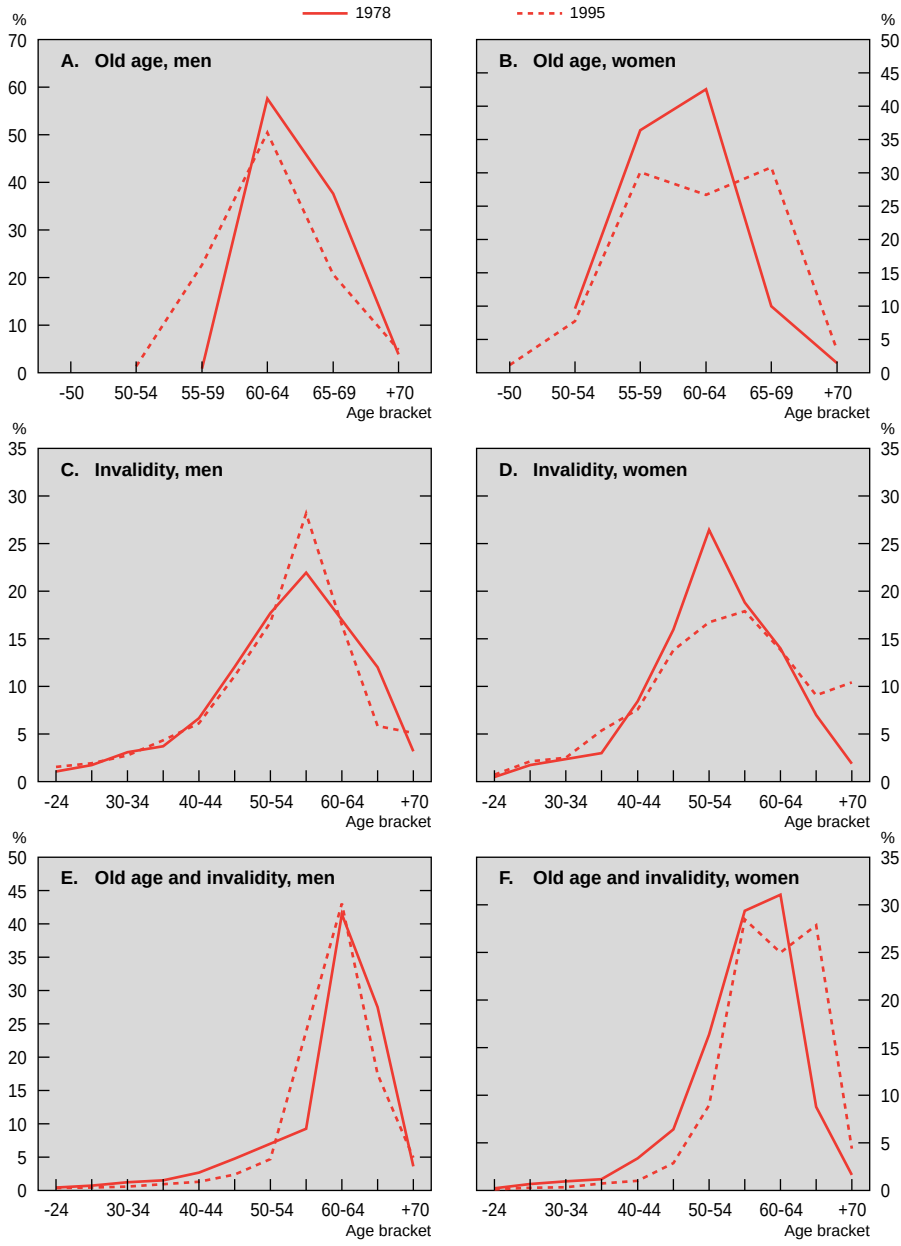

1. The pension formula for IKA provides higher replacement rates for lower incomes. Contributors are placed into 28 income brackets, with identical replacement rate formulas for the highest 15 income brackets. Most insured fall in the 13th income bracket. From 1992, the minimum pension was increased by 1 per cent for each year of contributions after the 15th year. For lower income brackets, the minimum pension exceeds the pension derived from the pension formula for 35 years contributions.

*Source:* IKA.

The few years for which private sector employees have actually contributed, especially for individuals retiring in the 1970s, likely reflects the impact of urbanisation and the belated increase in coverage that occurred in Greece, but also contribution evasion abetted by the relatively high minimum pension. Perhaps reflecting the latter, the age distribution for male old-age pensioners has shifted noticeably towards younger age groups during the past two decades, weakening the argument that non-qualification is a root cause for the low contribution period (Figure 14). Moreover, following the decline of the minimum pension in real terms since 1990, the number of retirees fulfilling only the minimum eligibility requirement fell, which provides further evidence for the importance of incentives to evade contributions.

Other factors also provide incentives for contribution evasion. First, the earning period used as the pension base is short – the shortest among OECD countries for private sector employees. Private employee pensions are based only

Figure 14. **AGE DISTRIBUTION OF IKA PENSIONERS**




*Source:* IKA.

on earnings over the last five years of employment (increased from the last two years in 1992), which introduces a bias towards under-reporting income in years falling outside this period.[43] Second, replacement rates are high for old-age primary pensions and are inversely related to the pension base, which also provides incentives to under-report income. For lower incomes, the gross statutory replacement rate for old-age primary pensions is near 90 per cent for 35 years' contributions while for higher incomes it is about 75 per cent – these are the highest replacement rates, relative to respective earnings, in the OECD.[44]

If national accounts data are used to estimate private sector wages, an admittedly rough estimation suggests a 20 to 30 per cent shortfall in contributions relative to their potential, which can only partly be explained by earnings such as overtime pay for which no contributions are required.[45] The stylised facts also support this notion. Reportedly, workers and their employers – especially in smaller firms and in the large construction sector – under-report working days to reduce the cost associated with social security contributions.[46] In the event, it would be surprising not to find evidence of contribution evasion in a system where income tax evasion is also considered to be high.

Small entrepreneurs and traders place themselves in low income contribution classes – significantly below the average income of the sector – but receive primary pensions which are broadly equal to pensions corresponding to dependent employment. In comparison to IKA pensioners, they make smaller contributions for more years, retiring closer to age 65.[47] Even subsequent to the raising of the minimum contribution class in 1992 toward the average income, the effective contribution rate relative to the average blue collar wage is only 8 per cent, compared with double that amount by private sector wage earners for their primary pensions (Table 13).[48] Though the effective replacement rate stands around 55 per cent and the base includes all contributing years – in contrast to the last five years for IKA pensioners – the pension is still far too generous relative to the low contributions in actuarial terms. The situation for self-employed professionals is similar, though both pensions (including supplementary and separation pensions) and contributions are higher and their funds are supported by earmarked taxes, which in certain cases exceed the revenues from contributions.

Farmers receive the lowest pensions, and receive them from age 65, though they often qualify earlier for a disability pension. They receive a flat pension,

Table 13. **Contribution and replacement rates in 1995**

Percentage points unless otherwise indicated

| | Contribution rate[1] | | | | Old-age replacement rates | | | |
|---|---|---|---|---|---|---|---|---|
| | Statutory | | Effective[2] | | Statutory[3] | | Effective[2] | |
| | Total[4] | Primary | Total[4] | Primary | Total[4] | Primary | Total[4] | Primary |
| Private sector wage earners (IKA) | 26 | 20 | 19 | 15 | 90-110 | 70-90 | 62 | 49 |
| Seamen (NAT) | n.a. | 15-23 | 28 | 21 | 100 | 80 | 65 | 53 |
| Small entrepreneurs and traders (TEBE)[5] | 20 | 20 | 8 | 8 | 90 | 90 | 54 | 54 |
| Professionals[5] | n.a. | n.a. | 13 | 5 | n.a. | 80 | 90 | 55 |
| Farmers (OGA)[5] | | | | | | | | |
| Old | 4 | . . | 1 | 0 | n.a. | n.a. | 20 | 16 |
| New | 7 | 7 | . . | . . | 70 | 70 | . . | . . |
| Civil servants[6] | 15.75 | 6.75 | 12 | 6 | 152[7] | 80 | 110[7] | 83 |
| Public enterprises and state-owned banks[8] | 42 | 33 | 26 | 21 | 100 | 80 | 108 | 88 |
| Total | . . | . . | 14 | 10 | . . | . . | 53 | 41 |
| *Memorandum:* | | | | | | | | |
| Post-1992 new entrants into work force | 26[9] | 20[9] | . . | . . | 80 | 60 | . . | . . |

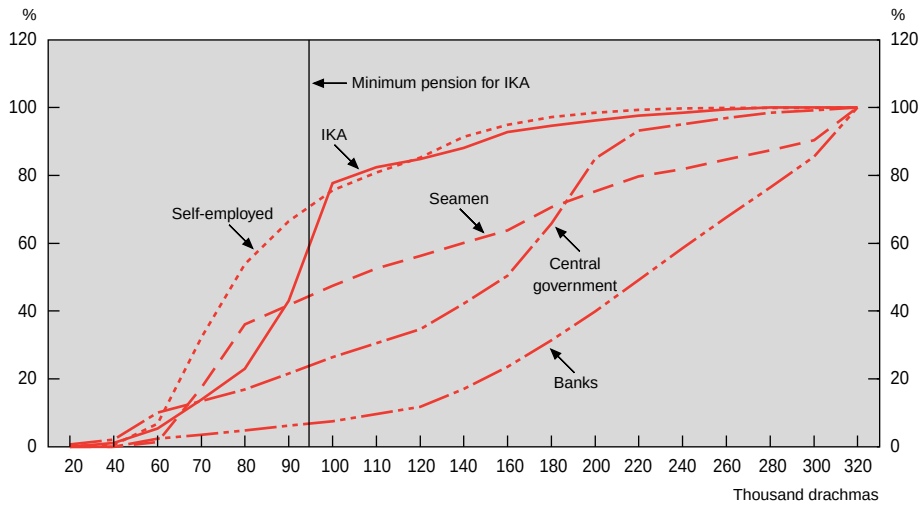
1. Employer and employee contributions.
2. The effective contribution and replacement rates are derived from the actual pension expenditure and contribution using national accounts data for the derivation of sectoral earnings.
3. For 35 years contributions.
4. Primary and supplementary pensions.
5. Contributions and pensions based on either fixed or voluntary income classes, which are often unrelated to actual earnings. Statutory contribution and replacement rates are based on these nominal values, and the median income class where applicable. Effective contribution rates are often biased upwards by the inclusion of receipts from earmarked taxes.
6. Excluding the participation of Government as employer. Employee contributions for primary pensions were introduced in 1993 and the rate raised to that in the private sector by 1995.
7. Includes two supplementary pensions.
8. Statutory rates are proxied by those of the Public Electricity Company (DEH).
9. Excludes government contribution amounting to 10 percentage points.

*Source:* Social Security Budget and OECD estimates.

currently equivalent to about 29 per cent of the minimum IKA pension. This fixed amount has lost approximately 70 per cent of its purchasing power since 1986. However, farmers do not make contributions towards their primary pensions, and their pensions are funded by the budget (1.2 per cent of GDP) and earmarked taxes (0.5 per cent of GDP). Their low supplementary pensions will be replaced by a new primary pension plan which will begin to collect contributions in 1998, with the Government providing two-thirds of the contributions.[49]

Civil servants and employees of public enterprises and state-owned banks receive the most generous pensions (Figure 15) and can retire at an earlier age, usually after serving 25 years (and significantly less for women hired before 1983). For neither category of employees is contribution evasion a real possibility. For civil servants, employee contribution rates for a primary pension have been similar to those in the private sector since 1995 (civil servants did not make contributions towards their primary pension prior to 1993). The employers' share of contributions is hard to define since the Government does not dedicate budgetary resources for civil servant pensions and it has a legal responsibility for the

Figure 15. **CUMULATIVE DISTRIBUTION OF PRIMARY PENSION LEVELS FOR MAJOR EMPLOYMENT CATEGORIES, 1995**




*Source:* Social Security budgets.

total pension outlay. Contribution rates in public enterprises and state-owned banks are higher than those in the private sector (33 per cent compared to 20 per cent in the private sector).[50] Another difference with the private sector is that public enterprises usually have enterprise specific pension funds (whose liabilities have to be recognised at the time of privatisation). Public sector pensions for both civil servants and employees of public enterprises and state-owned banks are generous because they are based on the last monthly salary before retirement. The effective replacement rate for the primary plus supplementary pension is estimated to be around 110 per cent for both public sector employees and civil servants, with civil servants receiving two supplementary pensions. The former receive the highest pensions due to their high wages, with only doctors' pensions coming close. In addition, civil servants receive a (tax-free) lump-sum separation payment upon retirement equivalent to two years of salary (for 25 years' work) raising their pensions further.

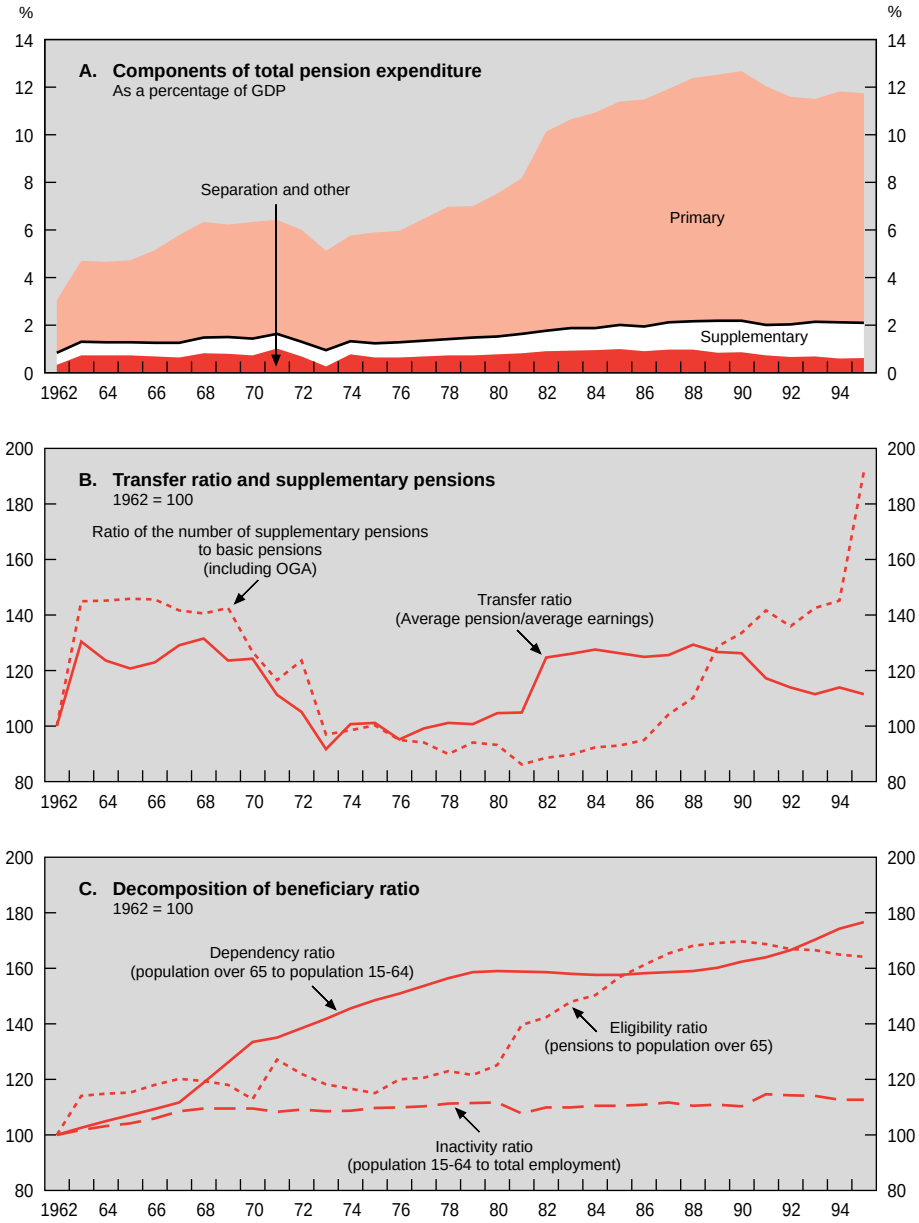
## *The unsustainability of the pension system*

### *Expenditure: a large increase in pensions and pensioners*

The microeconomic incentives imbedded in the system, in combination with the introduction of other measures that increased the system's generosity, resulted in a surge in aggregate pension expenditure from below 6 per cent of GDP in the mid-1970s to over 12 per cent of GDP in 1990 (Figure 16, panel A). These expenditure developments can be separated into an upward shift in the transfer ratio (the average pension divided by earnings per capita) at the start of the decade and a sharp and consistent increase in the eligibility ratio (the number of primary pensions divided by the population over the age of 65). The main policy change that raised the transfer ratio was the establishment of a link between minimum pensions and minimum wages during a period of rapid real wage growth at the lower end of the wage spectrum. A second factor raising the transfer ratio was maturing supplementary funds, which raised individual pension benefits in the latter part of the 1980s.[51] As a result, the transfer ratio rose by nearly 30 percentage points between 1978 and 1990 (Figure 16, panel B).

Nevertheless, a lowering of eligibility requirements was the main factor raising pension expenditure during the late 1970s and all of the 1980s. As a result, the eligibility ratio rose by over 50 per cent; to reach 1⅓ primary pensions for every member of the population above the age 65 in 1990 (Figure 16,

Figure 16. **EXPENDITURE ON PENSIONS**




*Source:* Social Security Budgets.

panel C).[52] The current level of 2.2 million primary pensions (comprising old-age, invalidity and survivor benefits) are equivalent to 60 per cent of employment, which is probably one of the highest percentages in the OECD. Policies that raised the eligibility ratio comprised the sharp increase in the minimum pension described above, better opportunities to obtain early retirement for the rising number of unemployed, easier access to invalidity pensions, and the provision of pensions for those who were falling short of qualifying for a pension (Box 3). The effect of these policies can be highlighted by developments in IKA.

Box 3. **Policies that raised pension expenditure during 1978-82**

**Transfer ratio**

The transfer ratio rose sharply in the period 1978-85 when the minimum pension was linked to an increasingly larger share of the minimum blue-collar wage (initially 60 per cent and subsequently 80 per cent), while the minimum blue collar wage was raised by 50 per cent in 1982. Other policies which raised the transfer ratio were: *i)* on an *ad hoc* basis, pensioners were placed in higher pension classes; and *ii)* Christmas, Easter and holiday salary bonuses, amounting to two months' salary, were added to the calculation of the monthly pension base in 1978, thus raising pensions by 16.6 per cent.

**Eligibility ratio**

Policy facilitated the provision of pensions for non-qualifiers and older age unemployed. The existence of many individuals who did not qualify for a pension reflected the extension of pension coverage to the entire population in 1982. Moreover, workers who migrated from rural to urban areas have difficulty in qualifying for a pension since farmers do not contribute toward a pension and thus have no pension rights to transfer to their new pension fund upon commencing work in urban areas. A broken employment history, primarily affecting women, is an additional factor which hampers qualification for a pension.

The main measures that eased pension eligibility for workers whose contribution period fell short of the minimum qualifying period were: *i)* permitting individuals over the age of 65 to purchase sufficient pension rights to obtain the minimum pension; and *ii)* introducing a special old-age pension for non-contributors (equivalent to a farm pension) for some 35 000 individuals. Similarly, pension rights were granted to many non-contributors, such as immigrants of Greek origin – arriving in Greece prior to 1984, especially from Eastern Europe – and members of the national resistance. In neither case were the beneficiaries required to make a payment towards these pension rights. Finally, agricultural pensions were automatically granted to farmers' spouses in 1981, increasing the number of pensions – albeit of a low individual level – by about 120 000.

Measures aimed at the unemployed comprised: *i)* the introduction in 1978 of the ''35 years'' rule, whereby anyone of age 58 with 35 years' contributions could receive a full pension; *ii)* the *ad hoc* expansion of the ''arduous and unhealthy employment'' category of contributors who are eligible for retirement five years earlier than the normal retirement age, albeit in exchange for somewhat higher contributions; and *iii)* a lax application of eligibility conditions for invalidity pensions.

During 1975-90, an increasingly larger share of retirements were made before age 60, with the earlier retirement of men accounting for the rise (Figure 17, panel A).[53] Women (especially those with under-age children) traditionally have had more generous eligibility requirements.[54] At the same time, about one-quarter of new pensioners received an invalidity pension, though the share has declined since the mid-1980s as pressures to provide pensions to unemployed workers eased (Figure 17, panel B).[55] Finally, about 40 per cent of private sector wage earners were classified as working under ''arduous and unhealthy employment conditions'', whereby retirement can be moved forward by five years. Spending developments during this period were relatively unaffected by deteriorating demographics. After having increased rapidly during the 1960s and 1970s, the dependency ratio remained fairly stable during the 1980s, which may reflect a low birth rate during the tumultuous period of the late 1910s and early 1920s.

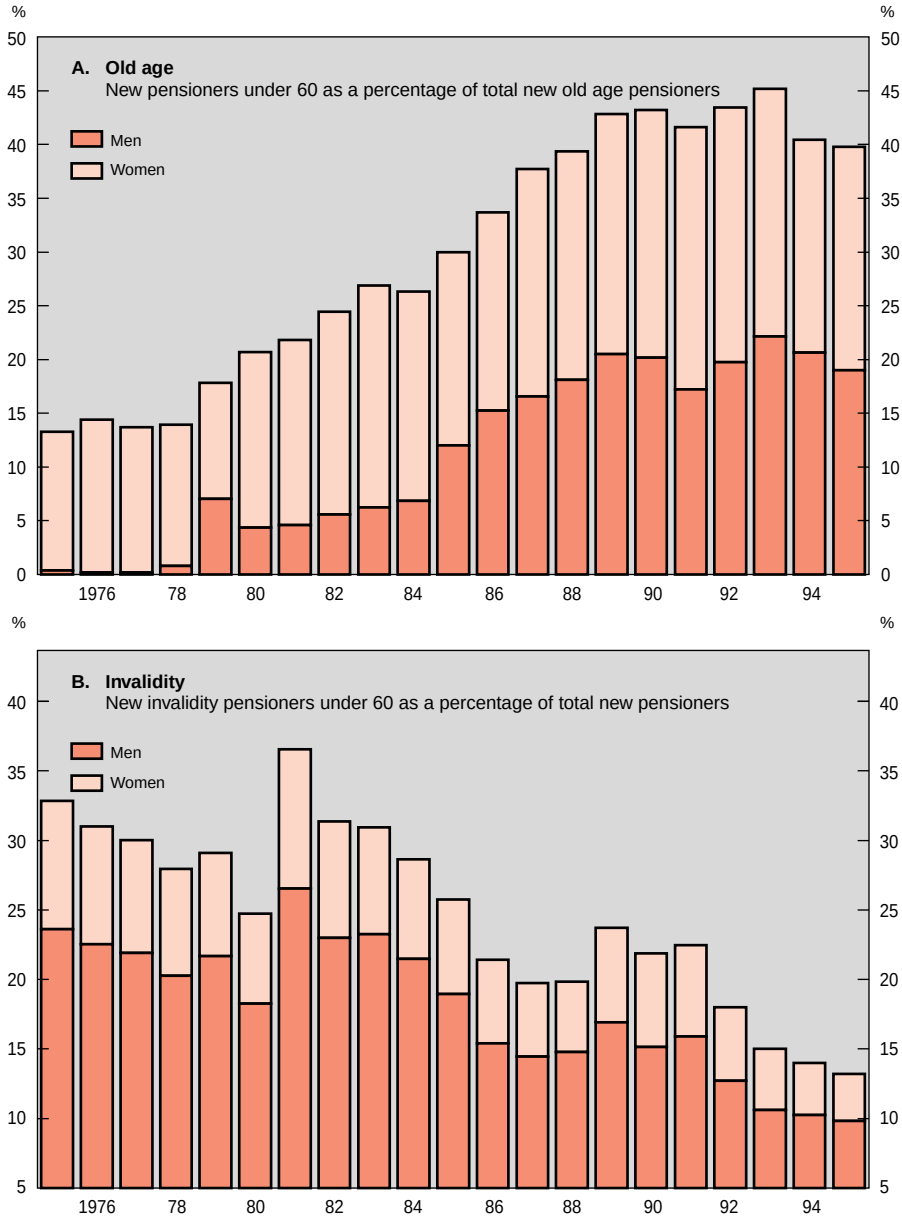
### *Revenues: rising salaries and contribution rates*

Though not keeping pace with expenditures, revenues from employers' and employees' contributions rose steadily as a per cent of GDP during the late 1970s and 1980s (Figure 18). During the period 1975-85, the underlying force was the sharp rise in real wages. Contributions as a per cent of GDP showed some further increase subsequently even though the labour share fell between 1985 and 1990, as the share of workers contributing to supplementary pension funds increased, and more than doubled between 1980 and 1990.[56] The increased coverage of supplementary funds initially provides extra revenues, but will soon lead to additional expenditures because eligibility requirements are low. A similar short-term source of contributions will arise from the introduction of the new primary fund for farm workers in 1997.[57]

### *The 1990-92 reforms: a stop gap measure*

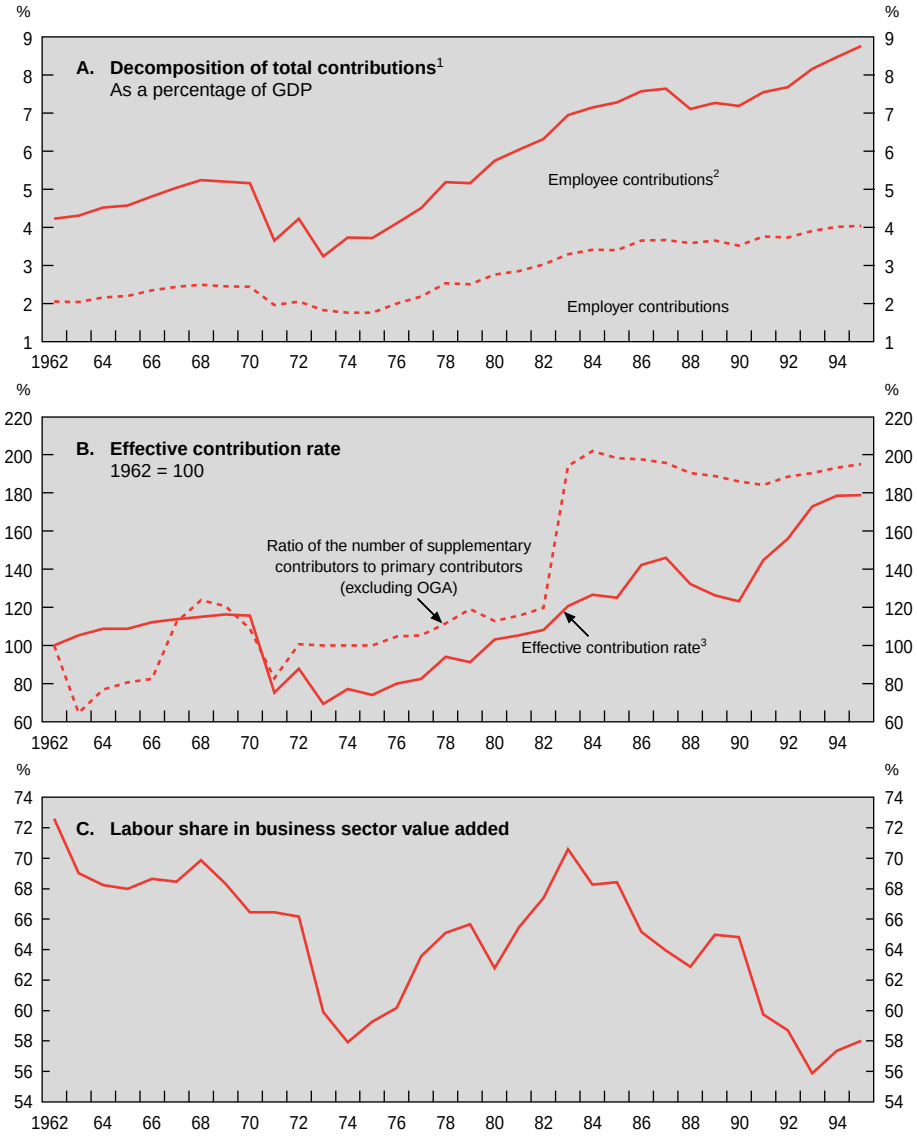
The 1990-92 reforms were required to stem an explosive situation. Total pension expenditures exceeded contributions by 5½ per cent of GDP in 1990 – compared with a shortfall of about 2 per cent of GDP in 1975 – despite a 3½ per cent of GDP increase in contributions during the corresponding period (Figure 19). To cover expenditures, transfers from the budget increased from about ½ per cent of GDP in 1975 to nearly 4 per cent of GDP in 1990. In addition, some of the larger funds became heavily indebted.[58] Following the introduction of the reforms, contributions increased and expenditures fell, each

Figure 17. **NEW IKA PENSIONERS BELOW AGE 60**




*Source:* IKA.

Figure 18. **CONTRIBUTIONS FOR PENSIONS**

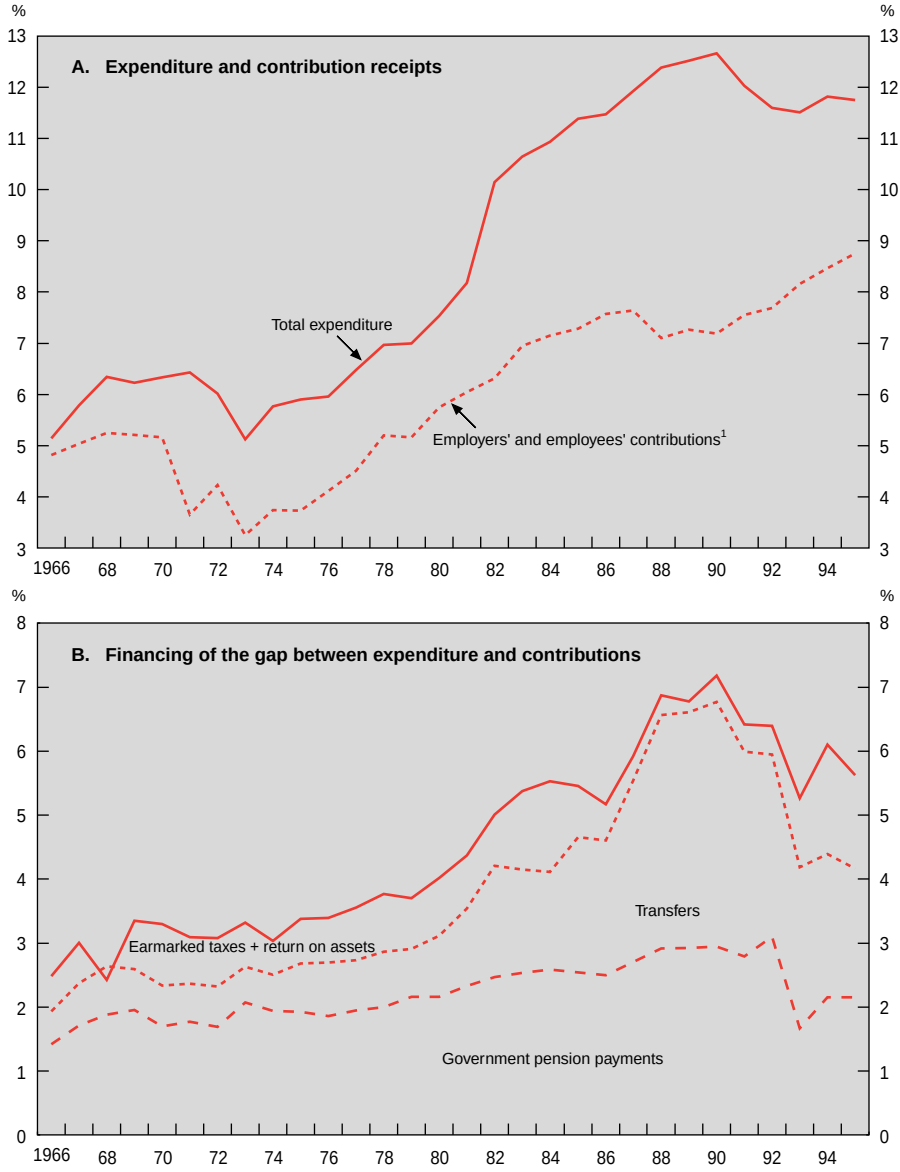



1. Including contributions for health care which amount to 2 per cent of GDP in 1994, excludes government contributions and transfers.
2. Includes total of self-employed contributions.
3. Contributions divided by total compensation.

*Source:* Social Security budgets.

Figure 19. **PENSION EXPENDITURE AND RECEIPTS**

As a percentage of GDP




1. Including total contributions for health care which amount to 2 per cent of GDP in 1994.

*Source:* Social Security budget.

by 1½ per cent of GDP. The former was achieved by a sharp increase in contribution rates which placed Greece among the countries with the highest non-wage labour costs in the OECD.[59] Regarding expenditure, the most important measure to break their fast upward trend was the de-linking of pensions from wages. Primary pensions in the private sector are now indexed to the increase in civil servant pensions, which have been used along with civil servant wages and the exchange rate policy to influence inflation expectations and thus increased in line with stringent government inflation objectives. As a result, during the 1990-95 period, average primary pensions (excluding farmers) on aggregate declined by over 20 per cent in real terms.[60]

The other measures to reduce pension expenditure varied across funds, but the main ones gradually reduced pension replacement rates and raised eligibility standards, especially in the public sector, and tightened procedures for granting invalidity pensions. (See Box 4 for a detailed description of the reforms.) The tightening of invalidity eligibility has proven relatively successful, and reduced the incidence of invalidity pensions to below 15 per cent of private sector retirees, from more than 20 per cent in the late 1980s. The reforms reduced many of the inequities in the system, especially those between private and public sector employees, but they created an important new differentiation, as all workers entering the work force from 1993 onwards are subject to less generous, though unified, rules. Entitlements have been moved in the right direction yet, according to OECD estimates, pensions remain too generous in actuarial terms – even for new workers – to bring the system back towards viability. The projections presented below indicate that the changes came too late and have been insufficient to restore the viability of the system.

The large loss in purchasing power arising from the de-indexation was not sustainable and 1996 legislation indexed minimum pensions to CPI inflation and provided for a means tested supplement to low income (non-agricultural) pensioners above the age of 65. This supplement is the first occurrence of the application of means testing in the Greek pension system. The supplement is a fixed amount equivalent to 12 per cent of the minimum IKA pension and will be henceforth indexed to inflation. Eligibility is based on uniform criteria, the most important of which is total pension receipts.[61] Some 380 000 pensioners (about 17 per cent of the total) have already received the supplement and its cost is estimated to be 0.2 per cent of GDP on an annual basis. Nevertheless, the potential for fraud will need to be faced through co-ordination with tax

## Box 4. **The 1990-92 reforms**

### Revenue measures

For private-sector wage earners, contribution rates were raised by 5.8 percentage points to 20 per cent. For civil servants, contributions were introduced and gradually raised to the level of the private sector by 1995. In the broader public sector, contributions were raised from a range of 14 to 25 per cent to 33 per cent by 1995 (for pre-1993 employees). For the self-employed, where contributions are based on a set of voluntarily chosen income classes (which had not been fully adjusted for inflation), the lower classes were eliminated and the earnings base for contributions was gradually raised towards the 1991 per capita GDP (adjusted subsequently by the increase in civil servant pensions). Finally, a progressive tax was placed on all pensions (except for farmers and seamen); amounting to 1 per cent for pensions exceeding Dr 100 000 per month, up to 5 per cent for pensions exceeding Dr 500 000.

### Expenditure measures

- Replacement rates: For IKA pensioners, the replacement rate will be gradually reduced by 14.2 per cent during the period 1992-2001 through the elimination of the double counting of the two months' bonus salary. Moreover, their pension base was increased from the last two to the last five years' salary (adjusted by the corresponding increase in civil servants pensions). For all pensioners, an overall limit was placed on the total value of all pensions per individual pensioner equivalent to four times the 1991 per capita GDP (adjusted by the corresponding increase in civil servant pensions). For all funds, but applying mostly to the broader public sector and professionals, the replacement rate of new retirees will be capped for primary pensions at 80 per cent and for supplementary pensions at 20 per cent (from 1998) of the applicable salary base. On the same date, separation payments will be capped at Dr 13.5 million.
- Indexation: Pensions in the private sector were no longer indexed to changes in the blue-collar minimum wage, but adjusted in line with civil servant pensions, which are in turn adjusted according to the government's inflation objective.
- Eligibility: For the private sector, pension eligibility conditions were raised from 13.5 years to 15 years' contributions for a minimum pension at age 65, and the minimum pension is increased by 1 per cent for each additional year of contributions above 15 years. In addition, the minimum retirement age (the ''35 year'' rule) will be gradually raised from 58 to 60 by 2002 (for men only).

For the public sector, a minimum pensionable age was introduced for those hired after end-1982 (60 for men and 58 for women), which will be gradually raised to those ages applying in the private sector (65 for men and 60 for women) during the period 1998-2007. Moreover, a minimum contribution period of 25 years was also introduced (which mainly affected women who previously could retire after 15 years' service and at age 42). In addition, the employment period in the public sector required to retire without an age limit will be gradually raised from 32 years to 35 years (by 1997 for women and between 1998 and 2004 for men), a back-loaded pension formula was introduced to

*(continued on next page)*

*(continued)*

provide a greater incentive to retire later, and transitional arrangements were set for public employees hired prior to 1982.

Eligibility for invalidity pensions was also tightened. The basic measures raised the minimum degree of invalidity required to receive a pension from 33 to 50 per cent and required verifications of past decisions.

**New system**

The reforms unified the pension rights and obligations for all public and private-sector employees entering the work force after end-1992, and made them less generous than those applying to the current generation of employees.

- Retirement age: At age 65 for men and women, and at age 55 for women with under-age children.
- Replacement rate: 80 per cent for a combined primary and supplementary pension (about 10 percentage points less than the current replacement rate for wage earners) and with a linear accumulation rule at a rate of 2.285 per year (compared with the progressive one applied by IKA currently). The assessment base is the average of the last five years' earnings (adjusted for inflation).
- Contributions: The new system introduces a tripartite system for contributions, with employer and employees' contributions remaining unchanged, but with government providing an additional contribution (10 percentage points or one-third of the total) Also, the cap on contributions for the post-1992 generation was eliminated.
- Dependency allowances: For those entering the workforce after 1993, the 1990-92 reforms eliminated the spouse dependency allowance and changed the child dependency allowance to 8 per cent, 10 per cent and 12 per cent, respectively for the first, second and third child, from 3 per cent for the spouse and 20 per cent, 15 per cent and 10 per cent, respectively for the first, second and third child.
- Multiple pensions: Contributions for a second pension are made on a voluntary basis by the employee for the full (employee and employer) amount, and eligibility for a survivor pension is subject to means testing unless the survivor is disabled.
- Right to work and simultaneously receive a pension: Pensions will be reduced by one-third if recipients continue to work. Recipients of minimum pensions will have their pension fully suspended.

authorities. If eligibility was broadened, the effectiveness of the measure as a targeted social policy will be reduced.

The 1990-92 reforms have succeeded in temporarily stabilising the finances of the pension system. Though a detailed decomposition of pension expenditure

and revenues for 1996 is not yet available, the shortfall between total (public and private) outlays and contributions is estimated to have been about 4½ per cent of GDP. It is, nevertheless, often believed that the pension system is in surplus following the implementation of the reforms, as the pension funds' cash balances are in surplus equivalent to about 2 per cent of GDP. This perception overlooks the transfers from the budget to deficit funds (2¼ per cent of GDP), that civil service and other government pension outlays – net of employee contributions – amounted to 2 per cent of GDP, and that funds received earmarked taxes of over 1 per cent of GDP. Moreover, a large part of the cash surplus arose from the increased return on fund assets following the liberalisation of interest rates in 1989.

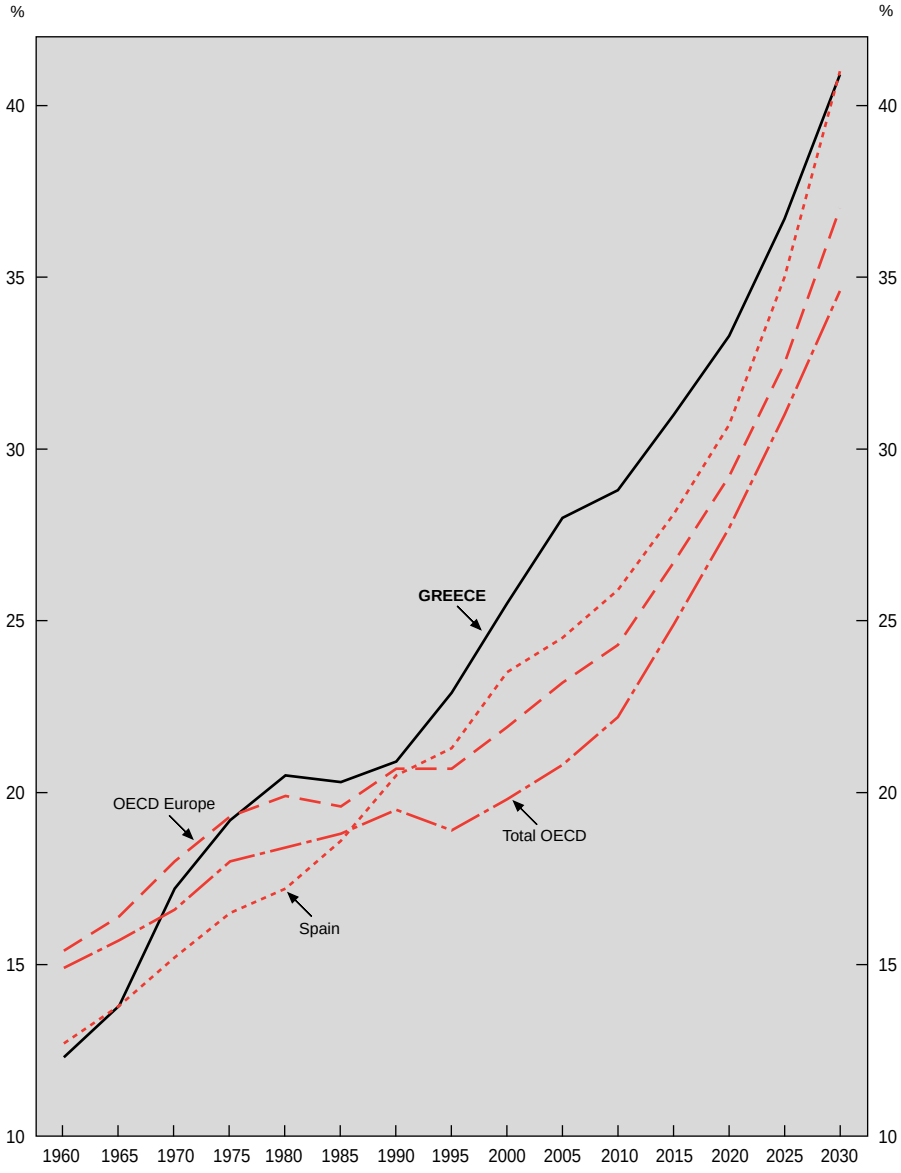
### *The future of the pension system*

Despite the further positive impact expected from the 1990-92 reforms, pension expenditures are set to increase substantially relative to contributions if no further reform measures are taken. Similar to projections for the other continental European countries and Japan, expenditure will be propelled mainly by the ageing of the population. In the case of Greece, demographic developments are even less favourable than those elsewhere due to the current relative longevity of the population (Figure 20). The elderly dependency ratio is expected to increase from about 25 per cent currently, to 40 per cent in 2030 and to a peak of 61 per cent in 2050. This development is largely pre-determined by past birth rates and could only be altered by further large immigration flows similar to the ones that occurred at the turn of this decade. However, several factors specific to Greece will add additional strain on the system. The system is still in the process of maturing, with replacement rates expected to increase as future pensioners become eligible for the full benefits of the system, especially those provided by supplementary funds and agricultural workers for their new primary pension fund. Further urbanisation will also put upward pressure, with a larger share of the labour force receiving more generous pensions than their farmer parents.

The OECD's pension simulation model, which attempts to capture these effects through to the year 2070, projects a rise of the expenditure to GDP ratio of 1.2 per cent of GDP every five years with expenditure more than doubling when it would reach 23 per cent of GDP by 2050 (Figure 21). (Annex I provides a detailed description of the projections.) With contributions projected to increase

Figure 20. **ELDERLY DEPENDENCY RATIOS**[1]

Per cent

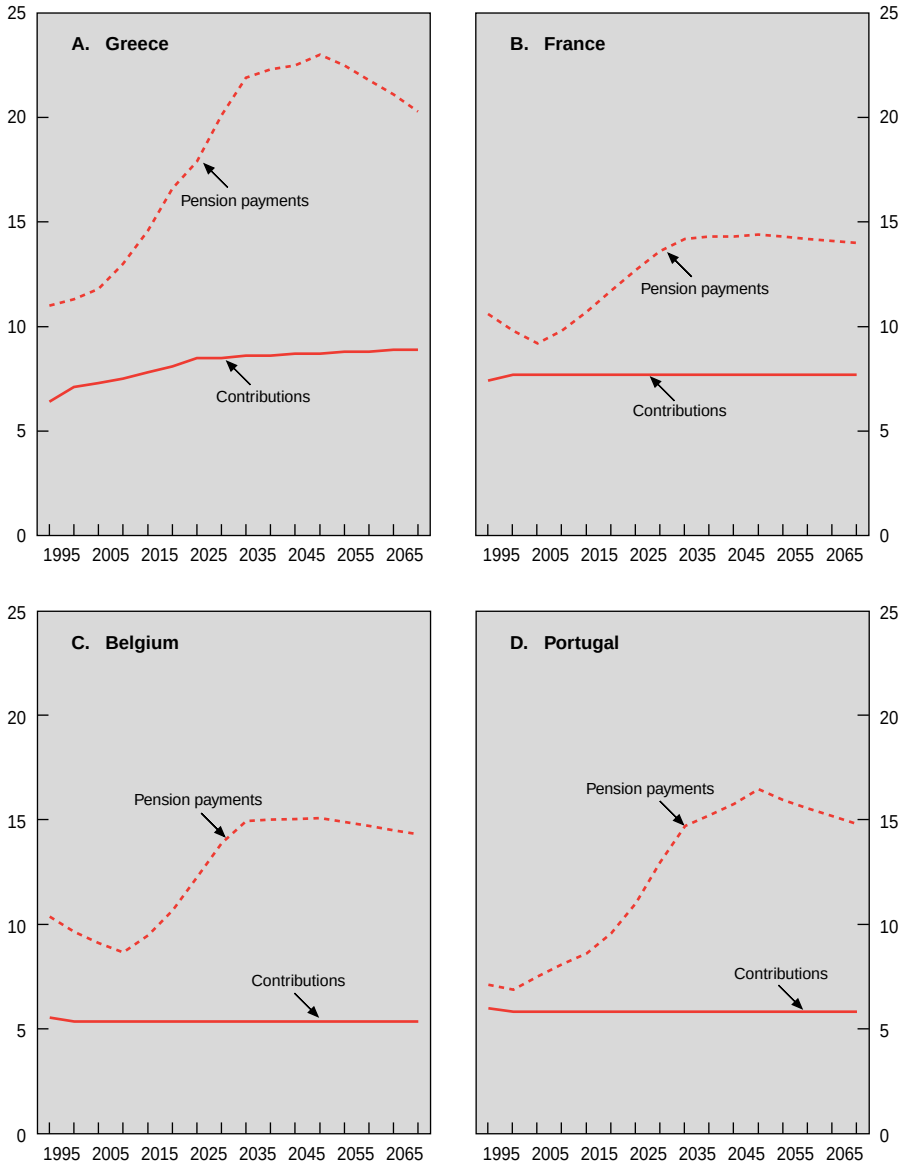



1. Population aged 65 and over as a per cent of working age population.

*Source:* World Bank.

Figure 21. **PENSION PAYMENTS AND CONTRIBUTIONS SIMULATION**

As a percentage of GDP




*Source:* OECD.

from 7 per cent of GDP to 9 per cent of GDP – mainly reflecting the assumption of a larger share of employment moving out of the lower income agricultural sector and longer contribution periods for private sector employees – the gap between total pension outlays and the current rate of contributions (excluding all government sources of funding) is projected to widen progressively from 4½ per cent of GDP in 1997 to about 14 per cent of GDP in 2050.[62] While Belgium, France and Portugal face similar dynamics, the projected situation in Greece is clearly worse.

Another indicator of the magnitude of the future gap on current contribution rates is the present value of net pension outlays as a percentage of GDP, which is projected to be about 196 per cent of 1994 GDP. This ratio exceeds by a wide margin similar estimates for all other OECD countries. OECD simulations for other countries found present values of net pension liabilities of 102 per cent of GDP for France, 153 per cent of GDP for Belgium and 109 per cent of GDP for Portugal.[63] The OECD methodology excludes all revenues except employer and employee contributions. Thus, the attainment of estimates for Greece consistent with other OECD countries requires excluding from the pension system's revenues: earmarked taxes, transfers from the budget, and government contributions. Inclusion of government contributions would raise revenues substantially, albeit by adding general taxation receipts to contribution revenue. The participation of the Government as employer would add revenues equivalent to 14 per cent of 1994 GDP, if it is assumed that the employer's share is the same as for the private sector. The Government's one-third share of the contributions for post-1992 entrants into the work force would raise revenue by the equivalent of 53 per cent of 1994 GDP, while the government's two-thirds share of farmer's contributions would add the equivalent of 18 per cent of 1994 GDP. In total, the present value of net pension outlays would thus be reduced to about 110 per cent of 1994 GDP. The projection is particularly sensitive to the assumptions for the real GDP growth rate and the discount rate.[64] Primary pensions account for 83 per cent and supplementary pensions for 17 per cent respectively of the net future liabilities, while the net future liabilities of separation pensions are equivalent to an additional 1 per cent of 1994 GDP. A ranking with respect to generosity as measured by the ratio of the present value of net outlays to contributions, places the farmers' fund first, the seamans' fund second, the general government

employees third (after adding an estimate for the government's participation as employer), and employees of public enterprises and banks (Table 14).[65]

The projections indicate that the introduction of additional pension reforms cannot be delayed. The gap between expenditures and contributions will have reached about 5½ per cent of GDP by the year 2010, primarily due to the transformation of the surplus currently achieved on supplementary pensions into a deficit exceeding 1 per cent of GDP. The gross financing requirement will be about ½ per cent of GDP greater as reserves are run down, and the concomitant revenue lost. Thus, the fiscal burden will increase by about 1½ per cent of GDP. Moreover, pension reforms should be started quickly as their implementation requires long lead times, and early implementation would provide large benefits by avoiding the snowball effect of a ballooning public debt.

With many OECD countries facing similar, though perhaps less pressing situations, two strategies for reform have emerged. Most countries have pursued reforms of their pay-as-you-go system, including France (1993), Spain (1996) and Italy (1992 and 1995), or have introduced multi-tiered pay-as-you-go systems, with a flat public pension and an earnings-based component, such as in the United Kingdom and Sweden. A few others have introduced more radical reforms, whereby all or part of the system was transformed into some form of defined contribution system, which is either fully-funded (Australia, Chile and Singapore) or multi-tiered (pursued in Central European countries).

Concerning reforms to pay-as-you-go systems, most countries have hesitated to raise contribution rates further, because of the adverse affect of higher payroll taxes on employment creation. A similar view is held in Greece as the 1990-92 pension reform raised contribution rates to the point where statutory payroll taxes are among the highest in the OECD for private sector employees. To ensure equilibrium, the effective contribution rate for the system as a whole would have to increase gradually from about 14 percentage points in 2000 to the obviously untenable 50 percentage points in 2050. If the adjustment was made up front it would not require such a large alignment, but it would nevertheless still be too large to represent a viable option. For example, a one-time and sustained increase in the average effective contribution rate to 1½ times the current rate in 2010 would reduce the present value of net pension liabilities by about one-third, to the equivalent of 137 per cent of 1994 GDP.

Table 14. **Implicit gross liabilities of the public pension system**[1]

Present value of expenditure minus contributions

| | Baseline scenario | | Alternative scenarios (as a percentage of 1994 GDP) | | |
|---|---|---|---|---|---|
| | As a percentage of 1994 GDP | As a percentage of total revenue | Increase in effective contribution rate by 50 per cent from 2010 | Decrease in effective replacement rate by 50 per cent from 2010 | Increase in effective retirement age by 5 years[3] from 2000 |
| IKA and other | 61 | 52 | 27 | 11 | 30 |
| NAT | 12 | 424 | 11 | 9 | 10 |
| Self-employed | | | | | |
| Small entrepreneurs and traders | 13 | 42 | 4 | 0 | 6 |
| Professionals[4] | 1 | 16 | –1 | –1 | 1 |
| OGA | 45 | 469 | 42 | 36 | 45 |
| Civil servants[2] | 53 | 296 | 48 | 37 | 43 |
| Public enterprises and banks | 11 | 62 | 6 | 5 | 5 |
| Total | 196 | 96 | 137 | 97 | 140 |

1. The OECD estimates exclude earmarked taxes, government transfers, tripartite financing and all other non-contribution based revenue. The model's assumptions are described in Annex 1.
2. Excluding the participation of Government as an employer. Including a government participation equivalent to twice the employees' contribution would reduce the present value of net liabilities to 42 per cent of 1994 GDP.
3. Except for the case of self-employed professionals and farmers, where the baseline scenario assumes retirement ages of 65.
4. The results for self-employed professionals are biased by the inability in many cases to separate out earmarked taxes from earnings based contributions.

*Source:* Social Security Budget.

On the other hand, the effective contribution rate for most funds is low as compared to the statutory one, reflecting the incentives for contribution evasion, including through the incentives provided by the high minimum pension in conjunction with the low eligibility requirement. For the system as a whole, the effective contribution rate for a primary pension is about half the average statutory one of about 20 per cent. The effective rate could be brought towards the statutory rate without raising contribution rates by stemming contribution evasion. Most importantly, the gradual replacement of the minimum pension by a means-tested minimum income for pensioners would eliminate a principal incentive for contribution evasion.

The elimination of this disincentive to contribute should help alleviate a fundamental problem of the system, namely the easy eligibility for a pension. Raising the years worked (and contributed) toward a pension is an extremely effective measure since it provides the double benefit of reducing the number of pensioners, while at the same time increasing the number of contributors. Other measures to raise the effective retirement age from its current level between 55 and 60 years would be to further tighten controls to arrest abuses of the early retirement opportunities, especially work in ‘‘arduous and unhealthy activity’’, and access to invalidity pension (though admittedly much progress has been made in controlling easy access to invalidity pensions). Extending the contribution period for five additional years prior to the receipt of a pension for all funds except those of farmers and self-employed professionals (effective from the year 2000) would reduce contingent liability by about 30 per cent. Such a delay in the retirement age could be achieved be raising the effective retirement age to the current statutory retirement age of 65 years. With life expectancy high and increasing, many countries are considering raising this limit to over 65 years.

Nevertheless, a portion of the gains from raising the effective retirement age would be offset by higher pensions once they are paid, arising from the generosity of the existing pension formulas. Thus, it is inevitable that reforms should strengthen the link between the value of individual pension payments and the supporting contributions by setting the pension base closer to earnings over a whole career. In Greece, the pension base for the private sector of the last 5 years of earnings is the lowest in the OECD, while the base used in the public sector of the last month’s salary (or the minimum wage at the time of retirement for the largest supplementary pension fund) is even more generous. Greece should

follow the example of other countries and use wages over a longer, if not the entire, working life to calculate pensions (as is already the case for the self-employed small entrepreneurs and traders who make up a large section of the population). Widening the pension base in this manner also reduces inequities between blue and white collar workers arising from the latter's more dynamic salaries, who as a result, benefit more from a back-loaded earnings base. Raising the contribution base to comprise a large share of lifetime income should also discourage contribution evasion. In order to achieve a gradual lengthening of the pension base, Greece should follow the example of other countries and soon start lengthening the pension base annually by increments.

In view of the longer life expectancy, as well as budgetary concerns, many countries have also reduced the rate at which pension benefits accrue in order to provide incentives for workers to delay retirements, and extended the period required to accumulate benefit rights for a full pension beyond 35 years (*e.g.* for France the limit is 38 years and for Germany 40 years). The accumulation rates in most OECD countries range from 0.5 per cent to 1.5 per cent per year while in Greece, the accrual rate remains very generous in almost all funds. Including supplementary funds, it ranges from a low of 2.3 per cent per year for workers entering the work force after end-1992 to over 3 per cent per year for public servants, while a full pension is usually obtained after 35 years' contributions. These benefit accumulation rates imply very high rates of return on contributions and the former would need to be reduced to about two-thirds as well as be based over a longer assessment period, according to OECD estimates, to eventually bring the system towards viability. A one-time reduction in the effective replacement rate in the year 2010 by 50 per cent would reduce the present value of the net pension liabilities by about one-half. Alternatively, to maintain the balance for the system would require a gradual reduction in the aggregate effective replacement rate from 53 per cent in 2000 to below 20 per cent in 2050. Similar to the strategy for lengthening of the pension assessment period, the accumulation rate should be gradually reduced for all funds. In addition, the accumulation rate could be made back-loaded, with higher rates of benefit accumulation provided towards the latter part of a career.

The indexation of pensions to price objectives of the government is one of the most potent means of controlling pension expenditure and provides one of the few ways of placing a share of the adjustment burden on current pensioners.

Despite the recent large loss in the purchasing power of pensions, this policy, in conjunction with the indexation of minimum pensions and the continued strict application of a means-tested supplement to protect the elderly poor, will probably need to be continued until other, more targeted, reforms to the pension system are introduced. However, for equity reasons the indexation of pensions should be identical across sectors and types of pensions.

In addition to changing the basic parameters of the current system, unification of the regulations would enhance equity and provide considerable economies of scale. The decision to unify the regulations was already accepted for the generation of workers who entered the labour force after end-1992. The concept of a unified system across sexes, sectors and levels of pensions (*e.g.* the supplementary pensions) could help make the need to reduce the generosity of pensions become more palatable. It would eliminate imbalances across sectors and generations, with the most generous benefits adjusting proportionally more. Similarly, the choice to subsidise pensions of women, and especially those with under-age children, and the pensions of the declining agricultural sector should at least be undertaken in a more equitable manner, emphasising means-testing. Such a policy would admittedly require an improved administrative capacity. Survivors pensions should also be means-tested, as will be the case for those that entered the labour force after end-1992. A unified fund also would lower costs both through administrative economies of scale and through improved monitoring to reduce abuses such as exceeding the global limit of the system on total pension receipts.

As a first step towards the unification of pension benefits, savings could be made by unifying primary and supplementary funds and their benefits within individual sectors. Moreover, since practically no fund is viable over the longer term, the proper functioning of a pay-as-you-go system suggests the pooling of all surpluses to cover the needs of deficitary funds. Raising the transparency regarding the financial position of the system in this manner may help reduce the abuses of the system. For similar reasons, earmarked taxes obtained over the years by powerful lobbies to finance their pensions should be transferred to the budget. Irrespective of decisions regarding the composition of the system, the speedy establishment of the register for contributors and pensioners would provide a strong weapon in combating abuses of the system (*e.g.* working during retirement while exceeding prescribed norms on earning).

The debate over the pros and cons of a funded versus pay-as-you-go system has often obscured the key issues of pension reform for countries, such as Greece, which have pension systems with large implicit liabilities in present value terms. Specifically, a successful reform needs to address the issue of the pensions expected by the current work force, but which will far exceed their contributions on present rules. These ''acquired rights'' count for about two-thirds of the total implicit debt, indicating that reforms must change these expectations whatever decision is made on systemic reform. In this regard, a switch from a pay-as-you-go system to a funded system could form part of a strategy to meet at least part of the pension expectations of current workers, at a reduced cost to the budget. In the case of Hungary, the generosity of the system was reduced for existing and future workers. At the same time, workers were allowed a choice between a reformed (*i.e.* less generous) pay-as-you-go system and a new multi-tiered system with the first tier proportionally identical to the reformed pay-as-you-go system and a funded second tier. It is expected that younger workers would be willing to voluntarily switch to the multi-tier system since they will earn a higher rate of return on their invested contributions, which would make up for the difference between an even less generous accumulation rate for past contributions (in the second tier) relative to the reformed pay-as-you-go system. The difference in the accumulation rates for past contributions between the multi-tier and reformed pay-as-you-go system (for the second tier) are the savings to the budget from this systemic change. However, the transfer of even a share of existing workers' contributions towards a contribution-based system will lead to a loss of revenue for the pay-as-you-go component of the system and thus result in a higher budget deficit over the medium-term. For Greece, taking this route would imply a strong commitment to significant additional fiscal consolidation as the debt burden is already very high.

## The public health care system

### *Introduction*

Reforms of the public health care system since the mid-1970s have focused on extending the coverage of health care and on improving its quality. Following the 1983 reform, which created the National Health Service, the entire population has access to some form of health coverage. Apart from the National Health

Service, which provides largely free services in hospitals and rural health centres, care is provided by Health Funds, which are occupation based, and for which contributions have to be paid. The health funds usually constitute part of parent pension funds. Finally, private health services also exist which are, in general, not reimbursed.

Despite the increase in the supply and quality of health care services, their provision is generally considered to have remained unsatisfactory.[66] An important source of the system's shortcomings concerns the quality of primary care. Primary care suffers from the virtual absence of family doctors, which makes it difficult to ensure prevention and continuity in care. Without an efficient referral system, patients have to search themselves for appropriate specialists. As a result, ambulatory care is often delivered by outpatient departments of hospitals and clinics, with patients facing long waits and other inconveniences, while scarce hospital resources are not used appropriately.[67] For instance, the distribution of health personnel by qualification is inefficient, with a serious shortage of well-trained nurses, an over-supply of specialised doctors and a lack of generalists.

The diminished yet still significant inequality in the provision and coverage of services is another important shortcoming of the health care system. Even though the number of doctors per inhabitant is higher than in the OECD on average (Table 15), service provision is very uneven. The majority of doctors are concentrated in the Greater Athens area with 5.7 doctors per 1 000 inhabitants, compared with only 1.6 in north-central Greece.[68] In addition, rural health clinics are understaffed. Partly due to the concentration of medical staff in large urban areas, a wide dispersion can be found in the regional distribution of per capita public expenditure on health.[69] Inequalities also arise from the large variation in the range of services provided and/or reimbursed by health funds which are only partly reflected in differences in contribution rates. Civil servants and banks' employees benefit from the most generous schemes (including broad access to private health care), while the self-employed have a more limited choice and are not covered for dental care. Finally, it is generally acknowledged that lack of efficiency and long waiting lists have resulted in unofficial payments as a frequent recourse to obtain faster access to – and better quality of – health care, seriously harming the official objective of equal access to health services.

The expansion of health services has allowed a considerable improvement in the health status of the population, with infant mortality, for instance, falling

Table 15. **Characteristics of the health care system**

| | Consumption of drug packages per head | Pharmacists per 10 000 inhabitants | Doctors per 10 000 inhabitants | Proportion of specialists | Beds per 1 000 inhabitants | Average bed occupancy (%) | Nurses per available hospital bed |
|---|---|---|---|---|---|---|---|
| United States | 6.5 | 6.2 | 25.3 | 50.9 | 4.1 | 66.0 | 1.6 |
| Germany | 13.5 | 5.5 | 33.6 | 59.8 | 9.7 | 83.3 | 0.7 |
| France | 52.0 | 9.7 | 29.4 | 49.4 | 8.9 | 81.2 | 0.4 |
| Italy | 21.1 | 9.6 | 16.5 | n.a. | 6.4 | 72.7 | 0.6 |
| United Kingdom | 9.3 | 5.9 | 15.6 | n.a. | 4.7 | 80.6 | 1.7 |
| Belgium | 10.4 | 13.5 | 37.4 | 40.5 | 7.6 | 83.5 | 0.8 |
| **Greece** | **26.5** | **7.8** | **38.8** | **55.7** | **5.0** | **70.0** | **0.7** |
| Ireland | 12.5 | 3.2 | 17.2 | 17.4 | 5.0 | 77.0 | 1.3 |
| Netherlands | 8.0 | 1.6 | 25.1 | 33.3 | 11.3 | 88.6 | 0.9 |
| Portugal | 19.8 | 12.8 | 29.9 | 66.1 | 4.1 | 71.0 | 0.7 |
| Spain | 14.8 | 10.1 | 40.8 | n.a. | 4.0 | 76.7 | 0.8 |
| Turkey | n.a. | 3.0 | 11.5 | 42.1 | 2.5 | 57.4 | 0.2 |
| EU average | 16.1 | 7.4 | 27.2 | 45.6 | 7.3 | 80.9 | 0.7 |
| OECD average | 16.0 | 6.8 | 24.3 | 49.2 | 7.3 | 77.3 | 0.7 |

*Source:* OECD Health Data.

sharply since the 1960s (Table 16). In addition, life expectancy has always been relatively high, which could partly be due to cultural factors (mainly the Greek diet). On the other hand, the health statistics also provide evidence that further improvements can be achieved and that the system is more curative than preventive. Despite the sharp fall, infant mortality is still among the highest in the EU,[70] the immunisation rate of children is low, and the number of avoidable deaths is considerable by OECD standards. Moreover, drug consumption per capita is high, partly reflecting doctors' use of prescriptions as a substitute for the provision of more complete health care.

Previous reform programmes (Box 5) have focused on increasing the provision of health care and have largely neglected a reform of financing. As a result, the payment system for public health care providers offers poor incentives, precludes cost control and hampers the improvement in the quality of services

Table 16. **Quality of health indicators**

| | | Life expectancy at birth[1] | | Life expectancy at age 60[1] | | Mortality | |
|---|---|---|---|---|---|---|---|
| | | Female | Male | Female | Male | Perinatal[2] | Infant[3] |
| **Greece** | **1960** | **70.7** | **67.5** | **18.9** | **17.0** | **2.6** | **4.0** |
| | **1995** | **80.3** | **75.1** | **22.8**[4] | **19.9**[4] | **1.0** | **0.8** |
| Ireland | 1960 | 71.8 | 68.5 | 18.3 | 16.3 | 3.8 | 2.9 |
| | 1995 | 78.5 | 72.9 | 21.1[5] | 17.1[5] | 0.9 | 0.6 |
| Portugal | 1960 | 67.2 | 61.7 | 18.6 | 15.9 | 4.1 | 7.8 |
| | 1995 | 78.6 | 71.5 | 22.0 | 18.0 | 0.9 | 0.7 |
| Spain | 1960 | 72.2 | 67.4 | 19.2 | 16.5 | 3.7 | 4.4 |
| | 1995 | 81.2 | 73.2 | 24.1 | 19.5 | 0.7[4] | 0.6 |
| Turkey | 1960 | 49.7 | 46.5 | 15.9 | 14.7 | n.a. | 19.7 |
| | 1995 | 70.0[4] | 65.4[4] | 18.1[6] | 15.8[6] | 5.3[5] | 4.5 |
| OECD average[7] | 1960 | 70.8 | 66.0 | 18.8 | 16.1 | 2.9 | 3.8 |
| | 1995 | 79.5 | 73.1 | 23.0 | 18.6 | 0.7 | 0.8 |

1. In years.
2. In percentage of live and stillbirths.
3. In percentage of live births.
4. 1994.
5. 1993.
6. 1990.
7. Non-weighted arithmetic average.

*Source:* OECD Health Data.

## Box 5. **The Greek health care system and reforms in historical perspective**

The first major step in building the Greek health care system was the establishment of IKA in 1934 which was to provide health and pension coverage to blue and white-collar workers in urban areas, and to workers employed in firms with more than 70 persons, *i.e.* about 30 per cent of the population. Thereafter, health insurance coverage was extended through the creation of numerous health funds whose membership was compulsory for the working population and based on occupation. In 1997, the health care insurance system encompasses 37 health funds with nearly universal coverage since health insurance is extended to the insured's family.

In conjunction with the extension of health insurance coverage, several reform programmes were implemented over the past three decades. Their main objective was to provide universal and equal access to health care services. The major reform programmes (1968, 1976, 1983 and 1992), sought to address persistent deficiencies of the system:

- inequalities in the provision of health care services (geographical and by health funds);
- the lack of a referral system and too little reliance on generalists;
- an inappropriate method of remunerating doctors and hospitals for services leading to inefficient practices by doctors, hospital personnel and the insured; and
- a large number of health funds and a lack of co-ordination between them.

Strong political opposition to these reforms (coming from doctors' unions and social insurance funds) impeded or mitigated some of the proposed changes. However, though only partially implemented, the 1983 and 1992 reforms introduced important changes in the health care system. The main characteristic of the 1983 reform was the prominent role given to the State in providing health care and substantially increased public health care expenditure. With the creation of the National Health Service (NHS) in 1983, primary and secondary care were to be provided mainly by public hospitals, health centres and publicly paid generalists. Four hundred health centres were to be built, but only those for rural areas (*i.e.* 170 health centres) were eventually established. Opening new private hospitals was forbidden. For those already functioning, the low level of fee-for-services (set by the Ministry of Health) forced most of them to close down or to be absorbed by the public sector. Doctors employed by the public sector were not allowed to have concurrently a private practice. To compensate for the resulting loss of income, their salaries were almost doubled. However, other provisions of the 1983 reform failed to be implemented. The development of a referral system, included in the 1983 legislation, has never been achieved. The proposed unification of the main insurance funds was not put into effect and existing health funds continue to operate under distinct regulations. Thus, large inequalities remained in the provision of insurance benefits and health care services.

Following the 1992 reform, market forces play a somewhat greater role, and patients' freedom of choice and private initiative were also emphasised. The prohibition of the opening of private hospitals was removed and doctors engaged in the NHS were given the choice to work on a part-time basis and concurrently in the private sector. This

*(continued on next page)*

*(continued)*

choice was rarely taken and the option was eliminated in 1994. The Government also raised the contribution rate (to 7.65 per cent of total salary) for the main health fund (IKA) and allowed for a gradual harmonisation in contribution rates across health funds. It also introduced a co-payment rate for patients visiting outpatient departments of hospitals (except for emergency care), raised significantly the *per diem* prices paid by the social security funds to hospitals and set the co-payment rate for pharmaceuticals at 25 per cent.

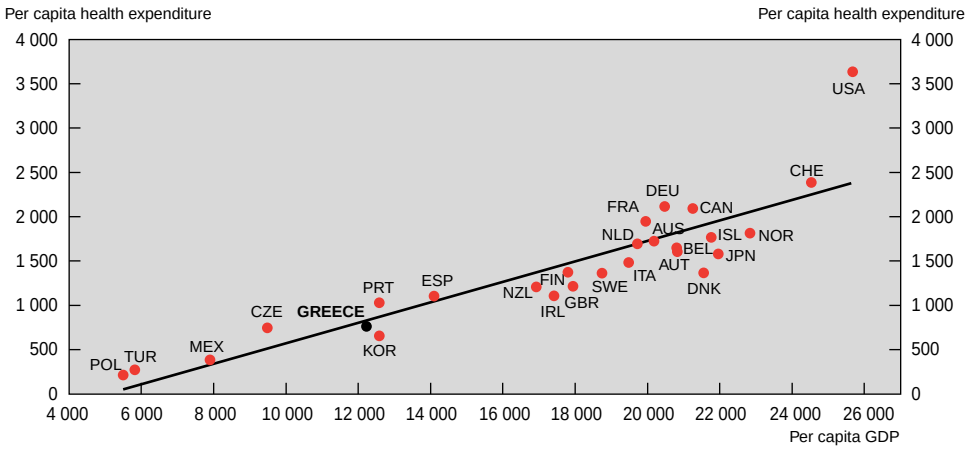
provided. The rising trend in health expenditure, which will be aggravated by the rapid ageing of the population, and the current objective of fiscal consolidation make reforms to the health care system and its financing a priority.

### *Financing health expenditure*

Health spending has risen steadily over the past three decades but with a clear acceleration since the early 1980s. As a result, public expenditure (health funds and subsidies to hospitals) is estimated to have reached about 5 per cent of GDP in 1996. Private outlays are likely to be significant, though estimates should be viewed with caution as they attempt to capture unofficial payments.[71] Accepting an estimate of 6½ per cent of GDP for total (public and private) expenditure, spending does not deviate significantly from that of other OECD countries, taking into account per capita income differences (Figure 22).[72] However, the composition differs. Reflecting the predominance of hospital care and pharmaceutical consumption, outlays on these two categories account for a larger share of public health expenditure than in other OECD countries.

Though the public health system was initially designed so that services would be mostly paid by social security funds, with only a small co-payment by patients, a significant share of hospital outlays are now funded directly from the state budget (Table 17). The change in the source of hospital financing arises from the compression of the price of hospital services, which are much lower than cost for most services. As a result, payments by social security funds covered less than 10 per cent of public hospital expenditure in 1990 from about

Figure 22. **HEALTH EXPENDITURE AND GDP PER CAPITA**[1]




The equation of the regression line is the following:
Health expenditure per capita = -582.68 + 0.12*GDP per capita
$R^2$ = 0.78 T: (-2.65) (9.58)
1. In 1995. Total expenditure on health care and GDP in purchasing power parity exchange rates.
*Source:* OECD Health Data.

30 per cent in 1981. The counterpart to this price control was artificially low health expenditure by social security funds and rising surpluses on health-related transactions, which were often used to cross-subsidise pension outlays. In the early 1990s, the Government raised hospital *per diem* prices by a cumulative amount of 600 per cent. Subsequently, payments from social security funds covered about a third of public hospital costs. As a result of these policies, total state budget transfers to hospitals increased from 0.6 per cent of GDP in 1981 to 2.5 per cent in 1990 before declining to an estimated 1.5 per cent in 1996. Also transfers to the health funds have risen, as the rise in contribution rates in 1992 was not sufficient to cover the higher payments to hospitals. Despite their increase in 1992, contribution rates remain low in international comparison (Table 18).

The current manner of financing the health system, which limits severely the role of ‘‘third party payers’’ (the health funds), and the fact that prices of services (fixed by government) are held at unrealistic levels, are major reasons for the imbalances described above. There exist other price distortions, with possible

Table 17. **Public health expenditure and its financing: key components**

| | 1981 | 1986 | 1990 | 1996 |
|---|---|---|---|---|
| | (As a percentage of GDP) | | | |
| Total public health expenditure[1] (1) + (2) + (3) + (4) | 2.7 | 3.5 | 4.7 | 4.9 |
| Social security funds' expenditure (1) | 1.8 | 1.6 | 1.7 | 2.7 |
| (Social security funds' revenues) | 1.7 | 2.0 | 2.4 | 2.2 |
| State payments for civil servant's health care[2] (2) | 0.3 | 0.4 | 0.4 | 0.6 |
| Public hospitals' expenditure | 1.4 | 2.2 | 3.1 | 2.7 |
| Private and social security funds' payments to public hospitals | 0.5 | 0.3 | 0.2 | 0.8 |
| General government transfers to public hospitals (3) | 0.6 | 1.4 | 2.5 | 1.5 |
| Investment expenditure (4) | – | 0.1 | 0.1 | 0.2 |
| | Expenditure by main category (as a percentage of total public health expenditure)[3] | | | |
| Hospitals | 50 | 61 | 67 | 55 |
| *Of which:* personnel costs | 26 | 33 | 38 | 28 |
| Social security funds | 66 | 45 | 35 | 55 |
| *Of which:* | | | | |
| Ambulatory care | 9 | 8 | 7 | 7 |
| Drugs | 12 | 12 | 11 | 18 |
| Hospitals | 28 | 13 | 5 | 18 |
| Cash benefits | 12 | 8 | 6 | 6 |
| Other funds | 4 | 3 | 2 | 2 |
| State payments for civil servants' health care | 8 | 7 | 6 | 3 |
| Investment | – | 3 | 2 | 3 |

1. Excludes health expenditure for military personnel.
2. Including funds of ministries not under the control of the Ministry of Labour and Social Affairs.
3. The sum is greater than 100 since it includes inter-institutional payments and private payments.

*Source:* Social Security budget, various years.

repercussions on efficiency. Pharmaceutical price increases for existing products have been constrained to rise less than inflation. However, pharmaceutical costs have continued to increase rapidly as existing drugs have been replaced with more expensive ''copies'' at higher prices. Since the pricing formula for new products favours imports, the pricing policy has stifled local production and

Table 18. **Contribution rates for health and maternity benefits in the European Union**

In July 1996

| | Employer's contribution rate | Employee's contribution rate | Ceiling (ECU) |
|---|---|---|---|
| Austria | | | |
| Employees | 3.4 | 3.4 | 2 892/month |
| Belgium | 6.15 | 4.7 | – |
| Germany | | | |
| Old Länder | 6.7 | 6.7 | 37 579/year |
| New Länder | 6.65 | 6.65 | 31 942/year |
| Greece (IKA)[1] | | | |
| People engaged up to December 1992 | 5.1 | 2.55 | 1 620/month |
| People engaged after December 1992[2] | 5.1 | 2.25 | – |
| France | 12.8 | 6.8 | – |
| Ireland | 7.75 | 8.5 to 12.0 | Employee: 26 284/year Employer: 31 540/year |
| Italy | | | |
| Employees in industry | 12.13 | 1.0 | – |

1. For benefits in kind only (excluding cash benefits).
2. The State also contributes 3.8 per cent on wages up to the ceiling of ECU 946 per month.

*Source:* EU (1996), Mutual Information System on Social Protection in the European Union.

resulted in a surge of imports. Finally, the differential between doctors' pay in the public and private sector, as well as the form of payment for public doctors (a salary unrelated to the quality or quantity of service) could be inducing some public doctors to transgress from their official duties in order to supplement their income.

## *Primary and ambulatory care*

According to the 1983 law, primary health care was to be provided by health centres in both rural and urban areas. In rural areas, 170 health centres have indeed been established. However, due to the low level of salary and difficult living conditions in rural areas, positions have been difficult to fill.[73] In 1993, less than half of the positions were covered and only about 25 per cent of doctors working in these centres were generalists. In addition to rural health centres, the

rural population has access to public clinics which are staffed by doctors doing one-year compulsory service after graduating from medical school.

In urban areas, health centres were never established. A plethora of services co-exist, with payment procedures differing among health funds. Members of the largest health fund, IKA, may resort to polyclinics owned by this fund. They may also visit salaried doctors under contract with IKA since its own facilities are not sufficient to meet all the demands of its members. TEBE, the main scheme for the self-employed, provides access to private doctors paid on a capitation basis. Other funds, mostly covering public sector employees, have contracts with doctors, paid on a fee-for-service basis.[74]

The method and level of remuneration of doctors for ambulatory care have created poor incentives which have undermined the quality of services. Most doctors are salaried and visits are normally free for fund members. Many of these doctors work on a part-time basis and reportedly use their part-time position to recruit patients for their private business. Doctors under contract with TEBE work on a capitation basis, but the fees are low and patients often have difficulty contacting a doctor. Regarding civil servants, the low fee-for-service has resulted in significant round-tripping with doctors' expenditure per head triple that of the private sector. As a result, while visiting a doctor in the public system or under contract with a social security fund is free in principle for virtually every Greek citizen, patients reportedly often have to make unofficial payments to facilitate an adequate provision of services. The poor quality of ambulatory care has resulted in a low level of preventive care and has led to the use of out-patient services of hospitals. In addition, doctors resort frequently to over-prescription of pharmaceuticals.

### *Public hospitals*

Public hospitals' spending has been propelled by several factors related to inefficient management and a poor incentive structure. While they do not face a hard budget constraint *ex post*, budgeting suffers from a lack of flexibility in the allocation of resources. Hospital management is constrained by line budgeting which limits budgetary flexibility. It should be noted that the creation of general manager positions in public hospitals, called for in the 1983 reform programme, was never implemented and hospitals are run by medical staff. The waste of resources is illustrated by the low average occupancy rate of beds in Greek

hospitals compared with long waiting lists for certain services. Beds are allocated by speciality or to individual specialists and some wards are full while others have empty beds. However, hospitals have little room for manoeuvre as the Ministry of Health determines the number of employees, the skill-mix for each public hospital and salary levels. Finally, hospitals do not use their market power for purchasing supplies and procurement procedures are considered poor, these shortcomings being magnified by collusion among suppliers. Corruption is reportedly also prevalent in this area.

The incentives for the effective management of hospitals are also skewed by job tenure and the absence of a link between remuneration and performance. Doctors in the NHS have civil servant status and are officially prevented from having a private practice.[75] This prohibition, combined with the ban on establishing new private hospitals between 1983 and 1992, has led to a surge in the creation of private diagnostic centres where doctors reportedly have indirect financial interests and refer their patients. On the other hand, many diagnostic centres play a legitimate role by helping bypass the long waiting lists in public hospitals for diagnostic tests. Nevertheless, the result is that patients often have to undergo private, expensive and sometimes unnecessary diagnostic tests which are paid for publicly. Indicative is the fact that there are 12.5 CT-scanners and 21.5 ultrasound scanners per one million inhabitants, while the corresponding ratios for the European Union as a whole are 5 and 13.5, respectively.[76]

Public hospitals, which are open to anybody, also have to make up for the deficiencies of the primary care system. People unsatisfied by the facilities provided by their health funds resort to out-patient departments of public hospitals. Visits to out-patient departments have more than doubled between 1981 and 1991. To curb the use of expensive hospital infrastructure, the 1992 law introduced co-payments. Patients are charged Dr 5 000 (US$20) for in-patient care and Dr 1 000 (US$4) for out-patient care. The impact on demand has been minor due either to the low rate or, as international experience suggests, due to the very low price elasticity of demand. Finally, the lack of a referral system, an underdeveloped preventive care system, as well as long waiting lists induce people to resort to emergency department care at public hospitals.

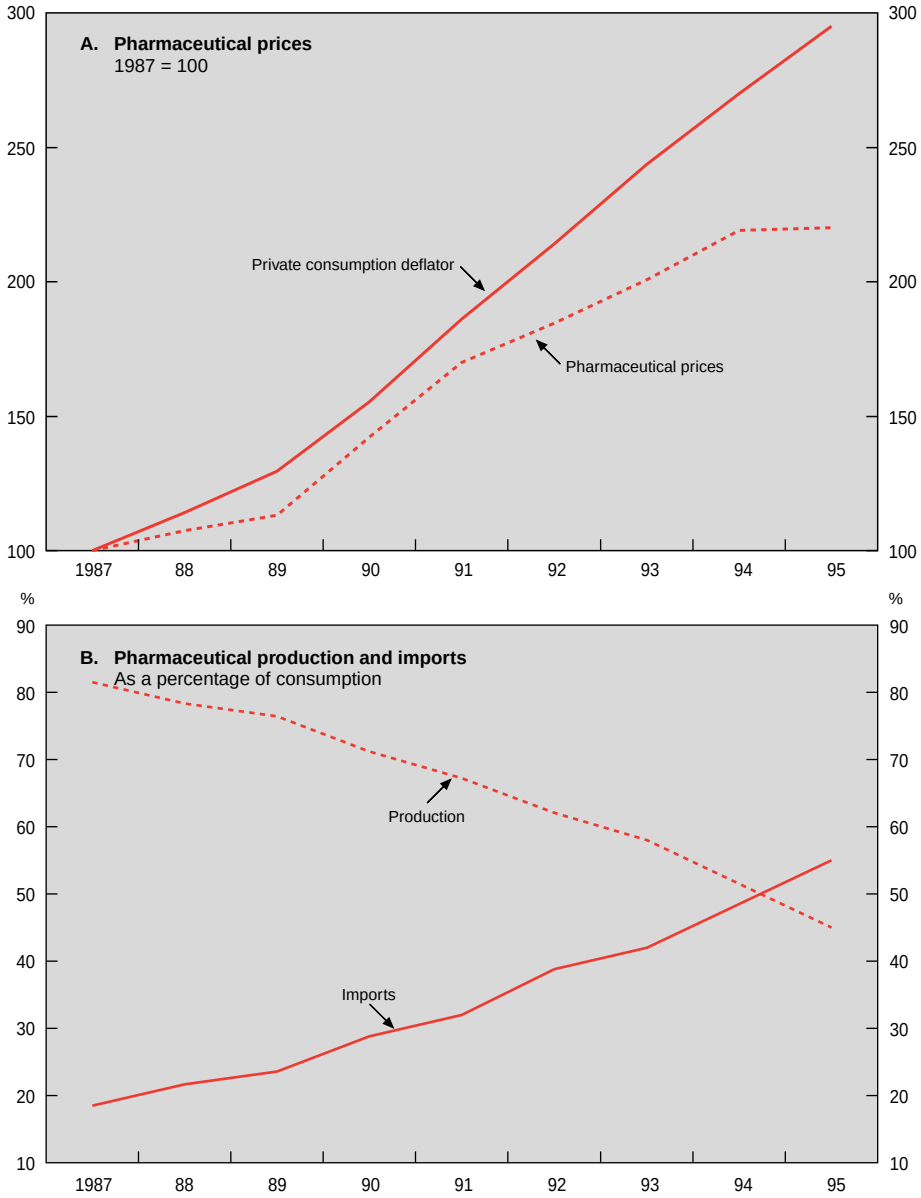
### *Pharmaceuticals*

The level of expenditure on pharmaceuticals as a percentage of GDP is the second highest in the OECD after Germany. Greeks spend 1.9 per cent of their GDP on drugs while the OECD average stands at 1.3 per cent of GDP. The high level of spending on pharmaceuticals reflects both a high level of consumption in volume terms and the high price of prescribed drugs.

Though an imprecise indicator of the volume of consumption, the number of drug packages bought annually per capita is the second highest in the OECD after France. It stood at 26.5 packages in 1994 while the OECD average was 14.4 packages a year. To curb consumption of pharmaceuticals, the co-payment rate on prescriptions was raised to 25 per cent in 1992, which is not far from rates elsewhere in the EU.[77] However, outlays on pharmaceuticals have remained on a rising trend. The high level of consumption seems to result mainly from doctors' prescribing practices. Legally, no pharmaceutical can be bought without prescription and, up to 1997, almost every drug was reimbursed by health funds, despite the existence of positive lists for two funds. These have apparently been circumvented as doctors can prescribe unlisted drugs with a justification. Doctors may also be induced by pharmaceutical firms to prescribe more than is necessary. There is no effective monitoring of doctors' prescribing practices since health reimbursements are managed by numerous health funds which, with the exception of IKA, do not have a computerised system. In addition, studies have underlined a great deal of wastage of drugs. A survey conducted by IKA showed that 30 per cent of drug packages dispensed were not used at all and in 18 per cent of the cases only one or two pills were taken.[78]

The cost of pharmaceuticals has soared despite an official policy of containing pharmaceutical prices below inflation. The restraint on prices for existing drugs (the official drug price index has lagged behind the CPI over the past twenty years, see Figure 23) has intensified incentives to continuously introduce more expensive drugs. To bypass controls on pharmaceutical prices, a common practice has been to create ''copies'' of the same drug under different brand names. This is reflected in the extensive use of expensive third generation antibiotics and the concurrent low use of generics. Moreover, the provision of generous margins to pharmacists (35 per cent) and wholesalers (7 per cent) have not promoted competition. Total mark-up and earmarked taxes result in a doubling of the ex-factory price for consumers. High profit margins help sustain the large

Figure 23. **PHARMACEUTICAL PRICES AND PRODUCTION**




*Source:* National Statistical Service of Greece, Pharmetrica and Ministry of National Economy.

number of pharmacies, which in turn fuels the excess consumption of pharmaceuticals. Finally, prices are kept high as neither over-the-counter nor generic drugs are common in Greece.

The way of setting mark-ups for drug price determination has also created a bias against domestic production as it makes it more attractive to import drugs rather than produce them locally. For domestically produced drugs, the pricing process is more restrictive as it is based on a mark-up on the cost of the basic ingredient. In contrast, for imported products, the base for the mark-up starts at a later stage, with the import price restricted to be no higher than the average of the three lowest prices for the drug in the EU (Table 19). As a consequence, the local drug industry has suffered, as evidenced by the dramatic loss in its domestic market share (Figure 23, panel B). In 1987, domestic production covered more than 80 per cent of domestic consumption, and this share had fallen to 35 per cent in 1995. The poor enforcement of patents for new drugs discourages domestic innovations and local production, and the difficulties for local producers are reinforced by poor property rights legislation.

In June 1997, a package of reforms was announced which aims to control pharmaceutical outlays better. First, the price of imported products will have to be equal to the lowest comparable price in the EU. Second, the margins of pharmaceutical wholesalers and pharmacists will be applied to the ex-factory price, rather than the wholesale price, resulting in their effective reduction by about 30 per cent. Earmarked taxes will also be reduced by a similar magnitude by reducing the rate of tax. Third, one positive list will be created to apply to all health funds. It will include daily treatment costs to reduce the sale of excess supplies. The authorities envisage implementing these reforms quickly.

### *The 1997 reform bill*

The wide-ranging reform bill submitted to Parliament in May 1997, if fully implemented, will address several important shortcomings of the system. The public character of the health system will be maintained by continuing the provision of largely free health care to all citizens. The bill aims to promote preventive health care through the development of primary care. In addition, it will endeavour to introduce modern management techniques in hospitals. The improvement of primary care will be undertaken through the introduction of a country-wide network of primary care providers. They will be jointly run by the

Table 19. **Mark-up on drugs**

Example for a drug with an ex-factory price of 100

| **Domestically produced pharmaceutical** | | **Imported pharmaceutical** | |
|---|---|---|---|
| Active ingredient | 41.0 | | |
| Other production costs | 19.2 | | |
| Administrative, commercial and financial costs | 28.7 | | |
| *Ex-factory cost* | 88.9 | | |
| Producers' margin (12.5% of ex-factory cost) | 11.1 | | |
| *Ex-factory price* | *100.0* | *Ex-factory price*[1] *= import price* | *100.0* |
| Social security funds of doctors (8.1% of ex-factory price) | 8.1 | Social security funds of doctors (6.5% of ex-factory price) | 6.5 |
| National drug institute and stamp duty (15.4% of ex-factory price) | 15.4 | National drug institute and stamp duty (15.4% of ex-factory price) | 15.4 |
| Social security of pharmacists (2.5% of ex-factory price) | 2.5 | Social security of pharmacists (2% of ex-factory price) | 2.0 |
| | | Importers' margin (12.5% of import price) | 12.5 |
| Wholesalers' profit margin (11.6% of wholesale price) | 16.6 | Wholesalers' profit margin (7% of wholesale price) | 10.3 |
| Regional discount (0.5% of wholesale price) | 0.7 | Regional discount (0.4% of wholesale price) | 0.6 |
| *Wholesale price* | *143.3* | *Wholesale price* | *147.3* |
| Pharmacists' profit margin (35% of wholesale price) | 50.2 | Pharmacists' profit margin (35% of wholesale price) | 51.6 |
| VAT (7% of wholesale price and pharmacists' margins) | 13.4 | VAT (7% of wholesale price and pharmacists' margins) | 13.8 |
| **Total** | **206.9** | **Total** | **212.7** |

1. Includes transportation expenses for imported drugs.

*Source:* Panagopoulos (1996) ''Greek pharmaceutical industry'' (in Greek), IOBE unit of sectoral studies, No. 137, Sept. 1996.

NHS and the social security funds. Regarding public hospitals, global budgeting will apply, diagnosis related groups (DRGs) will be gradually introduced to replace the uniform fee per patient, and professional managers will be hired. Other facets of the bill include, *inter alia*, the creation of a Public Health Service (supporting preventive health care activities) and a Central Laboratory, and the opportunity for additional training for doctors. It is expected to be introduced gradually over four years and cost Dr 100 billion (0.3 per cent of GDP) and has the potential for large cost savings. However, the full implementation of the legislation will be of critical importance. Medium-term objectives for the public health system are the unification of the social security funds and the providers of health care, with this legislation constituting a first step in that direction.

## *Conclusions*

Notwithstanding a rapid increase in expenditure and the establishment of universal coverage since the implementation of the National Health System, the overall performance of the public health system has remained unsatisfactory. The most important complaints relate to the quality of the service, including access to service, and the inefficient management of public hospitals and clinics. This situation has resulted in large private expenditure despite the existence of a largely free public service. The draft law presented to Parliament in May aims, *inter alia*, to eliminate some of the critical shortcomings of the system, especially through the introduction of a network of primary care, including the notion of a family doctor. In addition, professional management of public hospitals will be introduced and global budgets will provide greater flexibility to management. Its full implementation, including a more equitable geographic distribution of primary care doctors, will be an important step towards improving the Greek health care system.

However, as with past reforms, the draft law emphasises the improvement of the supply of services, and does not contain a major overhaul of the system's financing. In many other OECD countries, where the supply of public health services is no longer an issue, recent reforms have focused on establishing internal markets. Internal markets provide for competition between health care suppliers which should lead to a more efficient use and allocation of increasingly scarce resources and improve the quality of service. In Greece, in contrast, competition among suppliers has never been allowed and prices are fixed by the

government at unrealistically low levels for most services. As a result, social security health funds – the main purchaser of services – have little economic leverage *vis-à-vis* hospitals. Hospital management, in turn, has little budgetary flexibility, while soft budget constraints provide poor incentives to improve procurement procedures. Doctors have little incentive to maintain the quality of service as they are salaried and have status similar to that of a civil servant which provides a job guarantee. The effort to control drug prices has been bypassed by the private sector and, facilitated by the abolition of most trade barriers, has only resulted in the substitution of domestic production with imports.

The proper functioning of primary care is a necessity for the health system. The full implementation of the proposed network of family doctors – building on the existing system of primary care – whose referral should be required for access to secondary care should introduce an important gatekeeper role and limit the inefficient use of specialists at outpatient departments of hospitals. In the Greek context, the large number of doctors waiting for placement in the public health system provides an opportunity to fill the posts that are required at the primary care level. However, a critical component for realising such a primary care network is a rebalancing of incomes between generalists and specialists in large city hospitals. Remuneration of generalists should provide adequate incentives for them to practise outside large urban areas. Payment for services should involve incentives to maintain a high quality, and the experience of other countries as well as Greece suggests that a combination of two forms of payments – capitation and fee-for-service – works well as each by itself has drawbacks. In the United Kingdom, for instance, doctor's remuneration is based on capitation to maintain the quality of service and fee-for-service incentives for specific work (for more difficult cases). The system has been judged to perform reasonably well.

Regarding hospital care, the introduction of professional managers, global budgets and diagnostic related group payments should eliminate three obvious sources of inefficiency. The first two changes should help in rebalancing staffing, for instance by the hiring of more nurses, and raise transparency via the production of timely and modern accounts. These changes should also permit the use of more cost-effective procurement procedures, including the use of contracts with diagnostic centres. Nevertheless, a main source of the inadequate quality of service will remain. It emanates partly from the staff's job security and salaried

compensation which is unrelated to performance. Similar to primary care, a combination of two forms of payments (for instance, the combination of a fixed salary component with a fee-for-service payment depending on the number and type of patients treated) should raise service quality. In view of the existence of large unofficial private sector payments for public health care, permitting public sector doctors a limited private practice may introduce transparency and compensate doctors for salaries below their private sector counterparts. Performance based pay and the opportunity to work also privately should give an incentive to doctors to forfeit voluntarily their salaried, civil service status. In addition, increasing private co-payments could help dissuade unnecessary use of hospital care.

Reaping efficiency gains from global budgeting requires attaining a higher level of administrative capacity. Experience in other countries suggests that global budgets usually rise by similar amounts across hospitals *ex ante*, which creates medium-term imbalances and budgets are often overshot *ex post*. An attractive alternative would be to separate purchasers and providers of health services. International experience has also shown that competition is stimulated if patients are able to choose the health service and ''the money follows the patient''. This would, for instance, require that IKA and other insurers separate their hospitals from their insurance business. Resources would not be provided directly to hospitals who would be obliged to attract patients by the quality of services offered. Moreover, competition could be enhanced further if the State abstained from fixing the price of services for public hospitals. The European countries' experience with such management practices suggests that this approach is promising.

Another issue in the Greek context is whether the unification of the plethora of health funds would be beneficial. It would, on the one hand, result in a monopsony buyer of services and lower administration costs, but would eliminate a source of potential competition on the other. Competition would be possible with three or four funds, which would have to be separated from their parent pension funds. This could be easily implemented as the four largest funds currently provide coverage to a large majority of the population. However, this approach would require eliminating sectoral affiliation of the funds and should lead to the unification of benefits provided. This option would probably imply a scaling back of the generous treatment of public servants in the broader public

sector which is especially inequitable. The existence of several funds could also lead to ‘‘cream-skimming’’ where funds vie for the healthier population and avoid accepting the potentially more expensive persons in poorer health.

In light of the prospective review of the application of the EU structural funds, the large share of the health programme targeted to the construction of new hospitals (the other part is allocated to the renovation of older hospitals) could in part be better used in the improvement and expansion of the mobile health services. With a restructuring of the health care system, the number of hospital beds could shrink over the medium term, while frequent medical emergency transfers are required in the isolated mountain regions and islands.

The introduction of a unified positive drug list applicable to all funds is a large step towards reducing drug consumption. Moreover, the recently announced intention to reduce distortions between domestic and imported pharmaceutical products would reduce the penalisation of the domestic industry. Further measures to lower prices of pharmaceuticals could entail reducing or eliminating the large costs included in the ex-factory or imported price, such as the high margin for pharmacists and wholesalers and the earmarked taxes for doctors’ and pharmacists’ pension funds. Prescriptions by doctors should be monitored and abuses should be penalised. In addition, the use of generic and over-the-counter drugs should be encouraged. The new measures currently under consideration by the Government appear to move in this direction.

## Notes

1. National accounts have been revised in line with ESA accounting rules for the period 1988-91. Data for the period 1992-96 are preliminary estimates by the Ministry of National Economy.
2. A consumer loan granted by a bank can cover no more than 65 per cent of a purchase, while there was no limit previously.
3. For the first three years of the six-year programme, about 70 per cent of the projected resources have been utilised. Following practices in other EU countries, Greece has revised down the targets for the programme's initial years (primarily the first year). Without this adjustment about 50 per cent of the initially projected resources have been utilised.
4. Institute for Economic and Industrial Research (IOBE), *The Greek Economy*, Number 14, April 1997.
5. The 1996 annual labour force survey was carried out in June 1996 but detailed results were not released prior to June 1997.
6. Civil servants receive a catch-up component for the deviation of inflation from the government's inflation target. The catch-up for 1995 – which was paid in early 1996 – amounted to 2.1 per cent.
7. The weights in the CPI basket were revised to reflect the 1994 household survey. Based on the new weights, which – reflecting a higher income level – place lower weights on food products, the CPI increase has been revised down in 1996 from 8.5 per cent to 8.2 per cent, and the 12-month rate in December from 7.5 per cent to 7.3 per cent. It should be noted that the CPI continues to be based only on price surveys in large urban centres.
8. Only tobacco and alcohol taxes were raised, with the retail price of the former rising by 9 per cent and the average tax rising by 25 per cent for the latter. In addition, administrative prices rose by 5 per cent in January 1997 (year-over-year) reflecting restrained price increases in most public enterprises and utilities.
9. Including the impact of social security contributions.
10. An estimate of the degree of under-recording is provided by a comparison of Bank of Greece payments export data to the customs data from the National Statistical Service of Greece. The ratio of export merchandise values falls from 80 per cent in 1990 to 50 per cent in the first half of 1996. For imports, the ratio remains relatively stable over the same period at near 90 per cent.
11. OECD partner country import data also suggests a stagnation in exports to traditional markets, with merchandise exports in US dollar terms increasing by less than 10 per cent during

the three-year period 1994-96. The same data source reveals that imports of Greek products by Germany – Greece's largest export market – fell by 3 per cent in value terms during the same period.

12. OECD (1997), *Tourism Policy and International Tourism in OECD Member Countries 1995-1996*, Paris.

13. Bank of Greece, *Bulletin of Conjunctural Indicators*, No. 1, April 1997.

14. On a balance of payments basis, merchandise import volumes – which do not differ significantly from national accounts data – are estimated by the Bank of Greece to have increased by about 5 per cent in 1996, compared with 10 per cent in 1995.

15. The external debt to GDP ratio is biased downwards by the real appreciation of the exchange rate. Moreover, the ratio probably understates private sector debt. The Bank of Greece expects to introduce soon a new balance of payments methodology which will be based on a resident and non-resident distinction. Currently the accounts are based on the currency of transaction. The trade accounts will remain based on settlements data.

16. Following the pattern established in 1995, an additional Dr 200 billion of interest payments (0.7 per cent of GDP) was refinanced over a seven-year period. According to EC rulings, this amount may be excluded from the general government deficit as it substitutes for the lack of zero-coupon instruments in Greek financial markets.

17. These ''productivity'' bonuses were initially granted to civil servants in selected ministries from off-budget resources. Subsequently, as these bonuses were extended to more ministries, they were brought into the budget for purposes of transparency.

18. The corresponding outcomes of the central government budget (the consolidation of the ordinary and capital budgets) are a deficit of 9.7 per cent of GDP in 1996 compared with the targeted 8.5 per cent and an outturn of 10.5 per cent in 1995, with a primary surplus of 2 per cent compared with a target of 3 per cent and an outturn of 2 per cent in 1995 (see Table 8).

19. At the same time, capitalised interest and called guarantees equivalent to 1.0 per cent of GDP were also eliminated from the 1995 Budget outcome and moved ''below the line''. Another accounting change has been made to the 1995 accounts, further complicating comparisons with the 1996 Budget projection and 1996 Budget outcome. Back-pay awarded by court decisions to judges and civil servants in 1995, and to be paid over the period 1996-2000, was fully booked to the 1995 accounts. This amount is equivalent to 0.8 per cent of GDP and offsets the reduction in interest payment. Thus, the 1995 general government deficit remains broadly unchanged at 9.2 per cent of GDP, though the primary surplus has decreased by 0.8 per cent of GDP from earlier estimates, and primary current expenditures increased by the same amount. Without the adjustments to the 1995 accounts, current primary expenditures would have increased by ½ per cent of GDP in 1996, and the primary balance would have remained unchanged at 4 per cent of GDP.

20. The expansion of the consolidated gross general government debt during the past five years has been restrained by the large increase of inter-governmental debt and assets, as social security fund surpluses were used to purchase government securities.

21. The debt will increase further when the Government eventually takes over the debt of several enterprises in the process of restructuring, including the Athenian urban transport company and some of the ''problematic'' enterprises under the Industrial Reconstructing Organisation

(IRO) umbrella. Additional debt take-overs may arise from the clearing of payment arrears of public hospitals.

22. The most important reforms to tax expenditures comprise: *i)* the reduction in the exemption for imputed rental income from owner occupation from 200 square meters to 150 square meters for the first and second home taken together; *ii)* elimination of the deduction of lawyers' fees and the reduction of the deduction for private insurance premiums; *iii)* the reduction in the deduction limit for medical expenses for incomes above Dr 10 million if expenses exceed Dr 10 million; and *iv)* perhaps more as a symbolic gesture, the reduction of exemptions for a share of the income for various special groups (*e.g.* the salary of members of parliament, earnings of reporters, airline pilots, sailors, and signing bonuses of athletes).

23. The real estate tax will be assessed on property values; *e.g.* for a family of four the tax rate will start at 0.3 per cent for the portion of property in excess of a presumptive value of Dr 150 million.

24. Specifically, the so called ''lifetime'' pension for mothers in families with more than two children will be means tested and provided only to women over the age of 65, leading to estimated savings of 0.15 per cent of GDP. To help improve the demographic situation, the ''third child'' subsidy will be increased in amount and duration, and another set of benefits for families with more than two children will be linked to education. The additional cost of these last two measures is small.

25. M3 grew by 9.3 per cent compared with a target range of 6 to 9 per cent, and M4 (comprising M3 plus government securities held by the non-bank public with a maturity of no more than one year) grew by 12 per cent compared with a range of 9 to 12 per cent. Finally, total credit grew by 6 per cent compared with a target range of 5 to 7 per cent. Nominal GDP growth was 11.6 per cent in 1996.

26. Private sector credit denominated in drachma – which comprised over 80 per cent of total private credit at end-1995 – grew by 10.6 per cent during 1996 while foreign currency denominated credit grew by 56 per cent. The comparable growth rates in 1995 were 16.1 and 60.5 per cent, respectively. Foreign currency denominated credit extended by commercial banks to the private sector (*i.e.* excluding specialised banks) account for nearly 40 per cent of the total credit exposure to the private sector.

27. The required reserve ratio was raised by one percentage point to 12 per cent and the base on which required reserves are imposed widened to include all new (including renewed) bank liabilities *vis-à-vis* non-residents, independent of their maturity. The latter measure was aimed at curtailing derivative transactions that were being used to provide foreign currency denominated loans to domestic credit markets. Finally, a reserve requirement of 4 percentage points was placed on mortgage banks – which had been exempt until then – in a move aimed to curtail the boom in housing credit. See *OECD Economic Survey* for Greece 1996.

28. The application of the special bank tax on interest cost – rather than the loan value – provides an additional incentive to borrow in foreign currency, as foreign interest rates are lower than domestic ones.

29. The target range for M3, M4, and credit growth during 1997 were set, respectively, at 6 to 9 per cent, 8 to 11 per cent, and 4 to 6 per cent. The range for M3 remained unchanged relative to 1996, while those for M4 and credit have been reduced by one percentage point.

30. For each bank the criterion is their share of the total sum of credits, holdings of government securities, interbank deposits and their required reserves.

31. The Government is re-evaluating its social expenditure data so as to include several additional items and improve comparability with other countries.

32. There are an estimated 35 000 group policies, providing insurance for employees in large, often multinational, firms. Insurance companies' attribute the low demand for private pensions to the generosity of the public pension system.

33. The social security funds as a whole hold financial and real estate assets equivalent to approximately 8 per cent of GDP. The low level of savings reflects the deficit position of many funds and the placement of past accumulated surpluses in bank deposits which yielded negative real rates of return due to the existence of interest rate controls until 1989.

34. No one ministry is responsible for overseeing all funds, though the majority are under the supervision of the Ministry of Labour and Social Affairs.

35. The term ''contribution evasion'' is used to describe the phenomenon whereby individuals do not declare their working status, as well as the situation where contributions are in arrears.

36. Legislation does not forbid either the receipt of multiple pensions or working while receiving a pension. The most common cases of multiple pensions are reportedly: *i*) female spouses who receive their own and their husband's survivor's pensions (male workers can only receive a survivor's pension if they are invalids or have young children); *ii)* private sector salaried workers who had a second career as self-employed; and *iii)* public sector employees who had a second career in the private sector. In the second case, a slightly longer contribution period is required to qualify for a second pension. The total value of all pensions per individual was capped at four times the 1991 per capita GDP in 1992. Regarding the right to work while simultaneously receiving a pension, a pension is suspended only if the earnings from employment are more than twice the level of the minimum wage. For recipients of the minimum pension, the pension is fully suspended in the event the recipient works. However, these restrictions only apply to private sector employees (IKA).

37. The 1992 reform required the tax authorities to receive confirmation from a firm's social security fund that the requisite contributions have been paid prior to making any refunds. However, this measure's effectiveness is circumscribed without the fund's ability to verify wages and employment time. More effective was the introduction in the construction sector – where contribution evasion is reportedly most acute – of presumptive days of contributions.

38. Currently, two funds are suffering large losses due to the decline of their respective sectors; the seaman's fund and the fund for drivers of public vehicles, which have ratios of contributors to pensioners of 1:2 and 1.1:1, respectively. The consolidation of both these funds into other larger funds is an option under consideration.

39. The 1992 reform introduced a small pooling fund (LAFKA). It is funded by freezing the revenues from earmarked taxes received by certain funds at their nominal 1992 level and shifting the excess receipts to this pool. In 1996, LAFKA receipts were budgeted at Dr 25 billion compared to annual budget transfers to loss-making funds of Dr 750 billion.

40. The system also provides incentives for adverse selection. The large variation in pension benefits and eligibility requirements in conjunction with generous conversion provisions create incentives for individuals to switch pension funds towards those with higher benefit

levels relative to contributions. Such transfers of pension rights reportedly occur most often from the less generous funds for the small entrepreneurs and traders to IKA, which has a higher minimum pension and earlier eligibility. The transfer of pension rights to funds which have less onerous eligibility requirements is partly to blame for the severe financial difficulties of the seaman's fund, when lax transfer requirements during the period 1983-87 resulted in the granting of many full pensions after 15 years of contribution.

41. For example, the lawyers' fund receives a percentage from all contracts, the doctors' fund 6.5 per cent of the wholesale price of pharmaceuticals (hence taxing other social security funds), engineers a percentage of public works, and journalists a percentage of television and radio advertising costs. Part of the problem stems from the lack of a clear legal distinction between a contribution and an earmarked tax, especially for the cases of the self-employed.

42. The replacement rate formula for IKA is complicated. The replacement rates for the first 10 years of contributions range from 30-70 per cent and are inversely related to the 28 income brackets. For every year of service from the eleventh through twenty-fifth year, the replacement rate is raised by a supplementary 1 per cent. For every year of service above 25 years, the supplement is raised, and ranges from 1.5-2.5 per cent, with the premium applied to the higher brackets. The replacement rate is raised further by the double counting of two months' bonus salary (Christmas, Easter, and holiday). The bonuses are included in the pension base, but also continue to be distributed to pensioners. This double counting is being gradually phased out.

43. For the supplementary pension, the assessment base is even shorter, as the minimum blue collar wage at the time of retirement is used to determine the level for this pension.

44. Moreover, since the pension formula adjustments exceeded wage growth until 1990, replacement rates rose for unchanged real wages (*i.e.* effectively a negative bracket creep).

45. Data collected by IKA suggest that contribution evasion is mostly accomplished through an under-reporting of days employed. The degree to which salaries and daily wages are under-reported (in contrast to the number of days of contributions) is less clear, though the data suggest that this form of contribution evasion is also prevalent. Over the past two decades, most new pensioners are consistently located in income classes which imply daily wages in the last five years of work which are approximately 70 per cent of the average wages in manufacturing. Since at the end of one's working history wages are usually above average, one would have expected the wage used in the assessment base to be above, rather than below average. Earnings legally excluded from the assessment base comprise only extraordinary payments; *i.e.* (non-regular) bonuses or overtime.

46. The desire to maintain health coverage is not a disincentive to contribution evasion, since eligibility for full health benefits is achieved after only 50 days' contributions during the past year.

47. Their eligibility period is slightly longer than for IKA (20 years, which will gradually be increased to 25 years by 2007). Though no data exist on the contribution period on which TEBE pensions are based, the fact that the average pension exceeds the minimum by 20 per cent compared with 40 per cent for IKA suggests a lower number of pensioners retiring at the minimum eligibility period. Funds have different minimum pensions, with the one for IKA about 25 per cent more generous than that for TEBE.

48. For the few self-employed who entered the work force after end-1992, the effective contribution rates are higher.

49. The new fund will provide the opportunity for raising farmers pensions. However, it provides a high statutory replacement rate (70 per cent for a full pension) and easy eligibility conditions during the transition period. Moreover, the transformation of the farmers' fund into a full primary fund provides the option to transfer their pension rights to other primary pension funds if they change professions. Under the previous system, farmers did not accumulate pension rights which could be transferred to other pension funds. Such transfers may prove costly to other primary funds.

50. The contributions for employees in state-owned banks were made by the employer prior to 1990 and the subsequent introduction of contributions has been more than covered by real wage gains.

51. The transfer ratio was also propelled upwards by low labour productivity growth, which averaged below 1 per cent during the 1980s compared with 3 per cent during the previous decade.

52. The ratio exceeds 1:1 due to the institutional factors that permit and/or induce individuals to retire before the age of 65, including generous retirement provisions, women's earlier statutory retirement age, lower retirement limits for many public servants, access to invalidity and survivor pensions, and the indeterminate number of multiple pensions.

53. Individuals can retire up to five years earlier than the limit of 65 years, with a reduction in pension benefits of 6 per cent per year. This regulation also applies to retirement with a minimum pension.

54. The more generous eligibility regulations for women permit women in the private sector to retire at age 55 with 18.3 years' contributions if they have under-age children, and at age 60 with 15 years' contributions if married. The retirement conditions are far more generous in the public sector. In the civil service, married women hired after end-1982 and retiring before end-1997, can retire at age 58 with 17.5 years' contributions or without age limit after 24.5 years' contributions. For women with under-age children, and 17.5 years of contributions, the retirement age falls to 50. Women hired prior to end-1982 have even more generous age limits.

55. Eligibility for an invalidity pension requires only five years' contributions in IKA, while it is ten years for other funds. Perhaps reflecting the lower eligibility standard, IKA has the highest share of invalidity pensions (equalled only by OGA where invalidity pensions are nevertheless low).

56. The increase in the effective contribution rate also reflects the increase in coverage following the extension of the pension system to the whole population and the inflow of immigrants.

57. Upon its creation in 1982, the eligibility criteria for the largest supplementary fund (IKA-TEAM), covering over 50 per cent of private sector wage earners, was set at 1 000 days' contributions (equivalent to 3⅓ years) and increased by 175 days every year until it reaches 4 500 days in 2001. Thus, the fund already has close to 200 000 pensioners. On 1 January 1997, the eligibility criteria stood at 2 575 days (equivalent to 8.6 years). Similarly, the eligibility condition for the new farmers' primary pension has been set at five years, and will gradually rise to 15 years over 20 years. Moreover, contributions for supplementary pensions,

which it replaces, will count towards eligibility. It appears likely that recourse to the reduced eligibility criteria will be large since the majority of agricultural workers are near retirement. This situation reflects the declining importance of this sector since the second world war from approximately 50 per cent of the labour force to 20 per cent currently.

58. The Government eventually had to assume most of this debt which was issued with a government guarantee. In the early 1990s, the Government assumed debts equivalent to about 7½ per cent of GDP.

59. See OECD (1996), *Economic Surveys, Greece*, Paris.

60. Primary pensions for the self-employed are adjusted on an *ad hoc* basis for inflation and have exceeded the increases permitted by IKA by about 25 per cent since 1990, leaving their pensions broadly unchanged in real terms.

61. Eligibility for the monthly pension supplement requires a pensioner to fulfil all of the following cascading conditions: *i*) a gross pension income (from all pensions) which does not exceed Dr 110 000 per month; *ii)* total annual income from work and pensions which do not exceed Dr 1.4 million; *iii)* total annual taxable income which does not exceed Dr 1.8 million; and *iv)* total family taxable income which does not exceed Dr 2.8 million. The supplement is reduced progressively for pensioners receiving pensions between Dr 100 000-110 000.

62. The demographic projections do not include the immigrants which entered Greece in the early 1990s. In the short-run, their contributions would reduce the pension system's gross financing needs but in view of the generosity of the pensions system, in the long-run, their incorporation would increase the implicit debt of the system.

63. Fore D., W. Leibritz, D. Roseveare and E. Wurzel (1996), ''Ageing populations, pensions systems and government budgets: simulations for 20 OECD countries'', OECD Economics Department Working Papers, No. 168, Paris.

64. The baseline simulation assumes that productivity will grow by 1.5 per cent of GDP and a discount rate of 5 per cent. A lower productivity growth rate will improve the pension system's situation since the concomitant reduction in real wages will affect the more generous pensions proportionally more than contributions. A lower discount rate will obviously increase the present value of the liabilities. If productivity growth is lowered to 1 per cent or the discount rate lowered to 4 per cent the present value of net pension outlays change, respectively, to 185 and 264 per cent of GDP.

65. The generosity of the pension funds of self-employed professionals is underestimated due to the difficulty in separating out receipts from earmarked taxes from earnings-based contributions. For example, in the case of the engineer's fund (TSMEDE), all receipts, including a large amount from earmarked taxes, are considered to be contribution-based.

66. A survey of EU citizens on their health care system was conducted in 1992. It revealed that only 25 per cent of Greeks thought that the quality of health care received was good (compared with an average of 71 per cent for the 12 European countries). A large proportion (82 per cent) judged their health services as inefficient, and a majority (65 per cent) were willing to pay more for health care. Ferrera M. (1993), ''EC Citizens and Social protection: Main results from a Eurobarometer survey'', Commission of the European Communities, Brussels.

67. Ministry of Health and Social Welfare of Greece (1994), *Report on the Greek Health Services,* Athens.

68. Figures for 1992. Ministry of Health and Social Welfare of Greece (1994), *Study for a Plan and Organisation of Health Services,* Athens.

69. Ministry of Health and Social Welfare of Greece (1994), *Report on the Greek Health Services,* Athens.

70. Under-reporting of infant death in rural areas is considered a source of bias in the health statistics.

71. The reliability of the data for public and private health care expenditure is questionable. The OECD Health Data (using national accounts aggregates) estimate total expenditure at 4.3 per cent of GDP in 1991, with the public share equivalent to 3.3 per cent of GDP. The Greek Ministry of Health estimated total expenditure to be near 8 per cent with public expenditure near 5 per cent of GDP. Thus, the main source of uncertainty arises from private payments. Other data sources are more in line with the second estimate. The family expenditure survey data indicate a ratio of health expenditure to GDP of around 8 per cent in 1991 (see Ministry of Health and Social Welfare of Greece, 1994, *op. cit.*).

72. Another source of uncertainty regarding the amount of health expenditure arises from the poor accounts of hospitals and social security health funds, as well as the existence of large arrears to suppliers. They are cleared with a long delay by the Government. At end-1996, the outstanding stock of arrears to suppliers was estimated to be Dr 190 billion (0.6 per cent of GDP).

73. Nevertheless, there exists a waiting list of about 10 000 doctors to enter the public health care system, most of them specialists who prefer to work in the Greater Athens area.

74. If patients visit doctors in the private sector who are not under contract with the member's fund, the cost is fully covered by the patients.

75. Unofficially, they sometimes do and being employed at a well-equipped public hospital is viewed as a means for them to attract clients.

76. World Health Organisation (1997), ''Health Care System in Transition, Greece'', Preliminary report by the Regional Office for Europe, Copenhagen.

77. There are some exceptions for certain groups of people or illnesses for which the rate falls to 10 per cent or exceptionally to zero per cent for chronic conditions.

78. Ministry of Health and Social Welfare of Greece (1994), *Report on the Greek Health Services,* Athens.

## Glossary of acronyms

| | |
|---|---|
| CSF II | Second Community Support Framework programme |
| CPI | Consumer price index |
| DEP | National petroleum company |
| EFTE | Special bank tax |
| ESA | European system of accounts |
| ETBA | Greek Bank for Industrial Development |
| EMU | European Monetary Union |
| ETEBA | National Bank of Greece for Industrial Development |
| IKA | Social insurance institute |
| IKA-TEAM | Social insurance institute, supplementary insurance fund |
| IRO | Industrial Restructuring Organisation |
| LAFKA | Solidarity account for social security funds |
| NAT | Sailors veteran fund |
| OGA | National agricultural insurance institute |
| OTE | State telecommunications company |
| MHSW | Ministry of Health and Social Welfare |
| NHS | National Health Service |
| PSBR | Public sector borrowing requirement |
| TEBE | Insurance fund for craftsman and traders |
| TSMEDE | Fund of the organisation of engineers of public works |

*Annex I*

# Pension projections: a description of the model

This annex describes the construction of the pension model, whose basic design is based on Van Den Noord and Herd (1993).[1]

## Coverage

The model covers old-age, survivor and invalidity pensions, provided by primary and supplementary funds, as well as separation payments. Health care and other social benefits (and their corresponding contributions) are not included in the calculations. The model divides contributors and pensioners into the following seven groups:

- private sector dependent workers (IKA, IKA-TEAM, etc.);
- seaman (NAT);
- self-employed small entrepreneurs and traders (TEBE, TAE, etc.);
- self-employed professionals (doctors, lawyers, and engineers);
- farmers (OGA);
- general government employees (excluding war pensions and pensions for members of the national resistance); and
- public sector employees in public enterprises and state-owned banks.

Separate estimates are made for each of the above mentioned pension categories for each of the seven groups.

The revenues of the pension model exclude all receipts that are not contribution based such as earmarked taxes (where it was possible to separate out earmarked taxes from contributions), budget transfers to pension funds and earnings from funds' assets. In addition, in order to make the projections comparable to those undertaken by the OECD for other countries, the model excludes contributions that are made implicitly or explicitly by the State. The rationale for their exclusion is that the source of such revenues is general taxation. These receipts include the State's one-third share of the tripartite financing for the contributions of workers that entered the workforce after 1992, the State's two-thirds share of the contributions for farmers' pensions, and the State's implicit contributions as employer.

## Assumptions

**Demographics and employment growth:** The population projections for the period 2000 to 2070 are taken from the World Bank.[2] The main assumption underlying the projections are that fertility rates will return to replacement rates by 2030 and that life expectancy will increase by about 4 years during the same time horizon. The total labour force participation rate is assumed to be constant. The NAWRU is also assumed to be constant from 2000 onwards, implying that the unemployment rate thereafter remains at about 8 per cent.[3] The initial distribution of employment between the different sectors is based on the labour force survey of the Greek National Statistical Agency and national accounts data. The employment shares do not remain constant over the projection period for all sectors. The projections assume that the share of employment in the agricultural sector falls to 14 per cent by 2035 from 21 per cent in 1995. The employment share of the self-employed traders and small entrepreneurs and private dependent workers is each assumed to increase by one-half of this decline. For the other sectors, the employment share is assumed to remain constant throughout the projection period.

**Earnings and output growth:** The analysis is conducted in constant 1994 prices. Earnings are assumed to increase in line with labour productivity (1½ per cent per year for each sector). Output increases in line with productivity and employment.

## The model

**Pensioners and contributors:** The base year of the model is 1994 — the most recent year for which accounts could be collected for the main pension funds. Old-age primary pensions (the number of pensioners is not known) are assumed to grow in line with the population over the age of 60 or 65, depending on the sector, as well as movements in sectoral employment (lagged by the average number of working years in the sector). For the sectors where there is a discrepancy between the number of primary and supplementary pensions (primarily IKA), supplementary pensions are assumed to mature and equal the number of primary pensions. Survivors pensions are assumed to grow in line with the population age 50 and above, and invalidity pensions in line with the population between the ages of 45 and 60 years old. Separation payments are assumed to grow in line with the population age 60-65 years old. Contributors are assumed equal to the number of employed workers.

**Contributions:** For each fund (primary, supplementary or separation), contributions are the product of the sector specific contribution rate, the wage and the number of contributors. For each fund, the contribution rate is derived from 1994 data, using the above-described hypotheses for the number of contributors and the sectoral wage. Adjustments are made to the contribution rate to account for the 1990-92 reforms, including the increase in the contribution rates of government employees to those of the private sector, the unification of contribution rates for the post-1992 entrants into the labour force, the

increase in the presumed income categories of the self-employed traders and small entrepreneurs, and some increase in the contribution rates of IKA reflecting the reduction in contribution evasion as the minimum wage is kept constant in real terms (and thus provides a declining replacement rate relative to the pension arising from the pension formula). In addition, the contribution rate for farmers rises, reflecting the introduction of their new primary pension fund.

**Pension expenditure:** The pension outlay per individual is the product of the replacement ratio and the wage at the time of retirement. Since pensions are assumed to be indexed to inflation and not to wages, the pensions of older pensioners are lower than those of younger ones. Thus, total pension expenditure for each fund is the sum of outlays on three or four age groups of pensioners (depending if the retirement age for the sector is 60 or 65); those that have retired during the past five years, those that have retired between 5 and 10 years earlier, those that have retired between 10 and 15 years earlier, and those that have retired more than 15 years ago. The replacement ratio is derived from 1994 data by adjusting the average pension by wage and pension adjustments for the period 1975-94. The replacement rate is adjusted to account for the effects of the 1990-92 reforms, including the lower replacement rates for the post-1992 entrants into the work force, the gradual elimination of the double counting of the bonus months salary for IKA pensioners, and the reduction in supplementary pension replacement rates to 20 per cent starting from 1998. In addition, the replacement rate for IKA pensioners rises slightly reflecting the above mentioned reduction in contribution evasion following the reduction of the minimum wage relative to the average wage. Finally, the replacement rate for farmers also increases gradually reflecting the introduction of their new primary pension fund.

## Notes

1. Van Den Noord, P. and R. Herd, (1993), ''Pension liabilities in the seven major economies,'' OECD Economics Department Working Papers, No. 142, Paris.
2. Bos, E. *et al.* (1994), *World population Projections, 1994-95*, The International Bank for Reconstruction and Development/The World Bank, Washington D.C.
3. In view of the generosity of the individual pension, a higher labour force participation ratio and a lower unemployment rate would lead to a deterioration of the pension system's financial position, though the increased contributions would improve the system's finances in the short term.

*Annex II*

# Calendar of main economic events

## 1996

### April

Introduction of new legislation making the provision of financial investment services consistent with EU regulations.

### May

The European Court issued a ruling requiring VAT charges on all sea cruises.

The European Commission approved the building of the new Athens airport at Spata, which will benefit from EU financial support.

Legislation passed making it compulsory for corporations listed on the Athens Stock Exchange, and benefiting from public contracts, to convert all stocks from bearer to registered shares.

### June

The Bank of Greece announced an increase in primary reserve requirements to 12 per cent; a widening of the base on which reserve requirements are calculated; and a reduction in the share of the purchase price of consumer durables which can be credit-financed.

The Government introduced a new means-tested old-age pension supplement.

A new law was passed (Law 2414/1996) which permits public enterprises to become public liability companies (S.A.).

### July

The Bank of Greece lowered the default interest rate applied by commercial banks on all future overdue loans to 2½ points above the contractual interest rate.

Goodyear Tyre Company announced the closure of its plant in Thessaloniki.

**August**

Labour market legislation passed in Parliament.

**September**

National elections won by the ruling Socialist party lead by Mr. Costas Simitis.

**October**

Activities of two investment banks (ETBA and ETEBA) broadened to include consumer and housing loans, as well as foreign exchange operations for personal needs.

**November**

The 1997 Budget was presented to Parliament: the target for the central government deficit was set at 6.2 per cent of GDP as against 9.3 per cent of GDP in 1996.

A package of measures limiting credit institution market risk was introduced.

Remunerations of required foreign currency reserves reduced from 1.4 per cent to 1.2 per cent.

**December**

The Bank of Greece reduced its overdraft rate from 26 to 25 per cent and its discount rate from 17 to 16.5 per cent. It also cut the interest rate at which it remunerates mandatory commercial bank reserves from 12½ per cent to 11 per cent.

Sale of Bank of Attika to a pension fund (TSMEDE).

## 1997

**February**

Bank of Greece reduced its discount rate from 16.5 per cent to 15.5 per cent, and its Lombard rate from 21 per cent to 20 per cent.

**March**

The European Commission agreed to extend the time limit for ending State subsidies to Greek shipyards to December 1997.

The Bank of Greece introduced a new two-tier interest rate in the interbank market.

The Government agreed to a second partial privatisation of OTE, the Greek Telecommunication Corporation.

Introduction of new restrictions on shop opening hours.

**April**

The Council of EU Industry Ministers endorsed a proposal to write off Skaramanga shipyard outstanding debt.

*STATISTICAL ANNEX AND STRUCTURAL INDICATORS*

Table A. **National product and expenditure**

Billion drachmas

| | 1991 | 1992[1] | 1993[1] | 1994[1] | 1995[1] | 1996[1] |
|---|---|---|---|---|---|---|
| | At current prices | | | | | |
| Private consumption | 11 848.3 | 13 876.6 | 15 799.9 | 17 768.0 | 19 732.9 | 21 881.2 |
| Government consumption | 2 337.9 | 2 561.6 | 2 910.4 | 3 275.7 | 3 899.0 | 4 239.4 |
| Gross fixed capital formation[2] | 3 654.4 | 4 073.8 | 4 390.1 | 4 770.5 | 5 382.8 | 6 371.6 |
| Change in stocks and statistical discrepancy | 148.0 | 89.6 | 138.6 | 217.7 | 236.3 | 278.9 |
| Exports of goods and services | 2 619.3 | 3 111.8 | 3 422.0 | 3 986.6 | 4 360.5 | 4 617.5 |
| *less:* Imports of goods and services[2] | 4 381.2 | 5 033.9 | 5 555.0 | 6 259.4 | 7 122.7 | 7 829.1 |
| Gross domestic product at market prices | 16 226.9 | 18 679.5 | 21 106.1 | 23 759.0 | 26 488.8 | 29 559.5 |
| Net factor income from the rest of the world | 196.5 | 192.0 | 409.0 | 515.5 | 600.0 | 620.0 |
| Gross national product at market prices | 16 423.4 | 18 871.5 | 21 515.1 | 24 274.5 | 27 088.8 | 30 179.5 |
| | Constant prices of previous year | | | | | |
| Private consumption | 7 269.3 | 7 400.2 | 7 407.6 | 7 518.7 | 7 639.0 | 7 807.1 |
| Government consumption | 1 369.2 | 1 358.3 | 1 347.4 | 1 346.1 | 1 371.7 | 1 390.9 |
| Gross fixed capital formation[2] | 2 316.0 | 2 282.0 | 2 219.0 | 2 231.0 | 2 360.0 | 2 643.0 |
| Change in stocks and statistical discrepancy | 103.1 | 49.8 | 43.0 | 50.4 | 140.0 | 150.0 |
| Exports of goods and services | 1 768.0 | 1 921.8 | 1 933.4 | 2 082.2 | 2 119.7 | 2 117.6 |
| *less:* Imports of goods and services[2] | 3 010.8 | 3 158.4 | 3 193.1 | 3 324.0 | 3 530.1 | 3 749.0 |
| Gross domestic product at market prices | 9 814.8 | 9 853.7 | 9 757.3 | 9 904.3 | 10 100.3 | 10 359.5 |

1. Provisional data.
2. Excluding ships operating overseas.

*Source:* Data submitted by national authorities and OECD estimates.

Table B. **Gross fixed capital formation**[1]

Billion drachmas

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[2] |
|---|---|---|---|---|---|---|
| | At current prices | | | | | |
| **Total** | 3 654.4 | 4 073.8 | 4 390.1 | 4 770.5 | 5 382.8 | 6 371.6 |
| Dwellings | 1 115.3 | 1 131.1 | 1 222.9 | 1 234.3 | 1 274.4 | 1 619.7 |
| Other construction | 1 295.0 | 1 444.0 | 1 529.3 | 1 662.6 | 1 894.0 | 2 226.7 |
| Equipment | 1 244.1 | 1 498.7 | 1 637.9 | 1 873.6 | 2 214.4 | 2 525.2 |
| Private sector | 2 887.7 | 3 132.2 | 3 347.6 | 3 631.8 | 4 081.6 | 4 771.5 |
| Public sector | 762.5 | 904.9 | 1 016.5 | 1 098.9 | 1 263.0 | 1 538.3 |
| | Constant prices of previous year | | | | | |
| **Total** | 2 316.0 | 2 282.0 | 2 219.0 | 2 231.0 | 2 360.0 | 2 643.0 |
| Dwellings | 661.1 | 565.9 | 542.5 | 489.6 | 457.1 | 547.1 |
| Other construction | 775.1 | 770.3 | 720.4 | 725.0 | 769.6 | 847.0 |
| Equipment | 879.8 | 945.8 | 956.2 | 1 016.4 | 1 133.3 | 1 248.9 |
| Private sector | 1 842.6 | 1 781.8 | 1 715.9 | 1 726.2 | 1 817.7 | 2 017.6 |
| Public sector | 474.4 | 501.9 | 504.0 | 505.0 | 542.3 | 623.1 |

1. Data exclude investment in ships operating overseas.
2. Provisional data.

*Source:* Data submitted by national authorities.

Table C. **Income and expenditure of private sector**[1]

Billion drachmas, current prices

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[2] |
|---|---|---|---|---|---|---|
| Wages and salaries of non-agricultural branches | 5 369.5 | 5 938.0 | 6 594.6 | 7 523.0 | 8 659.4 | 9 806.2 |
| Entrepreneurial and property income[3] | 9 010.3 | 10 801.1 | 12 927.3 | 14 698.7 | 15 232.7 | 16 662.3 |
| Net current transfers from government | 2 468.7 | 2 848.5 | 3 321.7 | 3 875.1 | 4 337.8 | 4 849.7 |
| Net current transfers from the rest of the world | 572.3 | 649.6 | 808.1 | 886.3 | 951.5 | 1 037.1 |
| Income of households and private non-profit institutions | 17 420.8 | 20 237.2 | 23 651.7 | 26 983.1 | 29 181.4 | 32 355.3 |
| *less:* Direct taxes | 903.1 | 991.0 | 1 174.0 | 1 607.0 | 1 918.6 | 2 136.5 |
| Social security | 1 823.2 | 2 085.0 | 2 556.0 | 2 843.8 | 3 287.4 | 3 662.2 |
| Disposable income | 14 694.5 | 17 161.2 | 19 921.7 | 22 532.3 | 23 975.3 | 26 556.6 |
| Consumption | 11 851.4 | 13 877.3 | 15 800.6 | 17 770.1 | 19 733.6 | 21 884.6 |
| Saving | 2 843.1 | 3 283.9 | 4 121.1 | 4 762.2 | 4 241.7 | 4 672.0 |

1. Includes households, non-profit institutions and enterprises.
2. Provisional data.
3. Including agricultural income.

*Source:* Data submitted by national authorities.

Table D. **Government revenue and expenditure**

Billion drachmas, current prices

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[1] |
|---|---|---|---|---|---|---|
| **Central government ordinary budget**[2] | | | | | | |
| Current revenue | 3 661 | 4 588 | 4 971 | 5 862 | 6 728 | 7 440 |
| Direct taxes | 1 024 | 1 192 | 1 355 | 1 773 | 2 133 | 2 343 |
| Indirect taxes | 2 379 | 2 924 | 3 189 | 3 462 | 3 835 | 4 321 |
| Other | 258 | 472 | 426 | 627 | 760 | 776 |
| Current expenditure | 4 874 | 5 374 | 6 857 | 8 251 | 8 880 | 9 751 |
| Goods and services | 242 | 325 | 418 | 460 | 497 | 709 |
| Interest on public debt | 1 496 | 1 559 | 2 334 | 3 340 | 3 356 | 3 465 |
| Transfers and subsidies | 3 136 | 3 490 | 4 105 | 4 452 | 5 027 | 5 577 |
| Saving | –1 213 | –786 | –1 886 | –2 389 | –2 152 | –2 311 |
| **General government**[3] | | | | | | |
| Current revenue | 5 336 | 6 289 | 7 260 | 8 518 | 10 025 | 11 038 |
| Direct taxes | 903 | 991 | 1 174 | 1 607 | 1 919 | 2 137 |
| Households | 621 | 685 | 811 | 1 110 | 1 328 | 1 469 |
| Business | 282 | 306 | 363 | 497 | 591 | 668 |
| Indirect taxes | 2 247 | 2 769 | 3 022 | 3 341 | 3 796 | 4 221 |
| Social security contributions | 1 823 | 2 085 | 2 556 | 2 844 | 3 289 | 3 663 |
| Other | 362 | 444 | 507 | 726 | 1 022 | 1 017 |
| Current expenditure | 6 392 | 7 610 | 9 008 | 10 575 | 11 627 | 12 658 |
| Consumption | 2 338 | 2 562 | 2 910 | 3 276 | 3 899 | 4 239 |
| Interest payments | 1 529 | 2 190 | 2 699 | 3 381 | 3 271 | 3 388 |
| Transfers and subsidies | 2 526 | 2 858 | 3 399 | 3 918 | 4 457 | 5 031 |
| Business | 101 | 98 | 119 | 108 | 125 | 126 |
| Households | 2 469 | 2 849 | 3 321 | 3 876 | 4 359 | 4 887 |
| Rest of the world | –44 | –88 | –40 | –66 | –26 | 18 |
| Net saving | –1 057 | –1 321 | –1 749 | –2 057 | –1 602 | –1 620 |
| Net capital transfers (–) | 311 | 396 | 630 | 162 | 76 | –251 |
| Fixed investment (–) | 502 | 584 | 613 | 657 | 750 | 820 |
| Net lending | –1 870 | –2 300 | –2 992 | –2 876 | –2 428 | –2 189 |

1. Provisional data.
2. Administrative basis.
3. National account basis.

*Source:* Data submitted by national authorities.

Table E. **Industrial production**

1980 = 100

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|
| **Industrial production, total** | 110.8 | 112.8 | 110.1 | 108.9 | 108.0 | 104.7 | 106.1 | 108.0 | 109.3 |
| Mining and quarrying | 188.8 | 179.5 | 173.8 | 171.9 | 160.6 | 150.5 | 148.6 | 143.8 | 148.6 |
| Manufacturing | 103.2 | 105.6 | 102.6 | 101.7 | 100.6 | 97.1 | 98.2 | 100.3 | 100.9 |
| Food, beverages and tobacco | 117.0 | 126.6 | 119.9 | 127.1 | 135.0 | 133.5 | 137.3 | 142.0 | 139.6 |
| *of which:* Tobacco | 99.8 | 92.2 | 112.3 | 113.7 | 108.5 | 107.2 | 124.0 | 137.4 | 135.7 |
| Food | 114.8 | 126.7 | 112.6 | 123.8 | 134.9 | 131.6 | 130.2 | 132.8 | 132.7 |
| Other manufacturing | 100.2 | 101.2 | 98.9 | 96.1 | 93.1 | 89.2 | 89.7 | 89.6 | 91.0 |
| *of which:* Textiles | 101.3 | 99.1 | 95.4 | 86.7 | 79.3 | 74.3 | 73.9 | 69.9 | 66.7 |
| Chemicals | 125.8 | 132.4 | 133.3 | 126.8 | 122.6 | 127.4 | 129.9 | 144.0 | 155.3 |
| Non-metallic mineral products | 99.5 | 95.6 | 100.0 | 88.3 | 84.5 | 84.7 | 87.2 | 88.8 | 95.1 |
| Basic metals | 98.0 | 97.9 | 99.3 | 100.6 | 102.7 | 97.6 | 102.4 | 107.3 | 103.3 |
| Metal products | 90.5 | 83.8 | 74.4 | 73.6 | 74.8 | 68.5 | 67.0 | 70.1 | 69.1 |
| Consumer goods industries | 112.0 | 115.8 | 107.1 | 104.1 | 103.0 | 102.4 | 104.6 | 105.6 | 106.2 |
| Durable | 81.1 | 84.9 | 75.3 | 81.0 | 81.7 | 88.7 | 88.5 | 87.2 | 89.3 |
| Other | 115.3 | 119.1 | 110.4 | 107.7 | 105.8 | 103.8 | 106.4 | 106.9 | 107.7 |
| Capital goods industries | 83.7 | 82.9 | 92.1 | 92.1 | 92.1 | 84.5 | 83.2 | 88.6 | 88.7 |

*Source:* National Statistical Service, *Monthly Statistical Bulletin* and OECD, *Main Economic Indicators.*

Table F. **Prices and wages**

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|
| **Consumer prices (1994 = 100)**[1] | | | | | | | | | |
| Total | 41.6 | 47.3 | 57.0 | 68.0 | 78.8 | 90.2 | 100.0 | 108.9 | 117.9 |
| *of which:* Food | 38.7 | 46.1 | 54.9 | 66.6 | 76.0 | 85.6 | 100.0 | 108.5 | 116.1 |
| Beverages and tobacco | 32.8 | 37.9 | 48.1 | 58.0 | 71.6 | 84.7 | 100.0 | 104.8 | 115.2 |
| Clothing and footwear | 45.9 | 52.6 | 61.6 | 71.8 | 85.2 | 90.8 | 100.0 | 109.5 | 119.7 |
| Rent | 38.7 | 42.6 | 52.7 | 66.6 | 78.1 | 90.5 | 100.0 | 109.7 | 119.7 |
| Total less food | 42.3 | 47.1 | 57.0 | 67.6 | 78.9 | 90.8 | 100.0 | 109.1 | 118.3 |
| **Wholesale prices (1980 = 100)** | | | | | | | | | |
| Total | 360.2 | 408.5 | 473.7 | 553.0 | 615.6 | 689.0 | 748.7 | 807.1 | 856.7 |
| Imported goods | 427.9 | 481.3 | 541.7 | 620.6 | 698.9 | 783.9 | 856.1 | 915.8 | 930.8 |
| Industrial goods | 351.9 | 395.0 | 467.7 | 550.4 | 631.5 | 715.9 | 769.8 | 831.9 | 890.9 |
| Petroleum | 264.6 | 274.0 | 353.9 | 437.3 | 539.3 | 661.5 | 672.0 | 701.3 | 791.7 |
| **Wages (1985 = 100)** | | | | | | | | | |
| Hourly earnings in manufacturing[2] | 146.3 | 176.4 | 210.5 | 245.7 | 279.4 | 308.8 | 349.1 | 395.4 | 434.5 |

1. Break in series in 1994.
2. Wage earners in enterprises employing at least 10 persons.

*Source:* National Statistical Service, *Monthly Statistical Bulletin* and OECD, *Main Economic Indicators.*

Table G. **Foreign trade by commodity group**

US$ million

| | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|
| | Exports, fob | | | | | | |
| **Total** | 7 543.2 | 8 020.6 | 8 663.0 | 9 842.0 | 8 777.3 | 9 174.2 | 10 952.7 |
| Food and live animals | 1 567.9 | 1 627.7 | 1 922.3 | 2 085.3 | 1 736.7 | 1 834.9 | 1984.1 |
| Beverages and tobacco | 368.7 | 443.6 | 493.2 | 650.8 | 550.2 | 502.1 | 615.4 |
| *of which:* Tobacco unmanufactured | 254.6 | 312.1 | 345.4 | 394.2 | 349.6 | 284.3 | 308.8 |
| Crude materials, inedible, except fuels | 523.2 | 456.7 | 455.7 | 425.3 | 506.2 | 583.3 | 777.5 |
| Mineral fuels, lubricants | 410.4 | 588.0 | 773.4 | 518.9 | 688.5 | 944.7 | 719.9 |
| Animal and vegetable oils and fats | 355.6 | 302.5 | 222.2 | 550.7 | 314.2 | 335.3 | 605.2 |
| Chemicals | 300.1 | 314.9 | 338.0 | 372.7 | 409.7 | 389.8 | 531.8 |
| Manufactured goods classified chiefly by material | 1 913.9 | 1 889.6 | 1 952.5 | 2 167.7 | 1 763.7 | 1 895.7 | 2 357.9 |
| *of which:* Leather and leather products | 89.1 | 75.3 | 84.8 | 103.8 | 62.3 | 58.3 | 55.3 |
| Textiles | 494.4 | 500.1 | 533.3 | 520.2 | 446.0 | 418.5 | 520.7 |
| Iron and steel | 457.9 | 415.0 | 412.4 | 471.1 | 251.2 | 307.4 | 386.5 |
| Aluminium | 264.9 | 279.3 | 287.2 | 311.9 | 269.5 | 313.7 | 436.3 |
| Machinery and transport equipment | 248.7 | 337.5 | 372.3 | 472.3 | 549.9 | 610.8 | 878.7 |
| Miscellaneous, manufactured articles | 1 627.6 | 1 864.5 | 1 935.5 | 2 360.1 | 2 115.2 | 1 919.3 | 2 260.8 |
| *of which:* Clothing | 1 471.3 | 1 675.2 | 1 749.6 | 2 146.0 | 1 844.0 | 1 623.3 | 1 844.7 |
| Footwear | 55.4 | 59.0 | 32.7 | 41.2 | 49.6 | 33.9 | 38.4 |
| | Imports, cif | | | | | | |
| **Total** | 16 102.9 | 19 701.4 | 21 650.1 | 23 406.8 | 22 759.8 | 20 843.1 | 25 881.8 |
| Food and live animals | 2 230.8 | 2 506.9 | 2 379.3 | 2 762.6 | 2 561.1 | 2 680.8 | 3 399.0 |
| Beverages and tobacco | 243.2 | 323.7 | 366.5 | 464.1 | 493.4 | 439.5 | 532.6 |
| Crude materials, inedible, except fuels | 841.2 | 954.1 | 945.7 | 723.1 | 643.4 | 692.1 | 896.2 |
| Mineral fuels, lubricants | 1 037.3 | 1 530.5 | 2 097.3 | 2 301.3 | 2 430.7 | 2 056.4 | 1 890.8 |
| *of which:* Petroleum, crude | 781.5 | 997.2 | 1 368.9 | 1 718.3 | 1 733.0 | 1 427.5 | 1 329.9 |
| Petroleum products | 201.7 | 456.1 | 623.7 | 490.8 | 612.0 | 550.9 | 467.1 |
| Animal and vegetable oils and fats | 51.2 | 81.4 | 190.5 | 73.1 | 55.1 | 84.9 | 89.0 |
| Chemicals | 1 710.0 | 2 068.1 | 2 238.7 | 2 472.5 | 2 504.4 | 2 631.8 | 3 399.3 |
| Manufactured goods classified chiefly by material | 3 644.7 | 4 306.3 | 4 300.9 | 4 263.0 | 3 809.8 | 3 931.3 | 5 194.7 |
| Machinery and transport equipment | 4 967.1 | 6 120.4 | 7 124.1 | 7 969.6 | 8 012.5 | 5 951.8 | 7 111.2 |
| *of which:* Road motor vehicles | 1 240.4 | 1981.0 | 2 417.0 | 3 112.0 | 2 437.3 | 1 657.0 | 2 216.2 |
| Aircraft | 70.7 | 109.9 | 324.5 | 165.7 | 266.8 | 116.2 | 101.2 |
| Ships and boats | 805.1 | 527.1 | 886.1 | 819.7 | 1 486.0 | 586.8 | 429.6 |
| Other machinery and equipment | 2 850.9 | 3 502.5 | 3 496.5 | 3 872.2 | 3 822.6 | 3 591.8 | 4 364.2 |

*Source:* OECD, *Foreign Trade Statistics, Series C.*

Table H. **Balance of payments**

US$ million

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|
| Exports, fob | 5 612 | 5 933 | 5 994 | 6 365 | 6 797 | 6 009 | 5 035 | 5 219 | 5 783 |
| Imports, fob | 11 112 | 12 005 | 13 377 | 16 543 | 16 909 | 17 612 | 15 592 | 16 588 | 20 292 |
| **Trade balance** | –5 500 | –6 072 | –7 383 | –10 178 | –10 112 | –11 603 | –10 557 | –11 369 | –14 509 |
| Invisibles, net | 4 277 | 5 114 | 4 822 | 6 641 | 8 539 | 9 463 | 9 810 | 11 223 | 11 645 |
| *of which:* Tourism, net | 1 761 | 1 663 | 1 160 | 1 498 | 1 552 | 2 084 | 2 331 | 2 780 | 2 812 |
| Transportation, net | –409 | –343 | –516 | –545 | –688 | –604 | –298 | –404 | –737 |
| Migrant remittances | 1 334 | 1 675 | 1 350 | 1 774 | 2 115 | 2 366 | 2 360 | 2 576 | 3 071 |
| Receipts in convertible drachmas | 1 135 | 1 327 | 1 494 | 2 242 | 2 765 | 3 012 | 2 290 | 2 640 | 2 810 |
| **Current balance**[1] | –1 223 | –958 | –2 561 | –3 537 | –1 573 | –2 140 | –747 | –146 | –2 864 |
| Long-term capital | 1 387 | 1 438 | 1 941 | 2 975 | 3 587 | 2 191 | 5 351 | 6 668 | 4 750 |
| **Basic balance** | 164 | 480 | –620 | –562 | 2 014 | 51 | 4 604 | 6 522 | 1 886 |
| Short-term capital, including errors and omissions | 199 | 556 | –264 | –14 | –485 | –218 | –1 630 | –216 | 198 |
| **Balance on non monetary transactions** | 363 | 1 036 | –884 | –576 | 1 529 | –167 | 2 974 | 6 306 | 2 084 |
| Private monetary institutions short-term capital | 440 | 125 | 550 | 618 | 141 | –24 | 46 | 60 | –2 115 |
| **Balance of official settlements** | 803 | 1 161 | –334 | 42 | 1 670 | –191 | 3 020 | 6 366 | –31 |
| Change in reserves (+ = increase) | 806 | 1 194 | –342 | 50 | 1 662 | –194 | 3 016 | 6 366 | –31 |
| Gold | –316 | 226 | 36 | 74 | 42 | 128 | –26 | –45 | –8 |
| Currency assets | 906 | 1 053 | –377 | 54 | 1 645 | –314 | 2 940 | 6 360 | –36 |
| Reserve position in IMF | –13 | 4 | 0 | 0 | 0 | 0 | 0 | 12 | 0 |
| Special Drawing Rights | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 45 | 0 |

1. Includes receipts in convertible drachmas for services rendered.

*Source:* IMF balance of payments statistics and Bank of Greece.

Table I. **Money and banking**

Billion drachmas, end of period

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[1] |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 1. Money | | | | | |
| 1. **M1** (2 + 3) | 1 202.3 | 1 517.8 | 1 880.8 | 2 105.2 | 2 357.1 | 2 687.6 | 3 299.4 | 3 718.1 | 4 118.2 |
| 2. Currency in circulation | 735.5 | 969.3 | 1 162.0 | 1 255.5 | 1 410.1 | 1 512.0 | 1 687.7 | 1 863.6 | 2 050.0 |
| 3. Sight deposits | 466.9 | 548.4 | 718.8 | 849.7 | 947.0 | 1 175.5 | 1 611.7 | 1 854.5 | 2 068.2 |
| 4. Private | 237.7 | 295.6 | 421.4 | 487.4 | 558.2 | 711.7 | 1 105.9 | 1 285.4 | 1 349.3 |
| 5. Public entities | 173.3 | 184.0 | 245.6 | 286.3 | 295.8 | 370.2 | 388.1 | 453.3 | 596.3 |
| 6. Public enterprises | 55.8 | 68.8 | 51.7 | 76.0 | 93.0 | 93.6 | 117.7 | 115.8 | 122.6 |
| 7. **Quasi-money** (8 + 9 + 12) | 5 656.6 | 6 960.9 | 7 800.0 | 8 493.7 | 9 279.5 | 9 978.4 | 12 166.7 | 13 962.7 | 15 725.2 |
| 8. Savings deposits | 3 849.8 | 4 852.5 | 5 549.2 | 6 228.9 | 6 915.3 | 7 709.7 | 8 811.5 | 10 445.2 | 12 234.7 |
| 9. Time deposits | 1 658.1 | 1 939.3 | 2 055.9 | 2 040.1 | 2 071.7 | 1 958.3 | 3 008.2 | 3 170.9 | 3 162.9 |
| 10. Private | 1 634.8 | 1 902.2 | 2 024.1 | 2 010.5 | 2 052.6 | 1 943.9 | 2 994.2 | 3 119.2 | 3 127.0 |
| 11. Public entities | 23.4 | 37.1 | 31.7 | 29.5 | 19.1 | 14.4 | 14.0 | 51.7 | 35.9 |
| 12. Other deposits[2] | 148.6 | 169.1 | 194.9 | 224.7 | 292.5 | 310.3 | 347.0 | 346.6 | 327.6 |
| 13. Bank bonds | 384.7 | 481.2 | 594.0 | 598.7 | 673.7 | 703.5 | 838.4 | 570.8 | 92.0 |
| 14. Repos | 0.0 | 0.0 | 48.5 | 420.9 | 978.6 | 1 893.9 | 317.8 | 96.0 | 99.5 |
| 15. **M3** (2 + 4 + 8 + 10 + 13 + 14) | 6 842.5 | 8 500.9 | 9 799.3 | 11 001.9 | 12 588.5 | 14 474.8 | 15 755.5 | 17 380.2 | 18 952.5 |
| 16. Treasury bills | 412.3 | 467.9 | 1 017.4 | 2 384.2 | 3 308.7 | 3 904.4 | 4 981.7 | 5 466.7 | 6 631.1 |
| 17. Bonds up to one year | 0.0 | 0.0 | 144.6 | 122.7 | 204.3 | 187.8 | 412.2 | 42.5 | 0.0 |
| 18. **M4** (15 + 16 + 17) | 7 254.8 | 8 968.8 | 10 961.3 | 13 508.8 | 16 101.5 | 18 567.0 | 21 149.4 | 22 889.4 | 25 583.6 |
| *Memorandum items:* | | | | | | | | | |
| Total private deposits in drachmas | 5 811.7 | 7 146.5 | 8 109.5 | 8 865.8 | 9 690.5 | 10 555.3 | 13 126.3 | 15 093.9 | 16 962.2 |
| Total private deposits in foreign exchange | 1 501.6 | 1 870.2 | 2 240.1 | 2 701.4 | 3 445.0 | 4 117.4 | 4 439.0 | 4 999.1 | 5 140.8 |
| Total public sector deposits | 506.7 | 668.3 | 777.6 | 875.7 | 1 010.5 | 1 137.0 | 698.6 | 769.5 | 877.1 |
| *of which:* Deposits of public entities (Law 1611/50) | 174.4 | 259.2 | 317.7 | 343.3 | 422.3 | 494.2 | 0.0 | 0.0 | 0.0 |

Table I. **Money and banking** *(cont.)*

Billion drachmas, end of period

| | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[1] |
|---|---|---|---|---|---|---|---|---|---|
| | 2. Banking | | | | | | | | |
| **Total bank lending** | 4 558.8 | 5 614.9 | 5 925.9 | 6 442.6 | 7 164.5 | 7 558.2 | 8 287.5 | 10 021.2 | 11 612.8 |
| **1. By sector** | | | | | | | | | |
| Private sector | 3 176.3 | 3 917.2 | 4 457.1 | 5 134.7 | 5 828.2 | 6 008.9 | 6 824.7 | 8 328.0 | 9 840.2 |
| Agriculture | 782.8 | 767.1 | 919.2 | 1 031.6 | 1 121.0 | 841.0 | 899.7 | 1 031.0 | 1 149.4 |
| Manufacturing and mining | 1 378.9 | 1 697.4 | 1 764.6 | 2 002.5 | 2 200.0 | 2 191.3 | 2 437.9 | 2 788.9 | 3 115.7 |
| Short and medium term | 860.3 | 1 114.5 | 1 148.0 | 1 326.5 | 1 526.3 | 1 518.6 | 1 725.2 | 2 066.4 | 2 370.6 |
| Long term | 518.6 | 582.9 | 616.6 | 676.0 | 673.7 | 672.7 | 712.7 | 722.5 | 745.1 |
| Trade | 232.5 | 367.5 | 499.7 | 622.7 | 825.9 | 1 042.9 | 1 231.8 | 1 584.8 | 1 893.2 |
| Housing | 462.2 | 615.4 | 701.5 | 777.7 | 857.9 | 929.6 | 1 020.6 | 1 219.1 | 1 551.0 |
| Other | 320.0 | 469.8 | 572.1 | 700.2 | 823.3 | 1 004.2 | 1 234.6 | 1 704.3 | 2 130.9 |
| Public sector | 1 382.5 | 1 697.7 | 1 468.9 | 1 307.8 | 1 336.3 | 1 549.2 | 1 462.8 | 1 693.2 | 1 772.6 |
| Public enterprises | 666.1 | 793.9 | 636.9 | 497.3 | 432.4 | 497.4 | 543.1 | 581.0 | 682.6 |
| Public entities | 644.6 | 781.8 | 676.1 | 633.9 | 610.6 | 713.6 | 621.7 | 663.6 | 655.0 |
| Government purchasing agencies | 71.8 | 122.0 | 155.8 | 176.6 | 293.4 | 338.3 | 298.0 | 448.6 | 435.0 |
| **2. By lending institution** | | | | | | | | | |
| Bank of Greece | 7.2 | 9.3 | 8.8 | 7.5 | 4.3 | 3.9 | 4.9 | 14.1 | 19.9 |
| Commercial banks | 2 008.1 | 2 479.1 | 2 659.4 | 2 862.8 | 3 277.0 | 3 836.5 | 4 483.4 | 5 739.1 | 7 034.8 |
| Specialised credit institutions | 2 543.5 | 3 126.6 | 3 257.7 | 3 572.3 | 3 883.2 | 3 717.8 | 3 799.2 | 4 268.0 | 4 558.1 |

1. Provisional data.
2. Excluding deposits of public entities according to Law 1611/50.

*Source:* Bank of Greece and its *Monthly Statistical Bulletin.*

Table J. **Public sector**

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996[1] |
|---|---|---|---|---|---|---|
| | Per cent of GDP | | | | | |
| **Budgetary indicators:** | | | | | | |
| **General government accounts**[2] | | | | | | |
| Primary receipts (excluding interest) | 30.6 | 31.3 | 32.0 | 32.8 | 34.0 | 33.9 |
| Primary expenditure (excluding interest) | 35.0 | 34.3 | 35.8 | 33.7 | 34.7 | 33.3 |
| Primary budget balance | –4.3 | –3.0 | –3.8 | –0.9 | –0.7 | 0.6 |
| Net interest | –7.2 | –9.3 | –10.4 | –11.2 | –8.5 | –8.0 |
| General government budget balance | –11.5 | –12.3 | –14.2 | –12.1 | –9.2 | –7.4 |
| *Memorandum item:* Saving | –6.5 | –7.1 | –8.3 | –8.7 | –6.0 | –5.5 |
| **Structure of expenditure and taxes**[2] | | | | | | |
| General government current expenditure | 39.4 | 40.7 | 42.7 | 44.5 | 43.9 | 42.8 |
| Transfers | 14.9 | 14.8 | 15.5 | 16.0 | 16.4 | 16.6 |
| Subsidies | 0.6 | 0.5 | 0.6 | 0.5 | 0.5 | 0.4 |
| Tax receipts | 30.6 | 31.3 | 32.0 | 32.8 | 34.0 | 33.9 |
| Personal income tax | 3.8 | 3.7 | 3.8 | 4.7 | 5.0 | 5.0 |
| Corporate taxes | 1.7 | 1.6 | 1.7 | 2.1 | 2.2 | 2.3 |
| Social security contributions | 11.2 | 11.2 | 12.1 | 12.0 | 12.4 | 12.4 |
| Consumption taxes | 13.8 | 14.8 | 14.3 | 14.1 | 14.3 | 14.3 |
| *of which:* Value added tax | 7.3 | 7.4 | 7.7 | 7.4 | 7.3 | . . |
| | Per cent | | | | | |
| **Tax rates** | | | | | | |
| Income tax as a percentage of total tax | 18.2 | 17.0 | 17.4 | 20.6 | 21.3 | 21.3 |
| Personal income tax elasticity | 1.2 | 0.6 | 1.1 | 2.6 | 2.4 | 1.0 |
| Value added tax rate | 18.0 | 18.0 | 18.0 | 18.0 | 18.0 | 18.0 |

1. Provisional data.
2. National account basis.

*Source:* Data submitted by national authorities and OECD estimates.

*BASIC STATISTICS:*

*INTERNATIONAL COMPARISONS*

## BASIC STATISTICS: INTERNATIONAL COMPARISONS

| | Units | Reference period[1] | Australia | Austria | Belgium | Canada | Denmark | Finland | France | Germany | Greece |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | | |
| Total | Thousands | 1994 | 17 840 | 8 031 | 10 124 | 29 251 | 5 206 | 5 088 | 57 960 | 81 407 | 10 430 |
| Inhabitants per sq. km | Number | 1994 | 2 | 96 | 332 | 3 | 121 | 15 | 106 | 228 | 79 |
| Net average annual increase over previous 10 years | % | 1994 | 1.4 | 0.6 | 0.3 | 1.6 | 0.2 | 0.4 | 0.5 | 2.9 | 0.5 |
| **Employment** | | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 7 680 (93) | 3 737 | 3 724 (92) | 13 292 | 2 508 | 2 015 | 21 781 (93) | 35 894 | 3 790 |
| *of which:* Agriculture | % of TCE | 1994 | 5.3 (93) | 7.2 | 2.6 (92) | 4.1 | 5.1 | 8.3 | 5.1 (93) | 3.3 | 20.8 |
| Industry | % of TCE | 1994 | 23.7 (93) | 33.2 | 27.7 (92) | 22.6 | 26.8 | 26.8 | 27.7 (93) | 37.6 | 23.6 |
| Services | % of TCE | 1994 | 71 (93) | 59.6 | 69.7 (92) | 73.3 | 68.1 | 64.9 | 67.2 (93) | 59.1 | 55.5 |
| **Gross domestic product (GDP)** | | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 331.6 | 198.1 | 227.9 | 544 | 146.7 | 97.2 | 1 328.5 | 1 832.3 | 73.1 (93) |
| Per capita | US$ | 1994 | 18 588 | 24 670 | 22 515 | 18 598 | 28 181 | 19 106 | 22 944 | 27 826 | 7 051 (93) |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 327.9 | 162.3 | 204.2 | 596.7 | 107 | 82.5 | 1 111.8 | 1 601.7 | 118 |
| Per capita | US$ | 1994 | 18 382 | 20 210 | 20 166 | 20 401 | 20 546 | 16 208 | 19 201 | 24 325 | 1 450 |
| Average annual volume growth over previous 5 years | % | 1994 | 2.2 | 2.5 | 1.6 | 1.1 | 1.9 | –1.6 | 1.1 | 2.6 | 1.4 (93) |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 21.4 | 24.8 | 17.4 | 18.7 | 14.8 | 14.3 | 18.1 | 18.5 | 17.4 (93) |
| *of which:* Machinery and equipment | % of GDP | 1993 | 9.8 | 8.7 | 7.8 | 6.2 | 7.2 | 5.90 | 8.10 | 7.5 | 7.8 |
| Residential construction | % of GDP | 1993 | 5.2 | 6.3 | 4.6 | 6.1 | 2.9 | 3.7 | 4.9 | 6.5 | 3.6 |
| Average annual volume growth over previous 5 years | % | 1994 | 0.8 | 3.7 | 0.4 | –0.1 | –2.8 | –12.9 | –1 | 0.8 | 2.7 (93) |
| **Gross saving ratio**[4] | % of GDP | 1994 | 16.8 | 25.3 | 22 | 16 | 17 | 16.6 | 19 | 21 | 15.5 (93) |
| **General government** | | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 17.5 | 18.8 | 15 | 20.2 | 25.3 | 22.4 | 19.6 | 17.7 | 19.1 (93) |
| Current disbursements[5] | % of GDP | 1993 | 36.9 | 48.4 | 55.3 | 49 | 61.1 | 58.9 | 51.5 | 45.6 | 51.2 |
| Current receipts | % of GDP | 1993 | 33.5 | 48.6 | 50.1 | 43 | 58.3 | 52.5 | 46.8 | 45.7 | 40.2 |
| **Net official development assistance** | % of GNP | 1993 | 0.34 | 0.31 | 0.41 | 0.46 | 1.03 | 0.76 | 0.66 | 0.44 | .. |
| **Indicators of living standards** | | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 10 803 | 10 546 | 12 090 | 11 863 | 10 042 | 8 814 | 11 395 | 10 733 | 6 367 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 430 | 382 | 387 | 469 | 311 | 386 | 413 | 480[8] | 169 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 464 | 432 | 410 | 586 | 577 | 544 | 511 | 420[8] | 413 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 480 | 478 | 451 | 639 | 536 | 501 | 407 | 556[8] | 197 |
| Doctors, per 1 000 inhabitants | Number | 1993 | 2.2 (91) | 2.3 | 3.7 | 2.2 | 2.8 (92) | 2.6 (92) | 2.8 | 3.2 (92) | 3.8 (92) |
| Infant mortality per 1 000 live births | Number | | 6.1 | 6.5 | 8 | 6.8 | 5.4 | 4.4 | 6.5 | 5.8 | 8.5 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | 3 | 5.5 | 3.7 | 3.3 | 3.5 | 4.8 | 3.5 | 5.2 | 14.6 |
| Consumer prices | % | 1994 | 3 | 3.4 | 2.8 | 2.8 | 2.1 | 3.3 | 2.5 | 3.3 | 16.2 |
| **Foreign trade** | | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 47 363 | 44 881 | 137 259[7] | 165 358 | 41 850 | 29 514 | 235 337 | 422 243 | 8 958 |
| As % of GDP | % | 1994 | 14.3 | 22.7 | 60.2 | 30.4 | 28.5 | 30.4 | 17.7 | 23 | 11.5 (93) |
| Average annual increase over previous 5 years | % | 1994 | 5 | 6.7 | 6.5 | 7.1 | 8.3 | 4.9 | 5.6 | 4.4 | 3.4 |
| Imports of goods, cif* | Mill. US$ | 1994 | 49 731 | 55 071 | 126 006[7] | 148 297 | 35 932 | 23 091 | 220 508 | 376 566 | 21 111 |
| As % of GDP | % | 1994 | 15 | 27.8 | 55.3 | 27.3 | 24.5 | 23.8 | 16.6 | 20.6 | 30.1 (93) |
| Average annual increase over previous 5 years | % | 1994 | 4 | 7.2 | 5 | 5.4 | 6.1 | –1.3 | 3.5 | 6.9 | 5.4 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 7 730 | 11 523 | 9 505[7] | 8 416 | 6 203 | 7 304 | 17 986 | 52 994 | 9 924 |
| As ratio of average monthly imports of goods | Ratio | 1994 | 1.9 | 2.5 | 0.9 | 0.7 | 2.1 | 3.8 | 1 | 1.7 | 5.6 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

## BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

| | Units | Reference period[1] | Iceland | Ireland | Italy | Japan | Luxembourg | Mexico | Netherlands | New Zealand |
|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | |
| Total | Thousands | 1994 | 267 | 3 571 | 57 190 | 124 960 | 398 | 93 010 | 15 382 | 3 526 |
| Inhabitants per sq. km | Number | 1994 | 3 | 51 | 190 | 331 | 153 | 47 | 377 | 13 |
| Net average annual increase over previous 10 years | % | 1994 | 1.1 | 0.1 | 0 | 0.4 | 0.8 | 2 | 0.6 | 0.8 |
| **Employment** | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 138 | 1 168 (93) | 20 152 (93) | 64 530 | 162 (91) | 32 439 | 6 631 | 1 560 |
| *of which:* Agriculture | % of TCE | 1994 | 9.4 | 12.7 (93) | 7.5 (93) | 5.8 | 3.7 (91) | 25.8 | 4 | 10.4 |
| Industry | % of TCE | 1994 | 26.1 | 27.7 (93) | 33 (93) | 34 | 31.5 (91) | 22.2 | 23 | 25 |
| Services | % of TCE | 1994 | 65.2 | 59.7 (93) | 59.6 (93) | 60.2 | 64.8 (91) | 52.1 | 73 | 64.6 |
| **Gross domestic product (GDP)** | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 6.2 | 52 | 1 017.8 | 4 590 | 10.6 (92) | 371.2 | 334.3 | 51.2 |
| Per capita | US$ | 1994 | 23 199 | 14 550 | 17 796 | 36 732 | 27 073 (92) | 3 991 | 21 733 | 14 513 |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 5.1 | 54.3 | 1 068.4 | 2 593.7 | 11.7 | 673.3 | 285.9 | 57.3 |
| Per capita | US$ | 1994 | 19 271 | 15 212 | 18 681 | 20 756 | 29 454 | 7 239 | 18 589 | 16 248 |
| Average annual volume growth over previous 5 years | % | 1994 | 0.6 | 4.7 | 1 | 2.1 | 4.1 (92) | 3 | 2.3 | 2.5 |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 15.2 | 15.1 | 16.4 | 28.6 | 20.4 (93) | 20.7 | 19.3 | 20 |
| *of which:* Machinery and equipment | % of GDP | 1993 | 3.9 | 6.3 | 7.4 | 11.5 | .. | 9.4 | 8.6 | 9.3 |
| Residential construction | % of GDP | 1993 | 4.4 | 4.1 | 5.3 | 5.4 | .. | 4.9 | 5.1 | 4.9 |
| Average annual volume growth over previous 5 years | % | 1994 | –4 | 1 | –2.3 | 1.4 | 6.5 (92) | 7.7 | 0.4 | 5.8 |
| **Gross saving ratio**[4] | % of GDP | 1994 | 16.9 | 19.5 | 18.8 | 31.2 | 60.2 (92) | 15.1 | 24.4 | 20.7 |
| **General government** | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 20.6 | 16 | 17.1 | 9.8 | 17.1 (92) | 11.8[9] | 14.2 | 14.7 |
| Current disbursements[5] | % of GDP | 1993 | 34.9 | .. | 53.2 | 26.9 | .. | .. | 55.4 | .. |
| Current receipts | % of GDP | 1993 | 35.9 | .. | 47.1 | 32.9 | .. | .. | 54.5 | .. |
| **Net official development assistance** | % of GNP | 1993 | .. | 0.15 | 0.42 | 0.27 | 0.34 (92) | .. | 0.88 | 0.22 |
| **Indicators of living standards** | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 11 546 | 7 750 | 11 029 | 11 791 | 15 545 | 4 853 | 10 726 | 9 266 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 464 | 228 | 478 | 282 | 470 | 85 | 356 | 440 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 527 | 300 | 400 | 454 | 511 | 70 | 477 | 436 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 319 | 276 | 421 | 613 | 267 | 148 | 485 | 443 |
| Doctors, per 1 000 inhabitants | Number | 1993 | 3 | 1.7 (92) | 1.7 (91) | 1.7 (92) | 2.1 (92) | 1 | 2.5 (90) | 2 |
| Infant mortality per 1 000 live births | Number | | 4.8 | 5.9 | 7.3 | 4.5 (92) | 8.5 (92) | 18 | 6.3 | 7.3 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | .. | 4.6 | 5.9 | 2.4 | .. | 5.3 | 3.2 | 2.1 |
| Consumer prices | % | 1994 | 6.3 | 2.7 | 5.2 | 2 | 3.1 | 16.1 | 2.8 | 2.5 |
| **Foreign trade** | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 1 628 | 34 125 | 189 802 | 396 149 | .. | 60 882 | 155 084 | 12 169 |
| As % of GDP | % | 1994 | 26.3 | 65.7 | 18.6 | 8.6 | .. | 16.4 | 46.4 | 23.8 |
| Average annual increase over previous 5 years | % | 1994 | 2.7 | 10.5 | 6.2 | 7.6 | .. | 21.7 | 7.6 | 6.5 |
| Imports of goods, cif* | Mill. US$ | 1994 | 1 464 | 25 812 | 167 690 | 274 916 | .. | 79 346 | 139 800 | 11 859 |
| As % of GDP | % | 1994 | 23.6 | 49.7 | 16.5 | 6 | .. | 21.4 | 41.8 | 23.2 |
| Average annual increase over previous 5 years | % | 1994 | 0.7 | 8.1 | 1.9 | 5.5 | .. | 25.5 | 6.1 | 6.1 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 201 | 4 189 | 22 102 | 86 214 | .. | 4 301 | 23 655 | 2 540 |
| As ratio of average monthly imports of goods | Ratio | 1994 | 1.6 | 1.9 | 1.6 | 3.8 | .. | 0.7 | 2 | 2.6 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

## BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

| | Units | Reference period[1] | Norway | Portugal | Spain | Sweden | Switzerland | Turkey | United Kingdom | United States |
|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | |
| Total | Thousands | 1994 | 4 337 | 9 900 | 39 150 | 8 781 | 6 994 | 60 573 | 58 375 | 260 651 |
| Inhabitants per sq. km | Number | 1994 | 13 | 107 | 78 | 20 | 169 | 78 | 238 | 28 |
| Net average annual increase over previous 10 years | % | 1994 | 0.5 | –0.1 | 0.2 | 0.5 | 0.7 | 2.1 | 0.3 | 1 |
| **Employment** | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 1970 (93) | 4 372 | 11 760 | 3 926 | 3 772 | 19 664 | 25 044 (93) | 123 060 |
| *of which:* Agriculture | % of TCE | 1994 | 5.6 (93) | 11.5 | 9.8 | 3.4 | 4 | 44.8 | 2.2 (93) | 2.9 |
| Industry | % of TCE | 1994 | 23.1 (93) | 32.8 | 30.1 | 25 | 28.8 | 22.2 | 26.2 (93) | 24 |
| Services | % of TCE | 1994 | 71.3 (93) | 55.7 | 60.2 | 71.6 | 67.2 | 33 | 71.6 (93) | 73.1 |
| **Gross domestic product (GDP)** | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 103.4 (93) | 87 | 482.4 | 196.6 | 257.3 | 130.7 | 1 019.5 | 6 649.8 |
| Per capita | US$ | 1994 | 23 984 (93) | 8 792 | 12 321 | 22 389 | 36 790 | 2 157 | 17 468 | 25 512 |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 95.3 | 122 | 531.7 | 153 | 167.4 | 319.3 | 1 030.2 | 6 649.8 |
| Per capita | US$ | 1994 | 21 968 | 12 335 | 13 581 | 17 422 | 23 942 | 5 271 | 17 650 | 25 512 |
| Average annual volume growth over previous 5 years | % | 1994 | 2.1 (93) | 1.4 | 1.5 | –0.3 | 0.5 | 3.6 | 0.8 | 2.1 |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 22 (93) | 25.7 | 19.8 | 13.7 | 22.8 | 24.5 | 15 | 17.2 |
| *of which:* Machinery and equipment | % of GDP | 1993 | .. | .. | 5.7 | 5.7 | 7.5 | 10.3 | 7.3 | 7.7 |
| Residential construction | % of GDP | 1993 | .. | .. | 4.1 | 4 | 14.9[10] | 9.1 | 3.1 | 4 |
| Average annual volume growth over previous 5 years | % | 1994 | –3.93 | 2.7 | –1.2 | –7.6 | –0.4 | 5.1 | –2.1 | 4.6 |
| **Gross saving ratio**[4] | % of GDP | 1994 | 21.9 (93) | 24.2 | 18.8 | 13.7 | 29.3 | 22.5 | 13.5 | 16.2 |
| **General government** | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 22.1 (93) | 17.2 | 16.9 | 27.3 | 14.1 | 11.7 | 21.6 | 16.4 |
| Current disbursements[5] | % of GDP | 1993 | .. | .. | 43.7 | 67.3 | 36.7 | .. | 42.7 | 35.8 |
| Current disbursements[5] | % of GDP | 1993 | .. | .. | 40.1 | 59 | 36 | .. | 36.8 | 31.7 |
| Current receipts | % of GDP | 1993 | 1.23 | 0.36 | 0.32 | 1.33 | 0.49 | .. | 0.34 | 0.19 |
| **Net official development assistance** | % of GNP | 1993 | 9 826 | 7 780 | 8 412 | 9 240 | 13 730 | 3 617 | 10 942 | 16 444 |
| **Indicators of living standards** | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 378 | 260 | 307 | 418 | 441 | 29 | 361 | 568 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 515 | 273 | 340 | 687 | 603 | 143 | 445 | 553 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 423 | 187 | 400 | 468 | 406 | 175 | 434 | 814 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 3.2 (92) | 2.9 | 4.1 | 3 | 3 | 0.9 | 1.5 (92) | 2.3 (92) |
| Doctors, per 1 000 inhabitants | Number | 1993 | 5 | 8.7 | 7.6 | 4.8 | 5.6 | 52.6 | 6.6 | 8.5 (92) |
| Infant mortality per 1 000 live births | Number | | 4 | .. | 7.2 | 5.4 | .. | .. | 6.7 | 2.8 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | 2.7 | 9 | 5.6 | 5.7 | 3.9 | 73 | 4.6 | 3.6 |
| Consumer prices | % | 1994 | 34 645 | 17 072 | 73 129 | 61 122 | 70 467 | 18 456 | 205 170 | 512 627 |
| **Foreign trade** | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 30.9 (93) | 19.6 | 15.2 | 31.1 | 27.4 | 14.1 | 20.1 | 7.7 |
| As % of GDP | % | 1994 | 5 | 6.1 | 10.5 | 3.4 | 6.4 | 9.5 | 6.1 | 7.1 |
| Average annual increase over previous 5 years | % | 1994 | 27 345 | 25 967 | 92 182 | 51 730 | 68 126 | 22 976 | 227 026 | 663 256 |
| Imports of goods, cif* | Mill. US$ | 1994 | 23.3 (93) | 29.9 | 19.1 | 26.3 | 26.5 | 17.6 | 22.3 | 10 |
| As % of GDP | % | 1994 | 2.9 | 6.6 | 5.2 | 1 | 3.2 | 37.9 | 2.8 | 7 |
| Average annual increase over previous 5 years | % | 1994 | 13 033 | 10 627 | 28 475 | 15 929 | 23 790 | 4 911 | 28 094 | 43 350 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 5.7 | 4.9 | 3.7 | 3.7 | 4.2 | 2.6 | 1.5 | 0.8 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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